

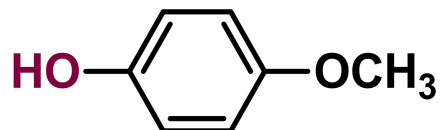
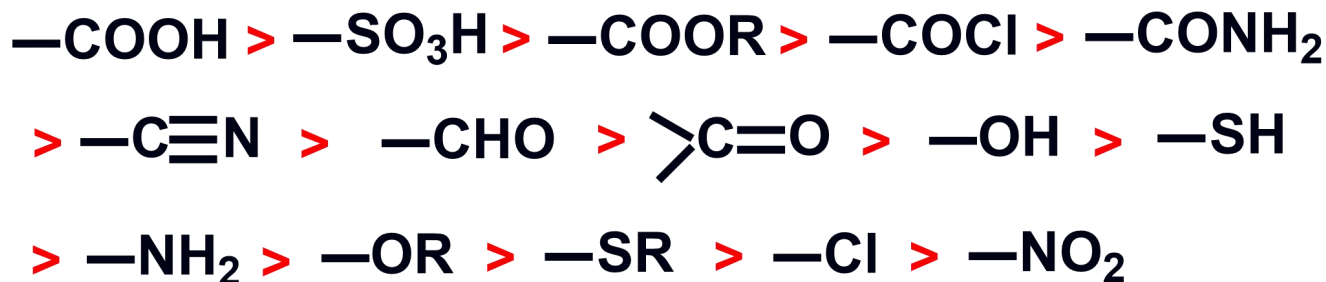
命名或写出下列化合物的结构（每题1.5分，共计15分）

- 写对母体给1分
- 认真画一下，注意找到最长的碳链
- 不要纠结，不会就过



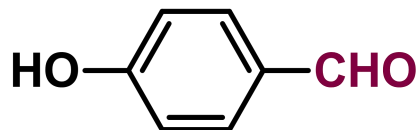
多官能团化合物的命名

按照“官能团的优先次序”，以较优的官能团为母体



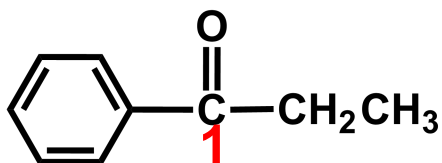
4-甲氧基苯酚

4-Methoxyphenol

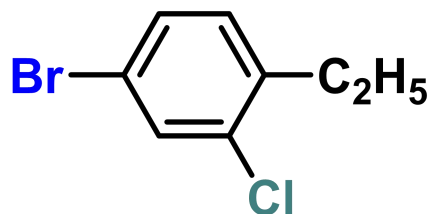


4-羟基苯甲醛

4-Hydroxybenzaldehyde

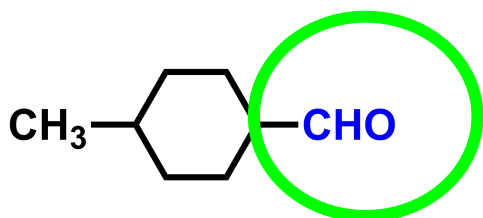


1-苯基**丙**-1-酮

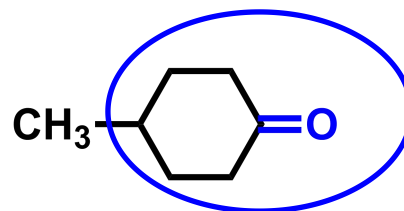


4-溴-2-氯-1-乙基苯

4-bromo-2-chloro-1-ethylbenzene

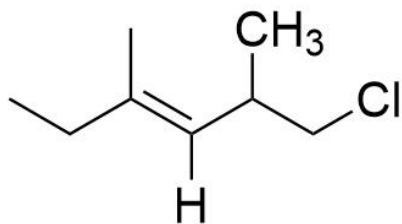


4-甲基环己基**甲**醛

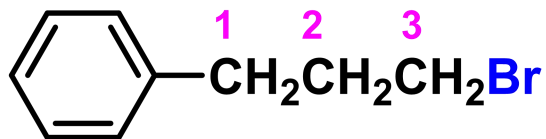


4-甲基**环己**酮

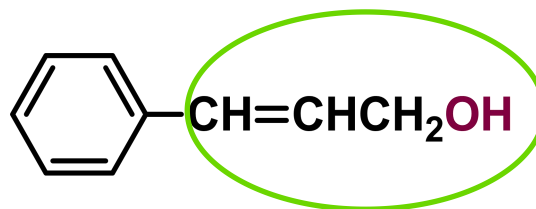
4-methylcyclohexanone



(E) -1-氯-2,4-二甲基己-3-烯

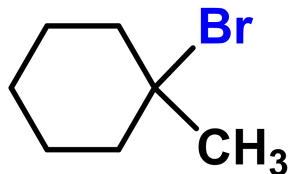


- **(3-溴丙基)苯**
- **3-苯基-1-溴丙烷**

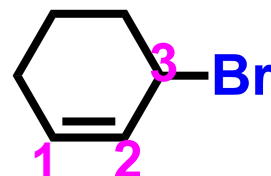


3-苯基丙-2-烯-1-醇

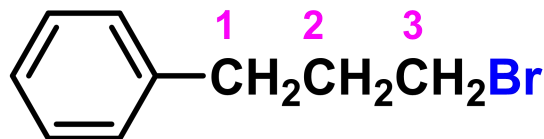
更多实例 (重要) :



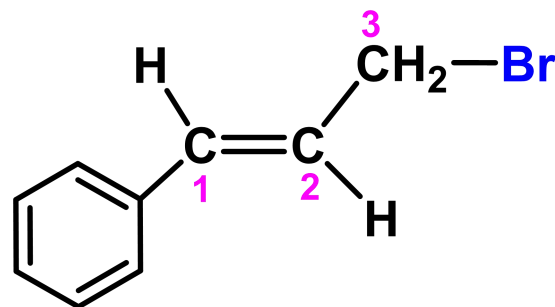
1-溴-1-甲基环己烷
1-bromo-1-methyl
cyclohexane



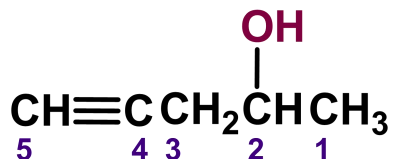
3-溴环己烯
3-bromocyclohexene



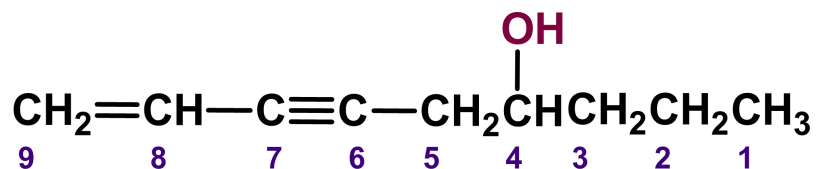
(3-溴丙基)苯
(3-bromopropyl)benzene



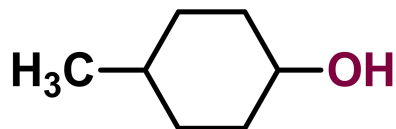
(*E*)-(3-溴丙-1-烯-1-基)苯
(*E*)-(3-bromoprop-1-en-1-yl)benzene



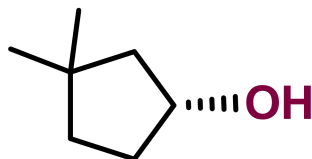
戊-4-炔-2-醇
Pent-4-yn-2-ol



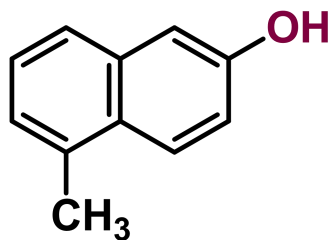
壬-8-烯-6-炔-4-醇
Non-8-en-6-yn-4-ol



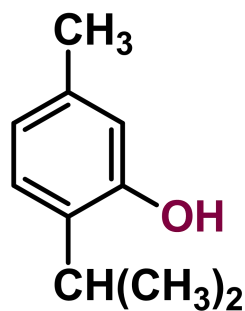
4-甲基环己醇
4-Methylcyclohexanol



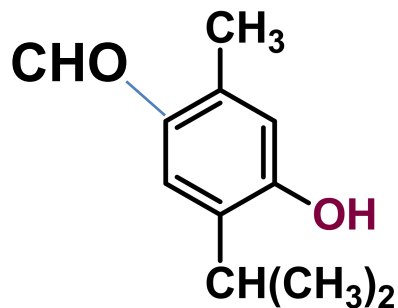
(*R*)-3,3-二甲基环戊醇
(*R*)-3,3-Dimethylcyclopentanol



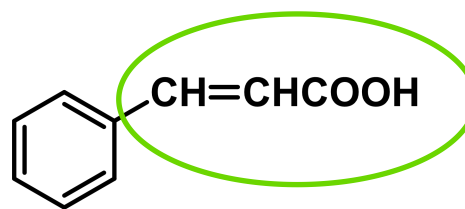
5-甲基萘-2-酚



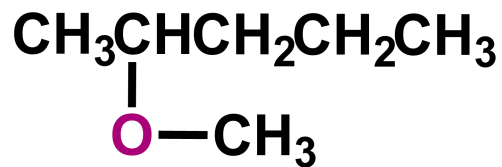
2-异丙基-5-甲基苯酚



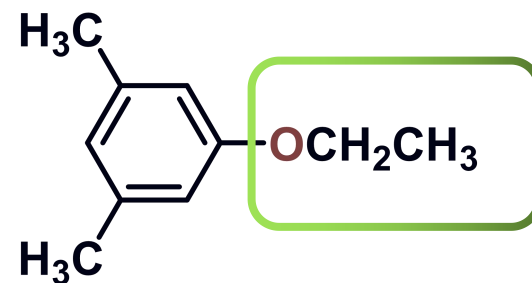
4-羟基-5-异丙基-2-甲基-苯甲醛



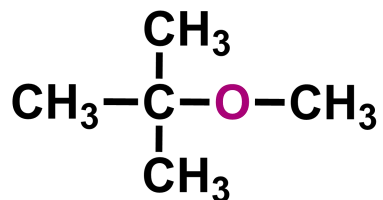
3-苯基丙烯酸



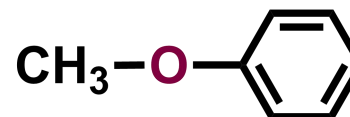
2-甲氧基戊烷



1-乙氧基-3,5-二甲基苯



叔丁基甲基醚

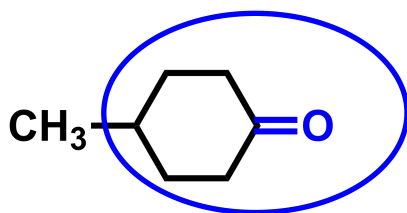


甲基苯基醚

脂环族醛和酮的命名:

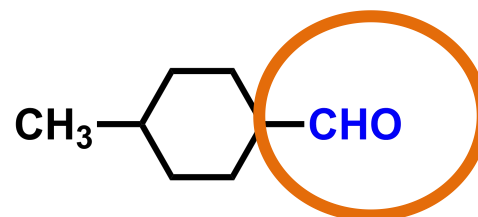


- 羰基在**环内的**——**环某酮**;
- 羰基在**环外的**——**环为取代基**, 脂环族醛——**烃名称加醛**。



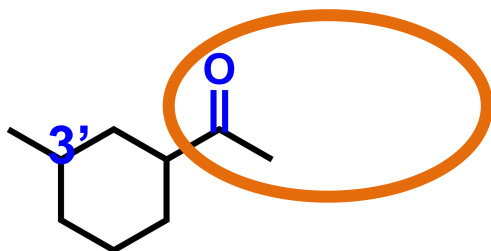
4-甲基**环己酮**

4-methylcyclohexanone



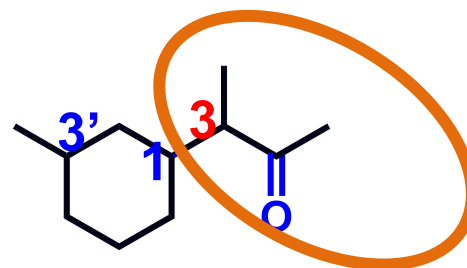
4-甲基环己基**甲醛**

4-methylcyclohexane carbaldehyde



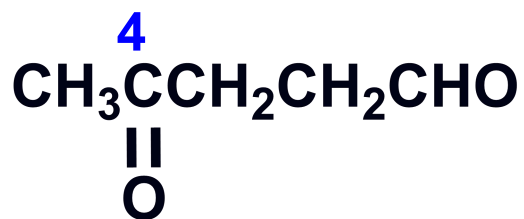
1-(3-甲基环己基)**乙酮**

1-(3-methylcyclohexyl)ethanone



3-(3-甲基环己基)**丁-2-酮**

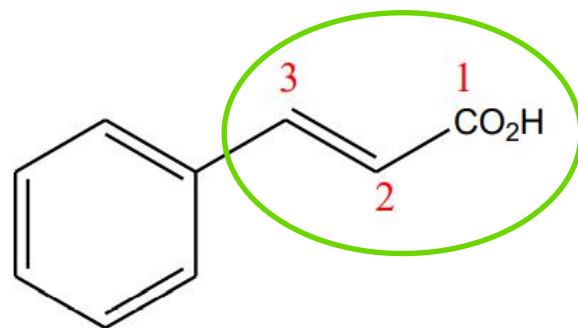
3-(3-methylcyclohexyl)butan-2-one



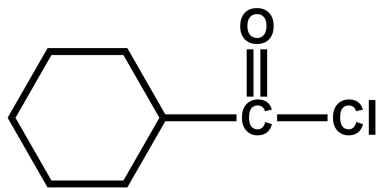
4-氧亚基戊醛

4-oxopentanal

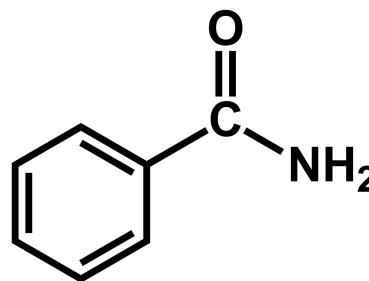
醛、酮同在, 醛为母体



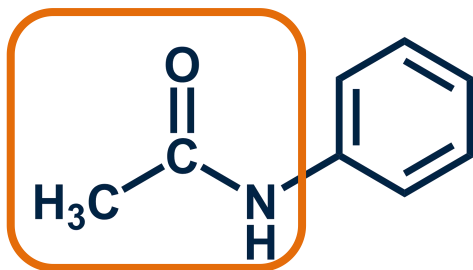
3-苯基丙烯酸



环己甲酰氯

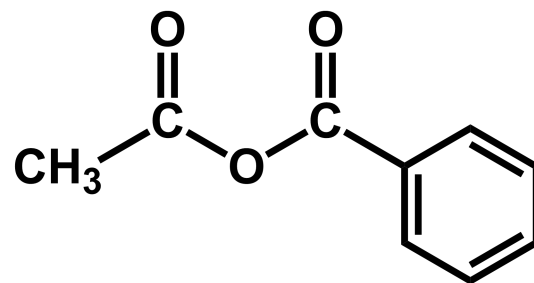


苯甲酰胺

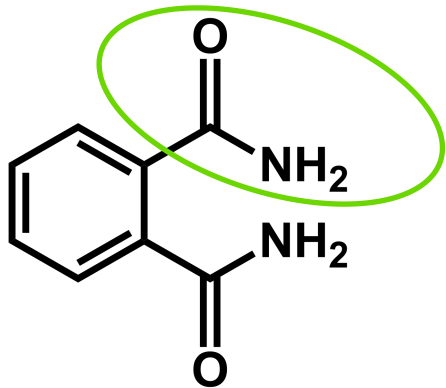


N-苯基**乙酰胺**

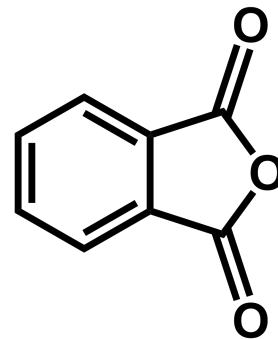
乙酰苯胺



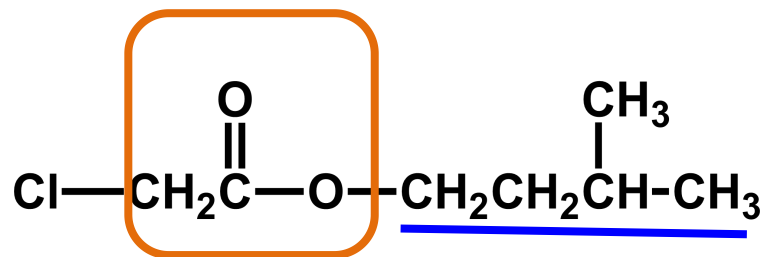
乙酸苯甲酸酐



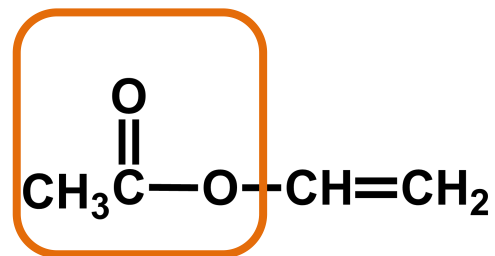
苯-1,2-**二甲酰胺**
邻苯**二甲酰胺**



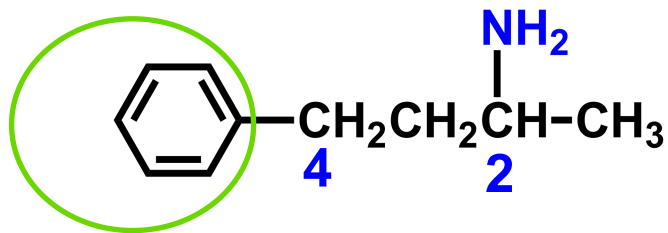
苯-1,2-**二甲酸酐**



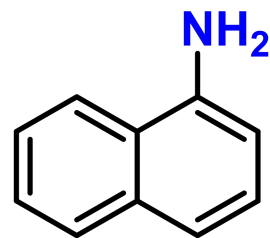
2-氯乙酸(3-甲基丁基)酯



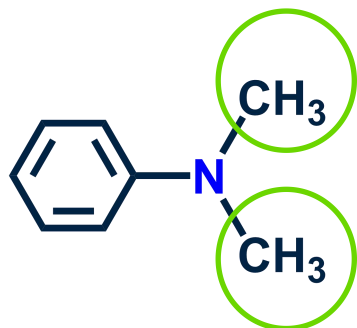
乙酸乙烯酯



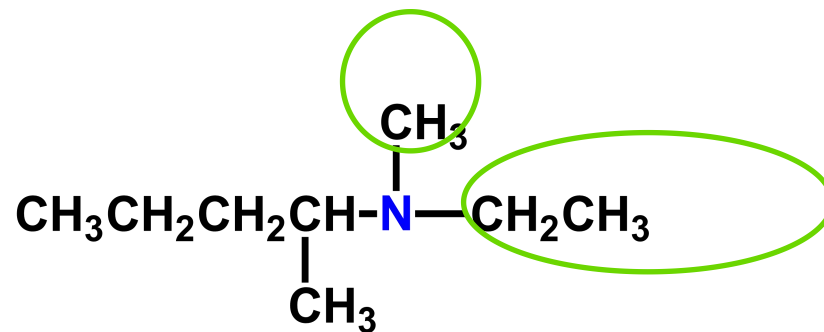
4-苯基丁-2-胺



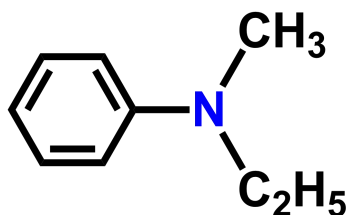
萘-1-胺



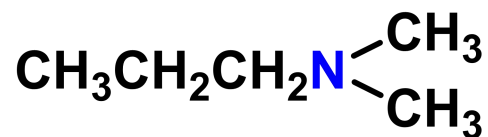
***N,N*-二甲基苯胺**



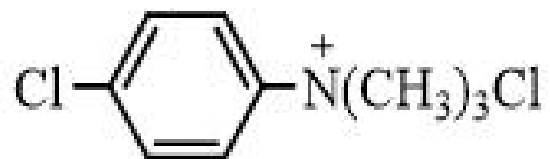
***N*-乙基-*N*-甲基戊-2-胺**



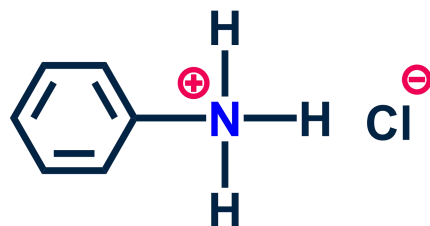
***N*-甲基-*N*-乙基苯胺**



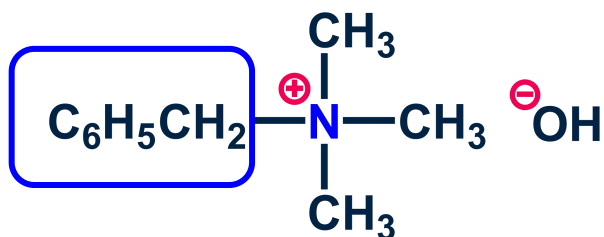
***N,N*-二甲(基)丙(基)胺**



氯化三甲基**对**氯苯基**铵**



氯化苯**铵**； 氯化苯基(三甲基)**铵**



氢氧化**苄**基(三甲基)**铵**

二、完成下列反应式 【每小题2分，共计24分】

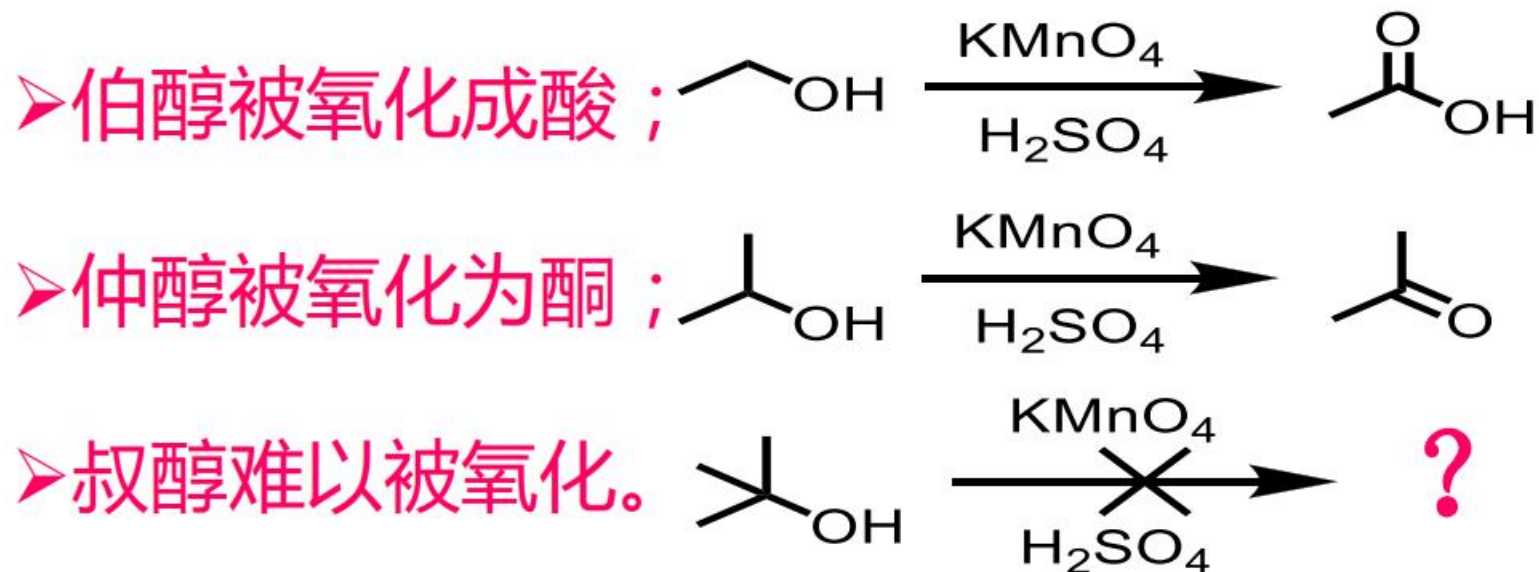
① 降低难度，基本是一步反应

② 注意反应条件

氧化反应

强氧化剂——生成酮/酸

◆ $\text{KMnO}_4 / \text{H}_2\text{SO}_4 / \text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$



常见的强氧化剂:

HNO_3 , KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ / H_2SO_4



鉴别伯醇与叔醇，或仲醇与叔醇

叔醇不反应



KMnO_4



紫红色退去，
变成棕色



$\text{K}_2\text{Cr}_2\text{O}_7$



Cr^{3+} 在溶液中
一般显浅绿色

强氧化剂——生成酮/酸



Jones试剂 —— CrO_3 / 稀 H_2SO_4 溶液



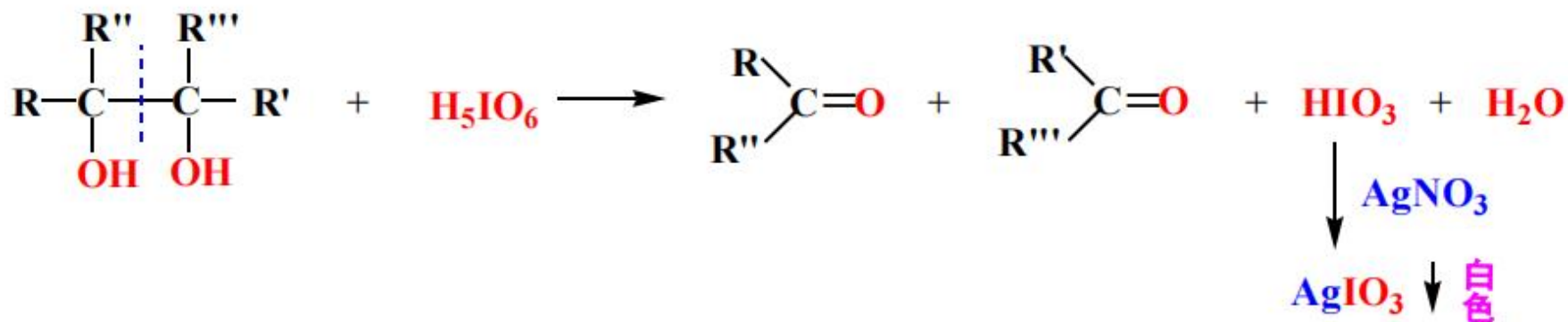
CrO_3 暗红色或暗紫色

Cr^{3+} 在溶液中
一般显浅绿色

邻二醇的氧化/鉴定，鉴别题

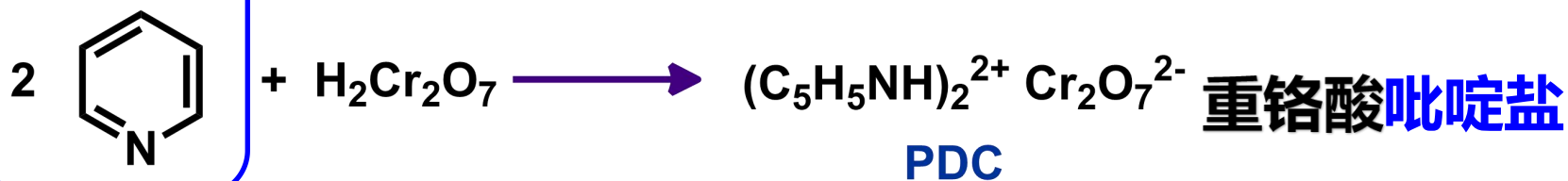
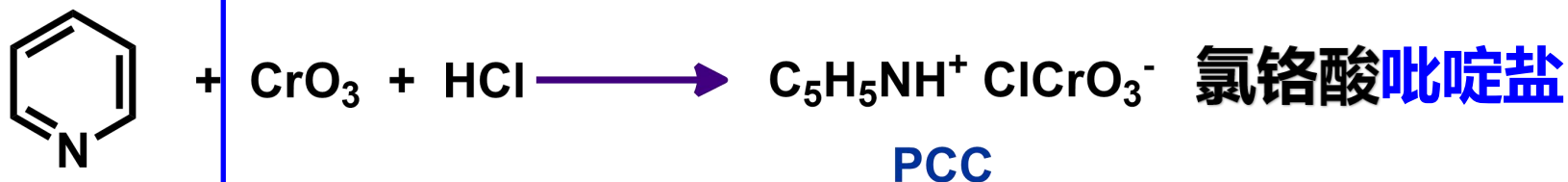
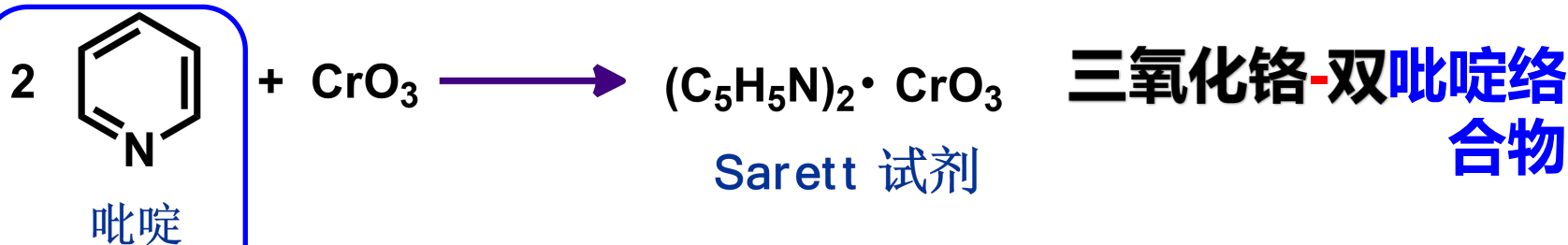


高碘酸/高碘酸钠 ($\text{H}_5\text{IO}_6/\text{NaIO}_4$) 氧化

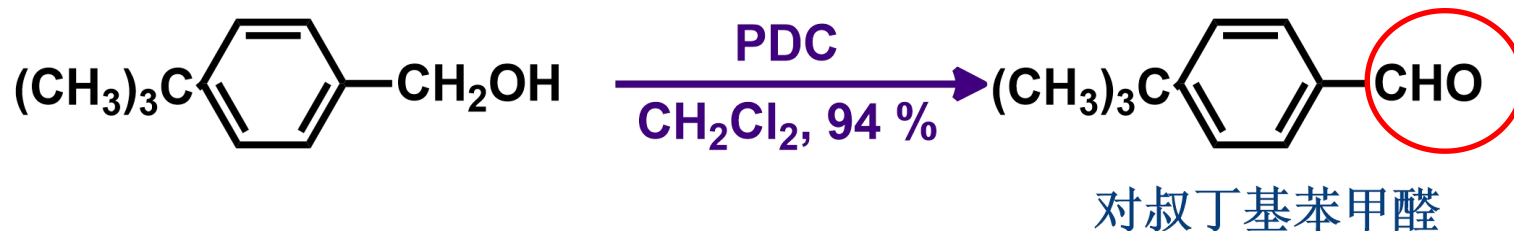
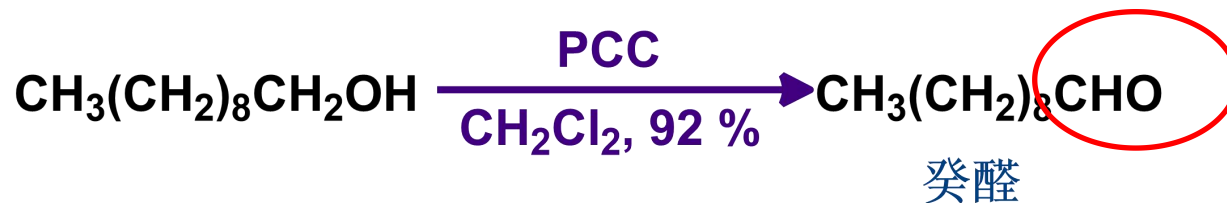


选择性氧化 (氧化性降低)

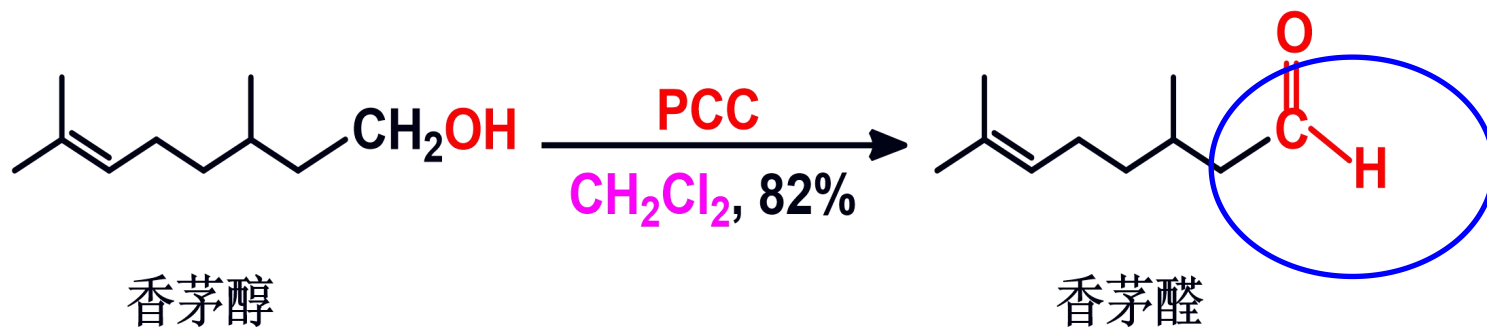
——化剂部分中毒 (络合)



选择性氧化醇,成醛 (II)



选择性氧化试剂，将伯醇氧化为醛，仲醇氧化为酮
分子中C=C, C=O, C=N等不饱和键，不发生氧化作用；
反应通常在CH₂Cl₂中于25°C左右进行，产率较高。



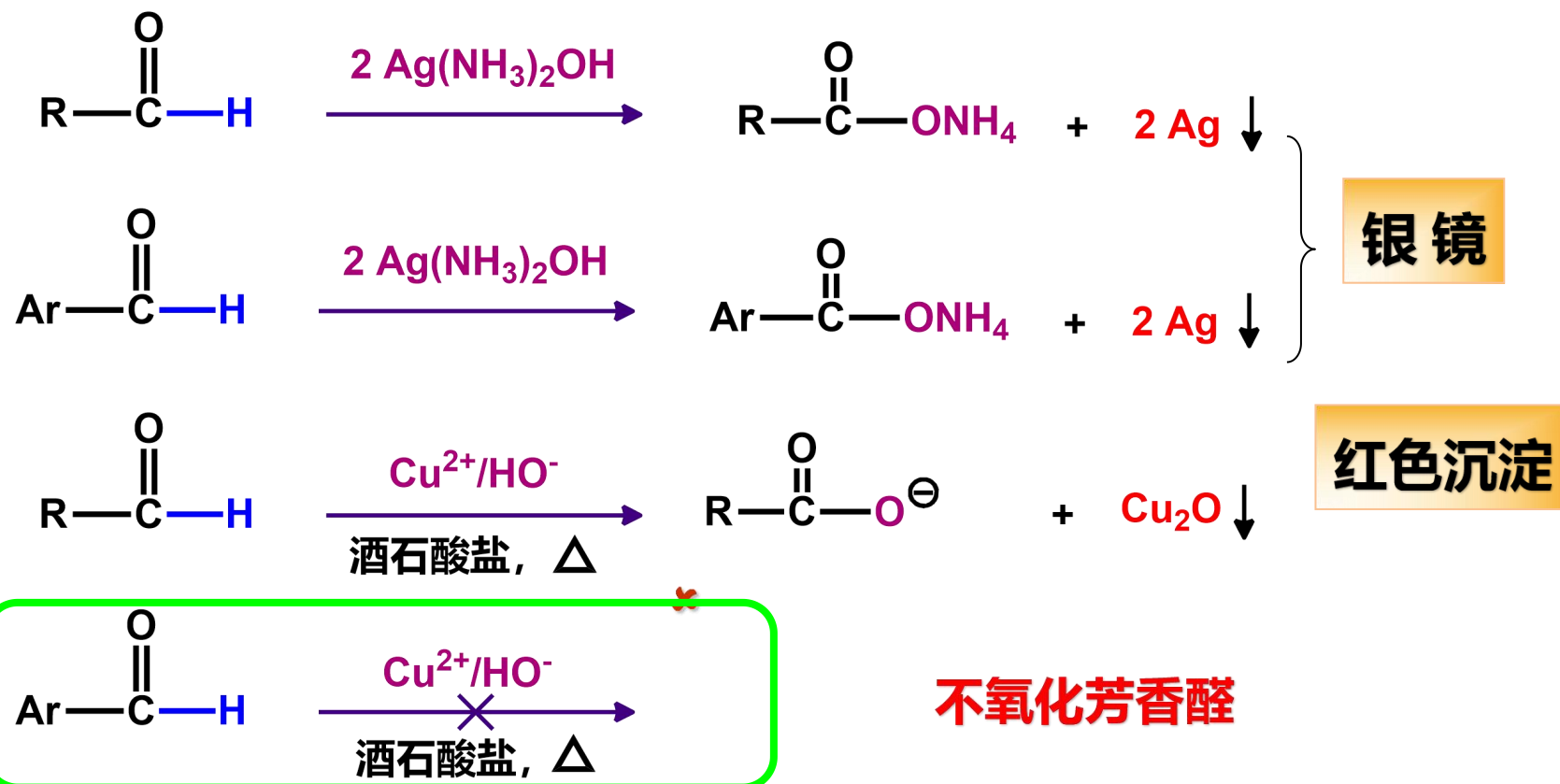
温和氧化剂: $\text{Ag}(\text{NH}_3)_2\text{OH}$ (Tollens 试剂, 银镜反应)

CuSO_4 /酒石酸钾钠碱性溶液 (Fehling试剂)

【特点】 弱氧化剂,将醛基氧化成羧基,均不氧化双键。

【区别】 Tollen' s(硝酸银的氨溶液)——氧化所有的醛;

Fehling' s(硫酸铜与酒石酸钾钠的混合液)——只氧化脂肪醛, 不氧化芳香醛。



还原反应

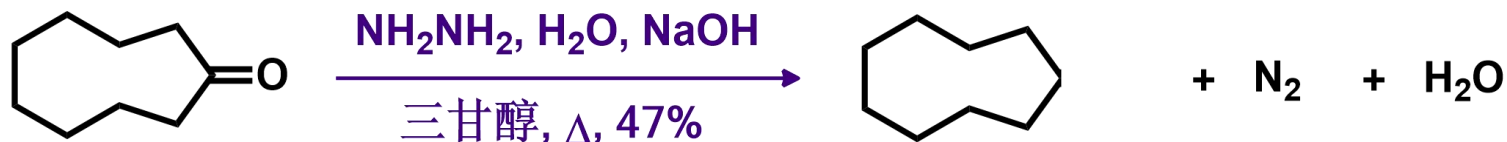
彻底还原----为亚甲基



1、克莱门森 (Clemmensen) 还原—酸性条件

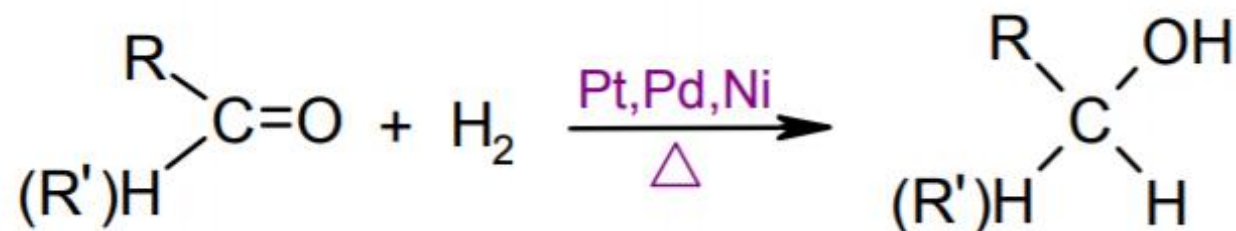


2、沃尔夫-吉斯尼尔-黄鸣龙还原—碱性条件



强还原剂——催化加氢

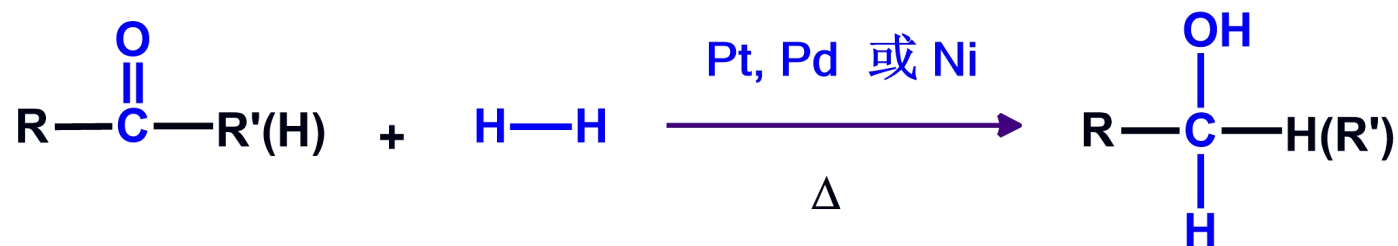
1、催化加氢还原法:



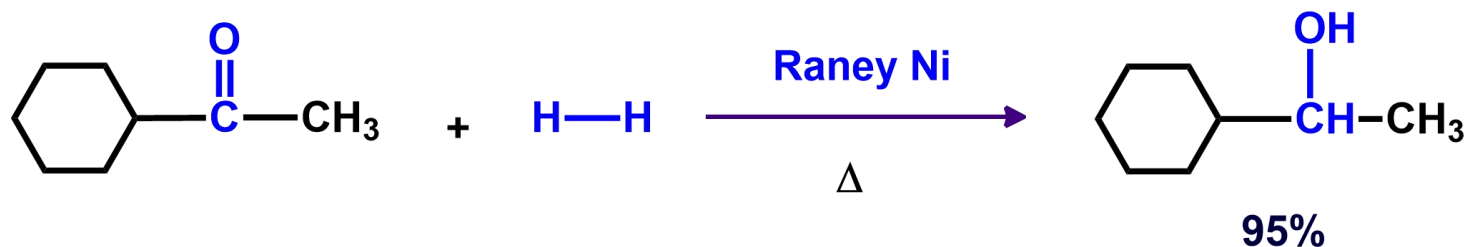
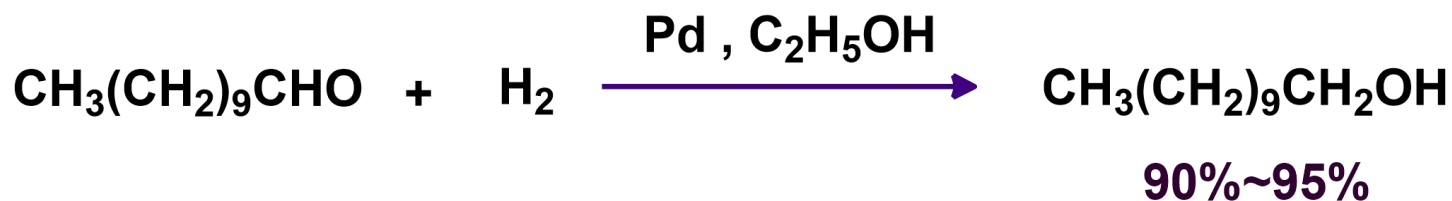
催化加氢是强还原条件，分子中如果含有**其它官能团**（如：-C=C-；-C≡C-；-C≡N；-NO₂；-X；-COOR等）**也同时被还原**。

如果羰基两侧的立体环境不同，催化剂通常从空间位阻小的一侧被吸附，**顺式加氢**。

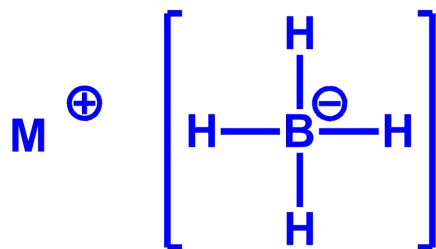
强还原剂——催化加氢



例:

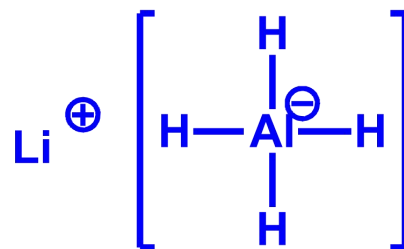


强还原剂--用金属氢化物还原 (选择性还原)



M = Na, K

硼氢化钠(钾)

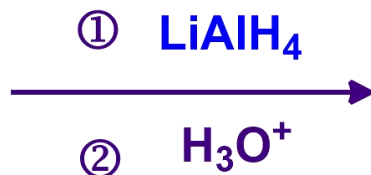
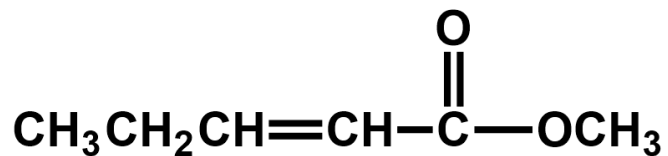
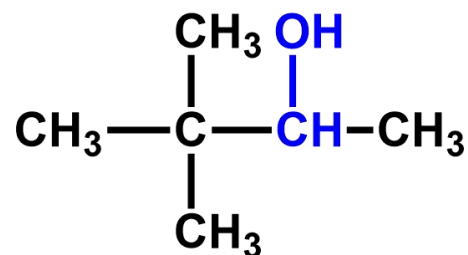
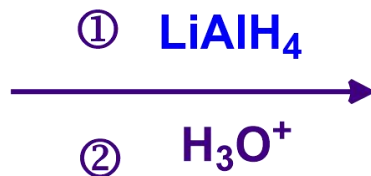
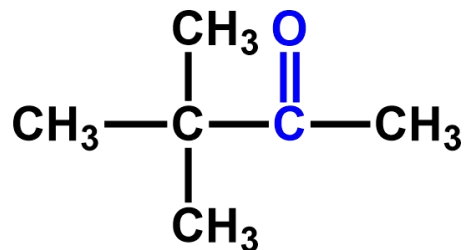
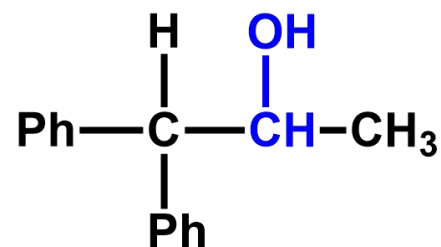
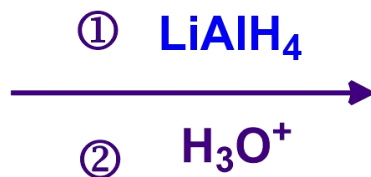
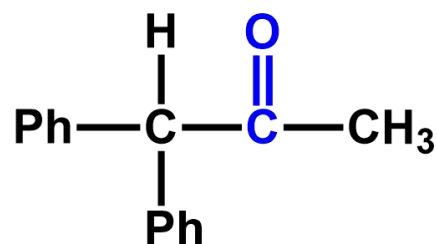


四氢铝锂

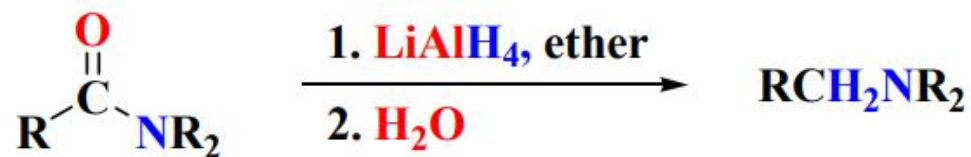
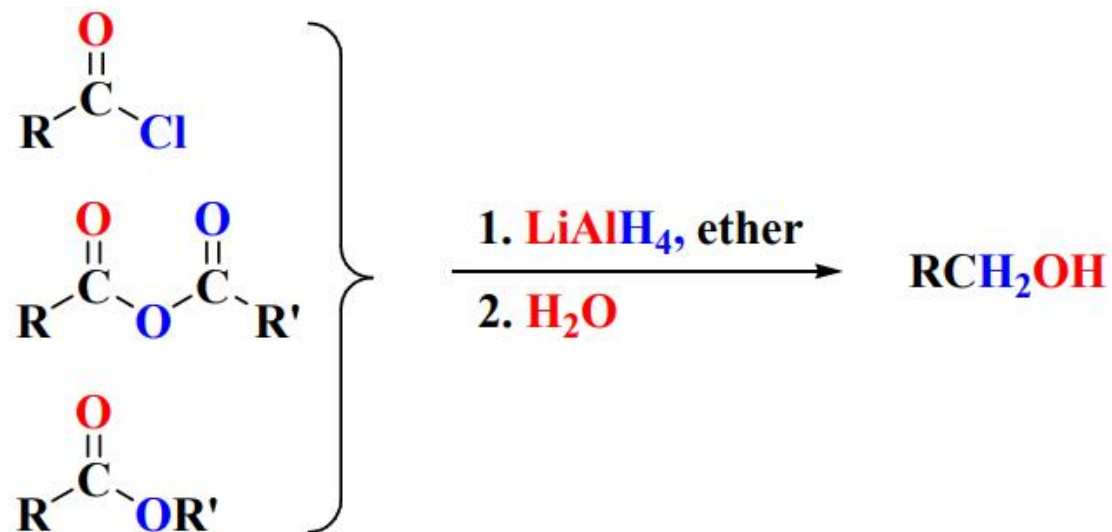
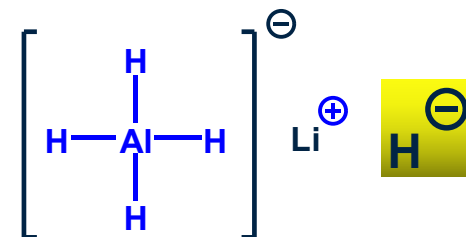
还原特点:

- NaBH₄ 可还原: 醛、酮 (稍弱) ;
- LiAlH₄ 可还原: 醛、酮、羧酸、酯、酰胺、腈、硝基
- 都不能还原: 碳碳双键、叁键

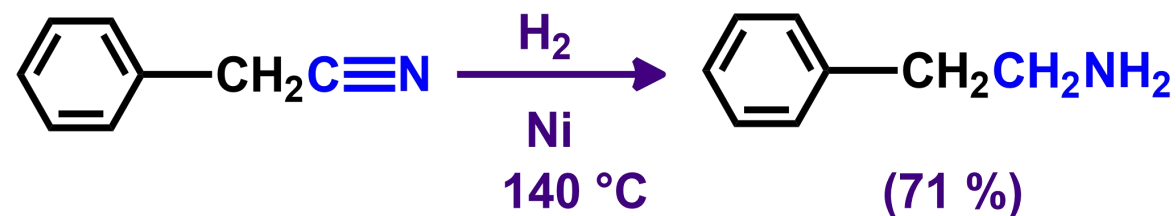
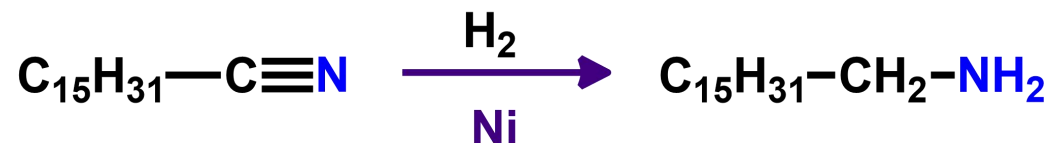
➤ LiAlH_4 还原反应



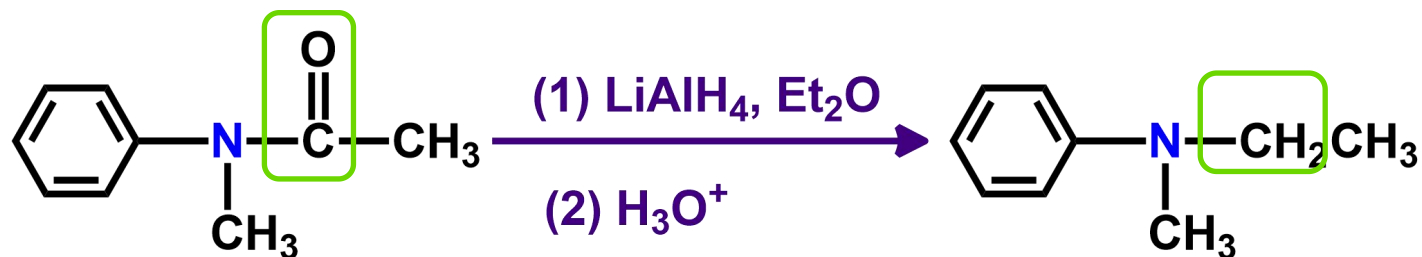
**LiAlH₄还原，酰卤、酸酐和酯被还原为醇；
酰胺被还原为胺**



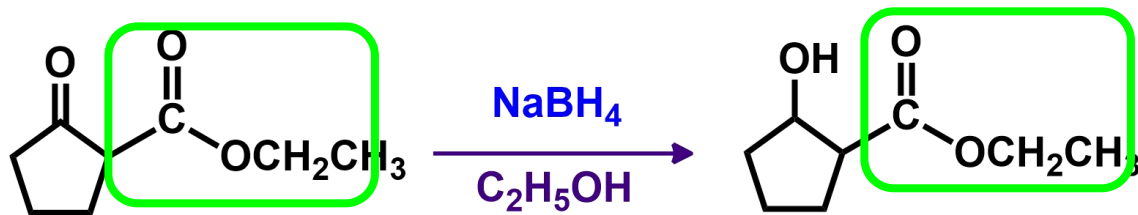
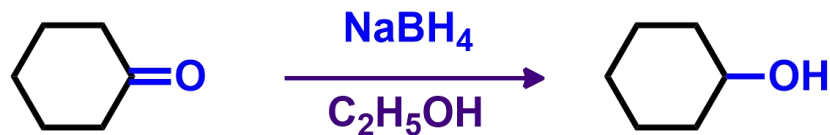
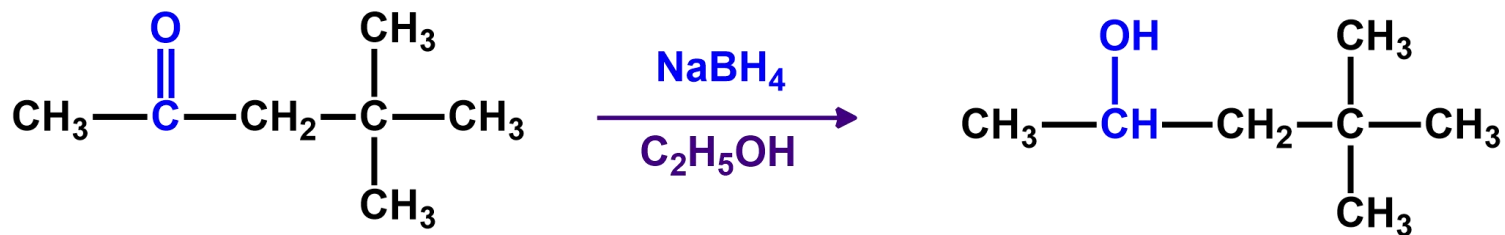
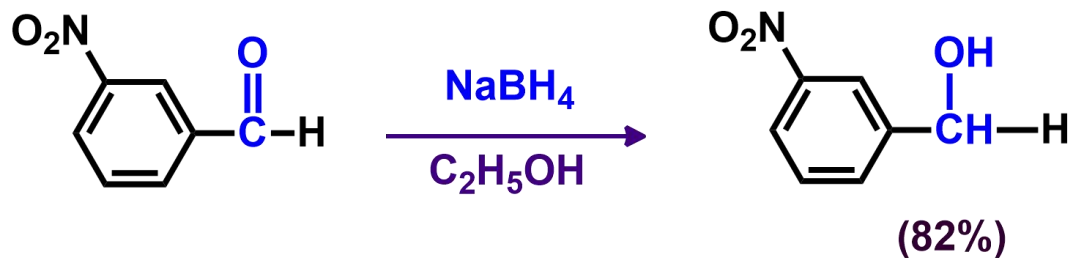
腈可以催化氢化或用 LiAlH_4 还原为伯胺。



酰胺通过 LiAlH_4 还原。

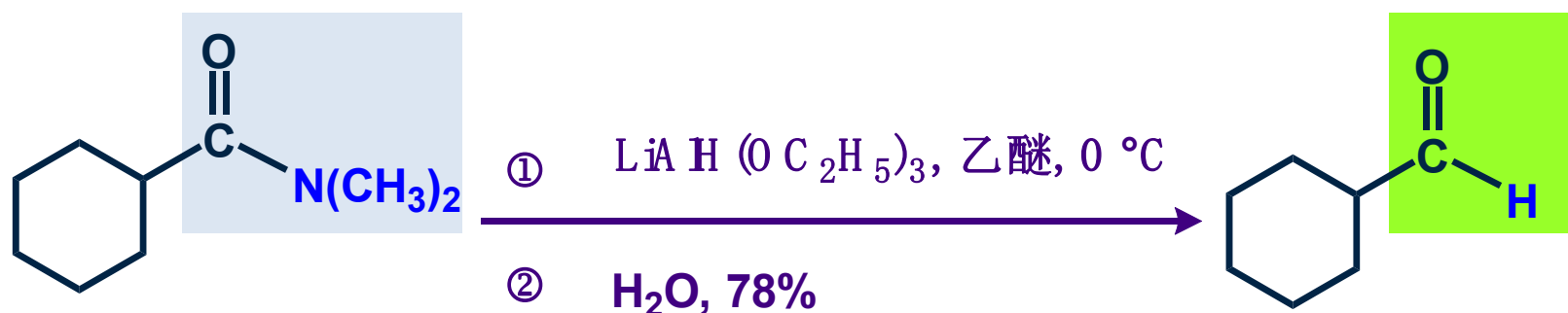


➤ NaBH_4 还原反应

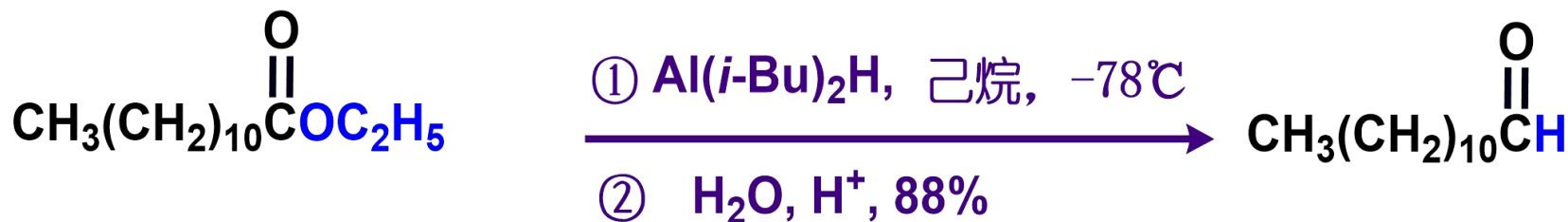
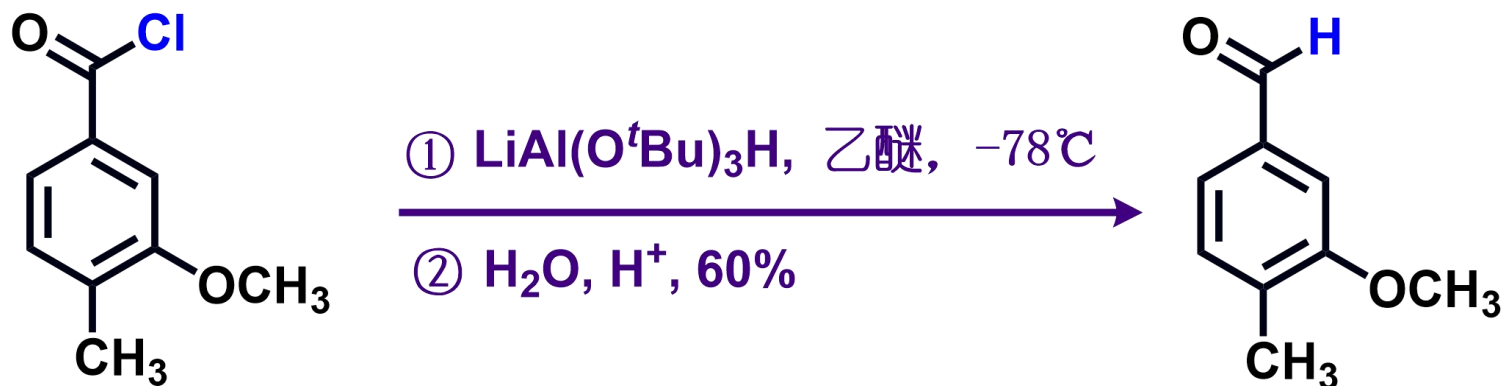


还原剂强，但是体积大

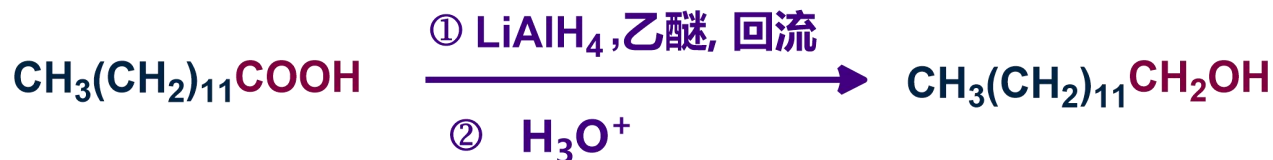
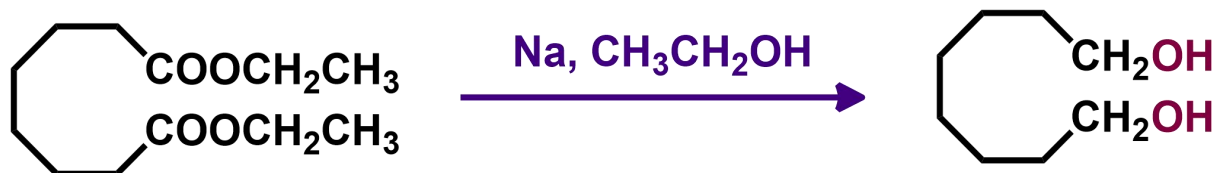
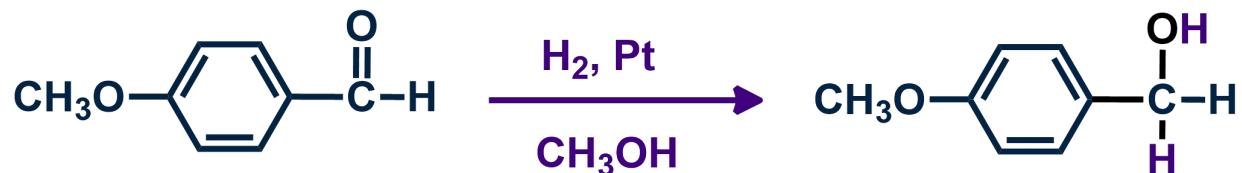
烷氧基氢化铝锂的反应—温和的还原剂—醛



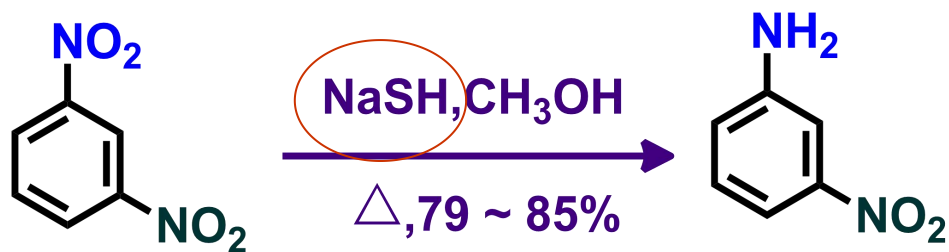
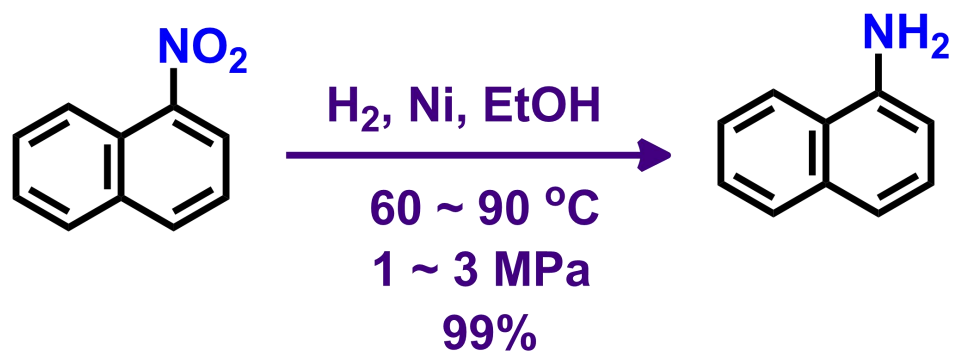
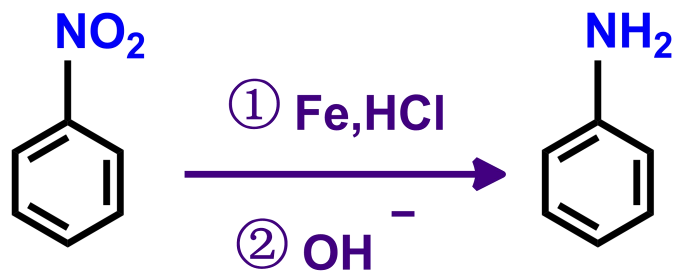
烷氧基氢化铝锂的反应—温和的还原剂—醛



醛、酮、羧酸和羧酸衍生物的还原，制备醇



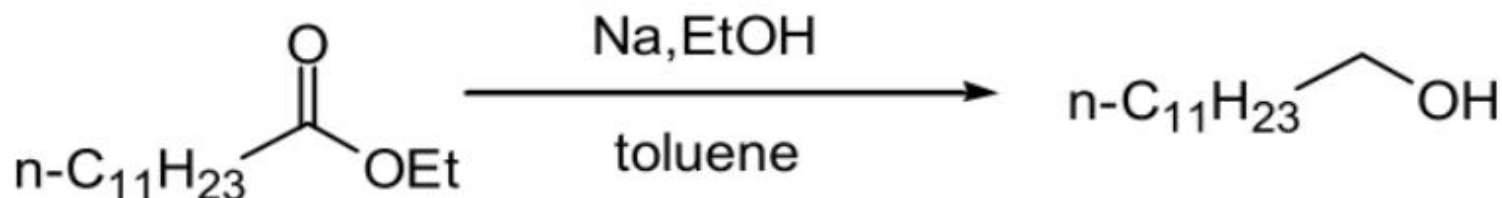
硝基还原，成为NH₂



金属钠-醇还原 (提供电子)



酯与金属钠在醇溶液中加热回流，可被还原成相应的伯醇，此反应称为Bouveault-Blanc (鲍维特-勃朗克) 反应。



机理：酯从钠获得一个电子还原为自由基负离子，然后从醇中夺取一个质子转变为自由基，从钠得到一个电子变成碳负离子，消除烷氧基成为醛；醛再经过相同的步骤还原成醇钠，醇钠再得到一个质子变成相应的醇。

(罗森蒙德) 还原—选择性还原—化剂部分中毒



- 催化剂活性减弱
- 催化剂部分中毒

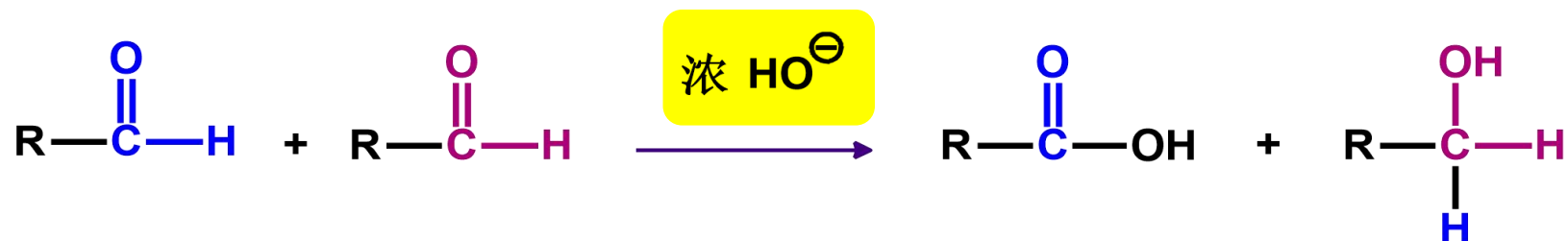


制备醛的一种好方法.

歧化反应

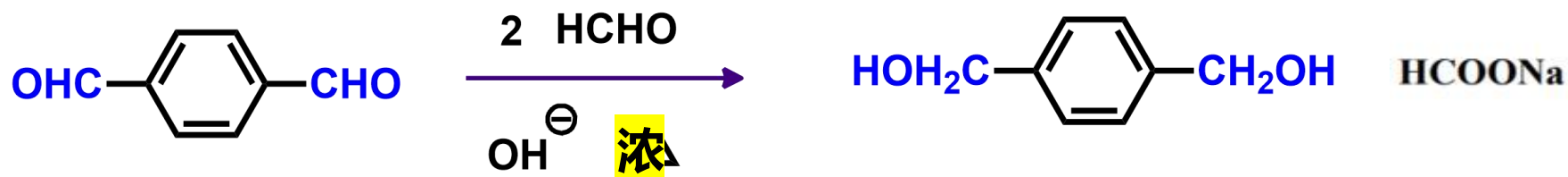
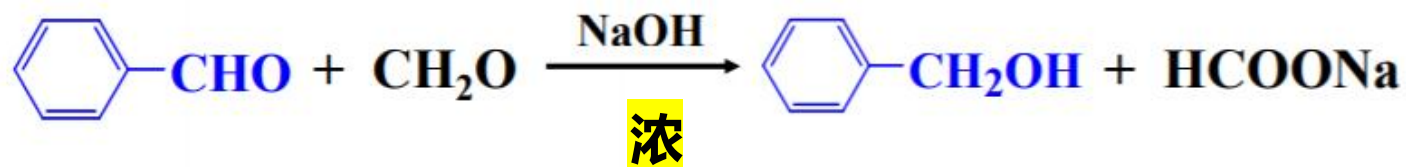
没有 α -氢的醛，在浓碱作用下：

一分子醛被氧化为酸，另一分子被还原成醇



R = 没有 α -H的烃基，芳基

交叉坎尼扎罗反应中，**甲醛总是被氧化。**

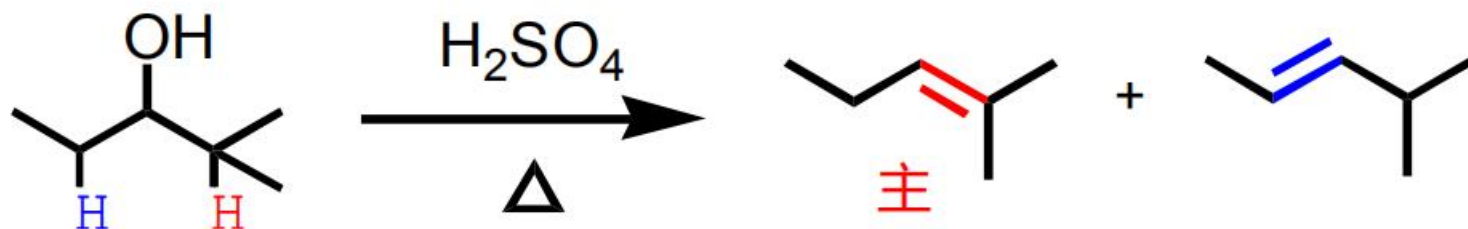


消除反应

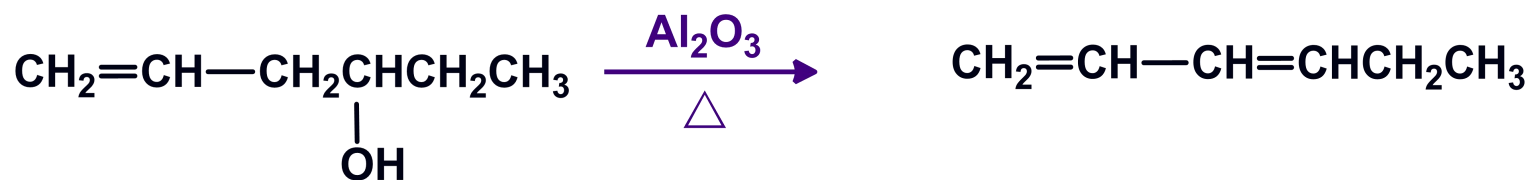
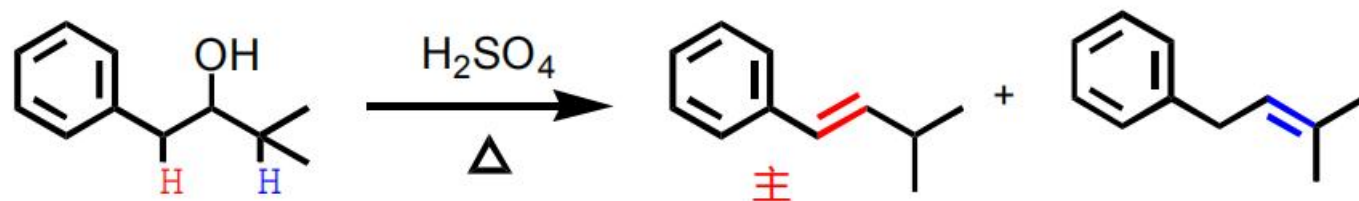
醇脱水遵循 Saytzeff 规则



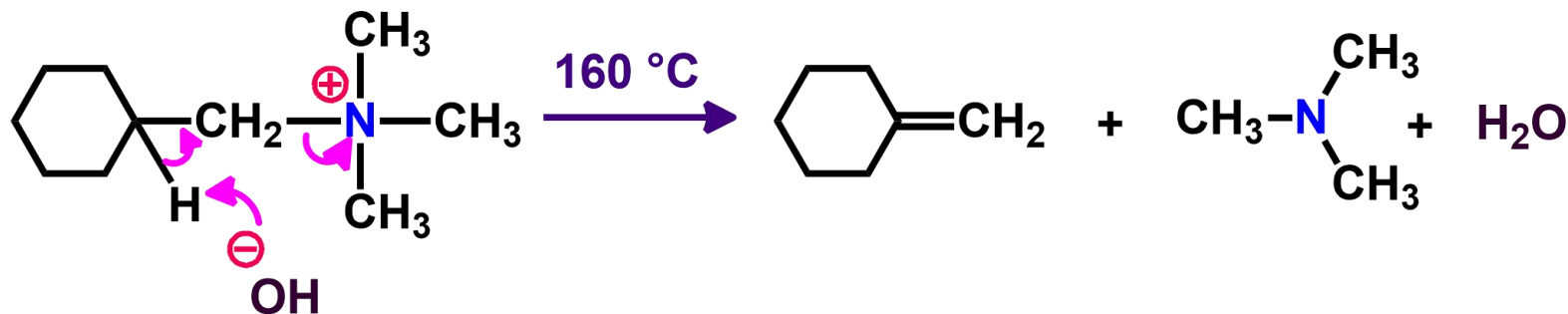
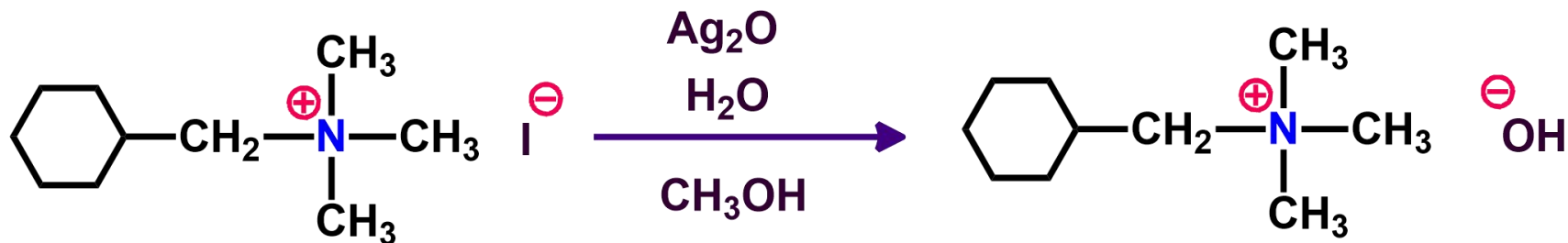
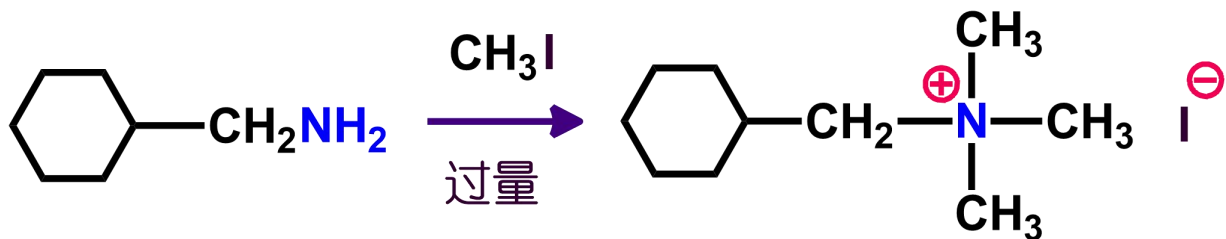
- 醇脱水得烯，遵循札依采夫规则。



- 优先生成共轭体系。

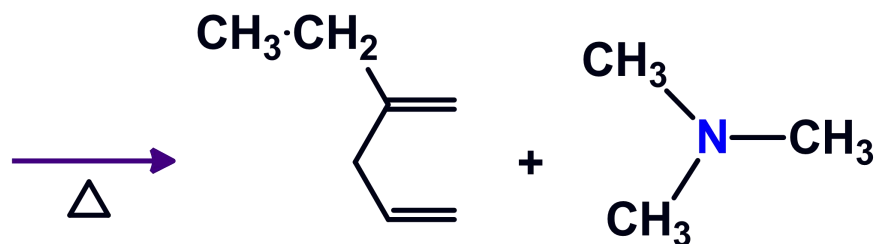
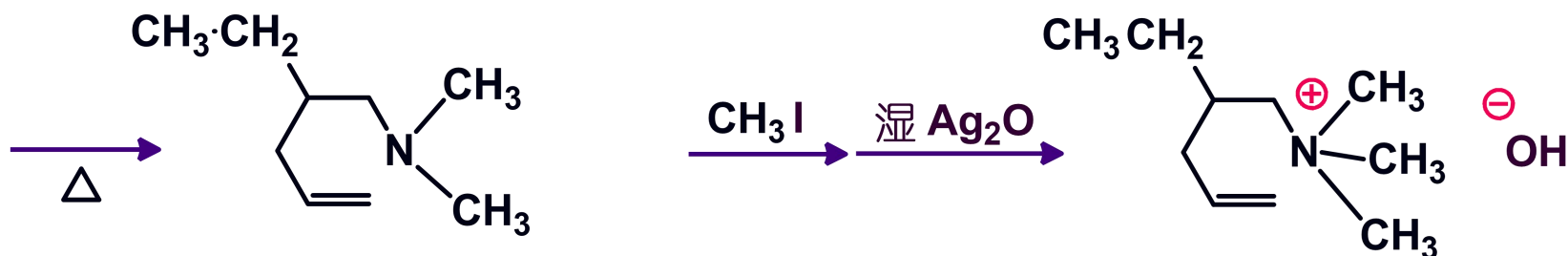
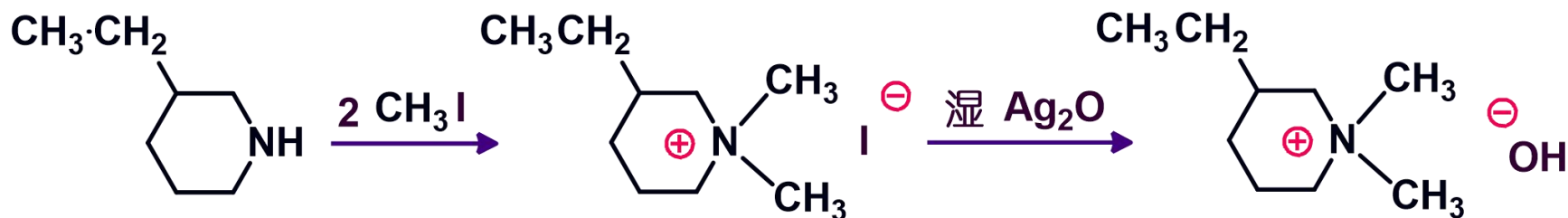


Hofmann消除反应



β -H消除

Hofmann消除反应



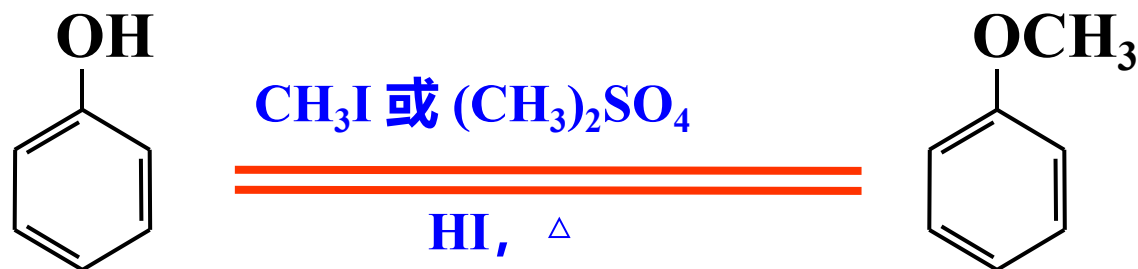
基团的保护反应

保护醇羟基



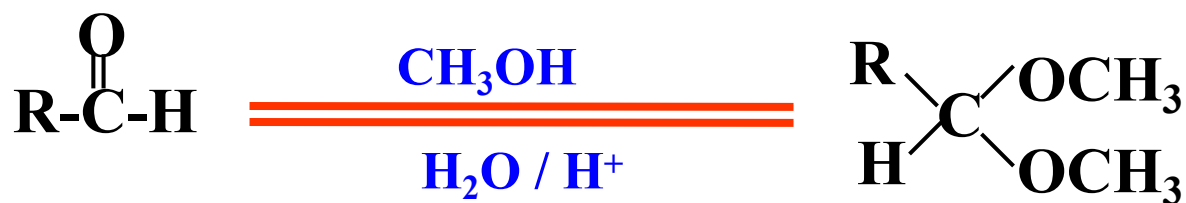
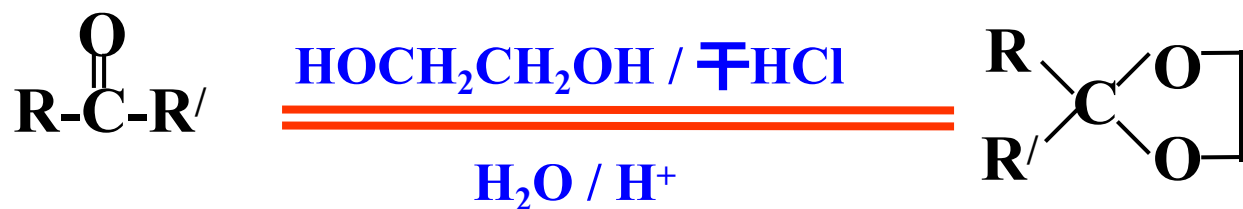
可逆反应，有机合成中可用于**保护醇羟基**。

酚羟基的保护



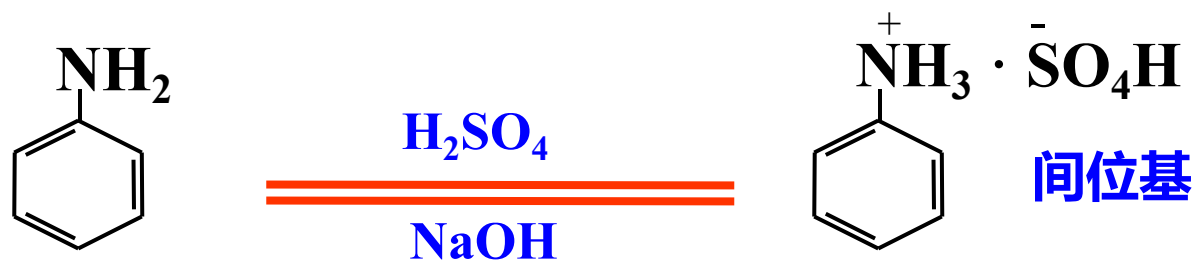
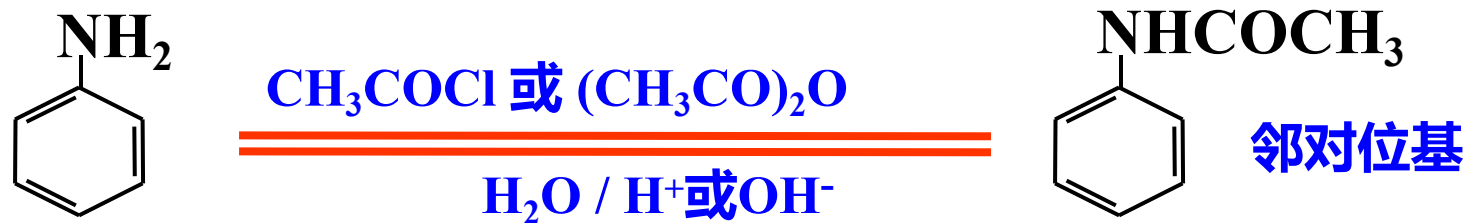
苯甲醚：对碱、氧化剂、还原剂等稳定，但对酸不稳定。

醛、酮、羧基的保护



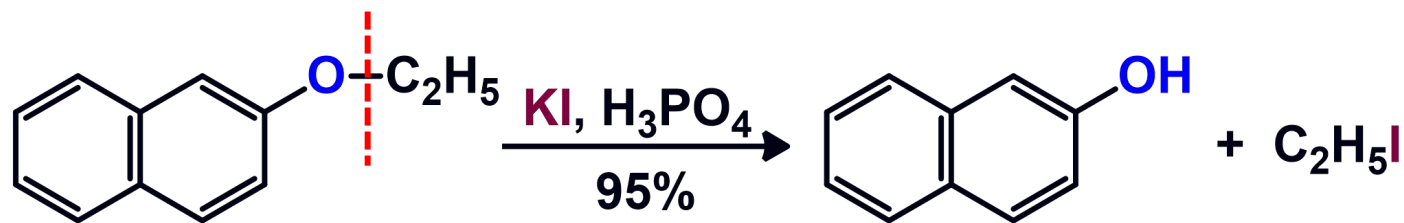
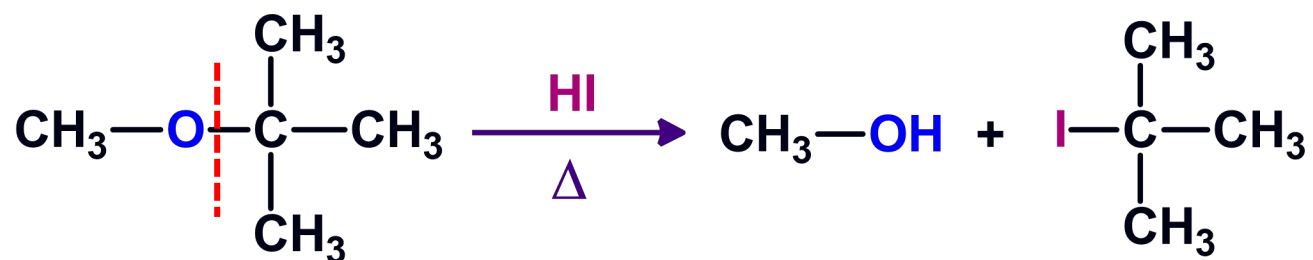
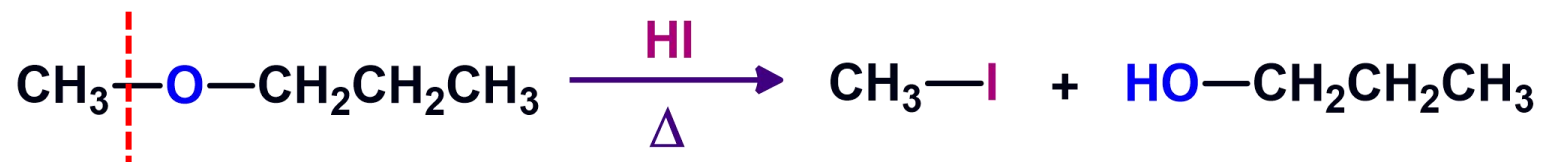
缩醛（酮）：对还原剂、氧化剂、碱、格氏试剂稳定，但对酸不稳定。

氨基的保护

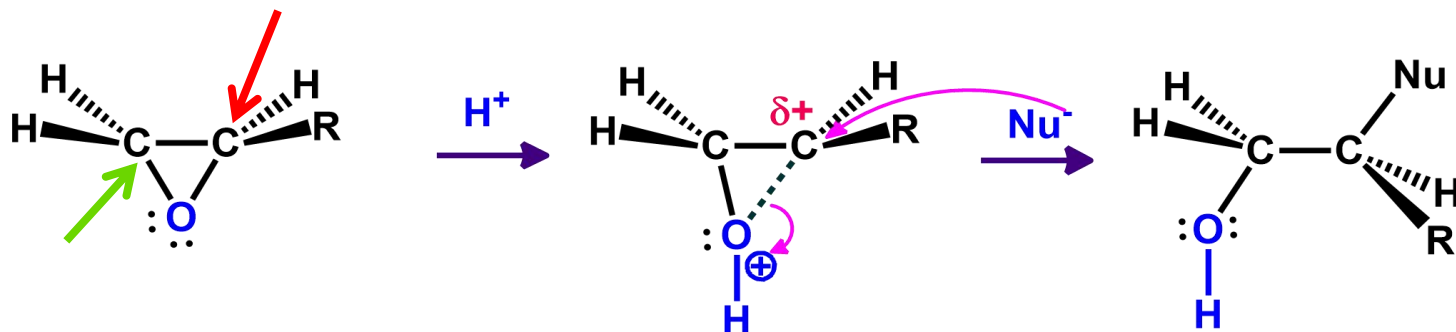


反应取向问题

不对称醚键的开裂取向？



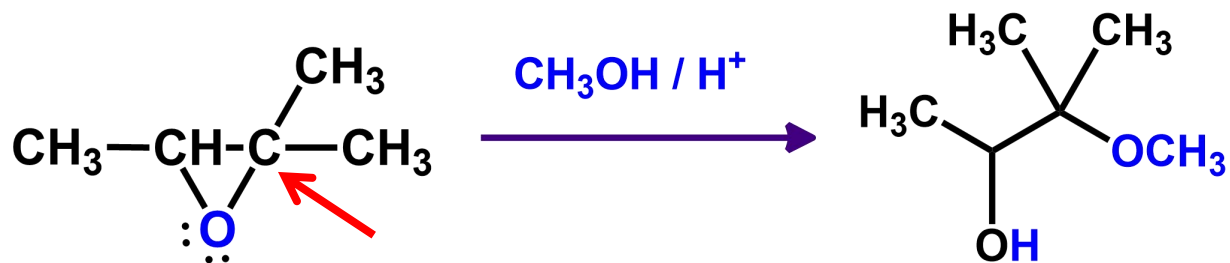
不对称环氧化合物，酸性条件下开环



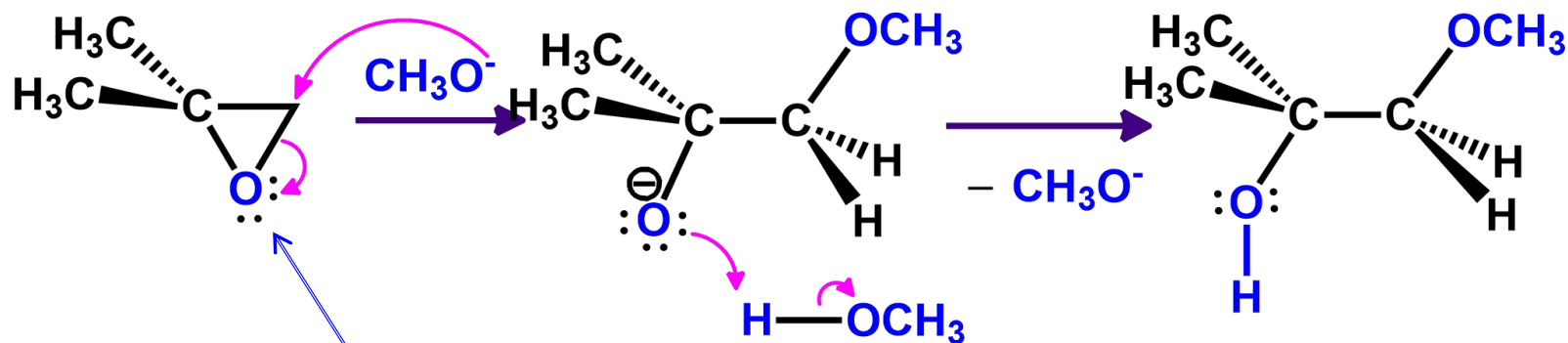
反应取向：亲核试剂优先进攻取代较多的碳，类 $\text{S}_{\text{N}}1$ 。

立体化学：反式开环

实例：

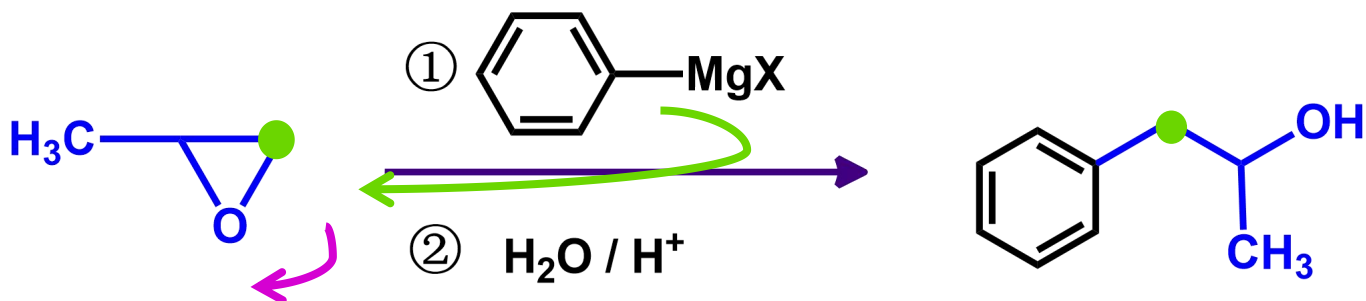


不对称环氧化合物，**碱性条件下开环**：

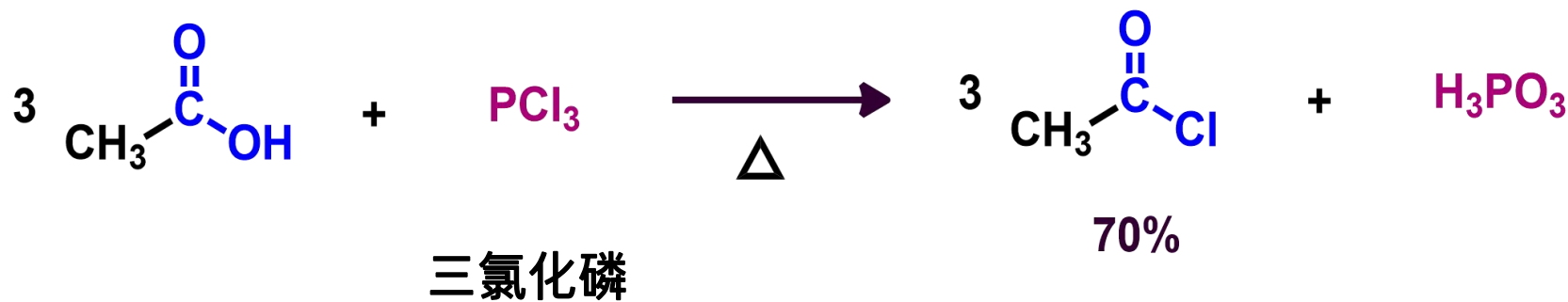
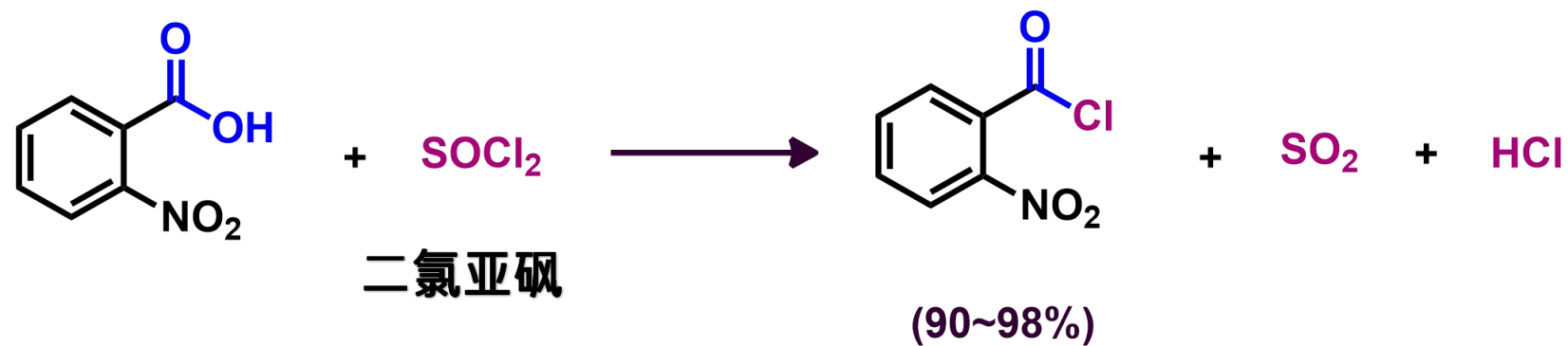


位阻小有利

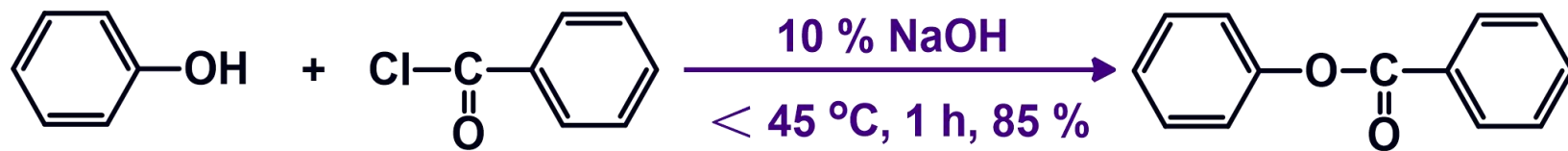
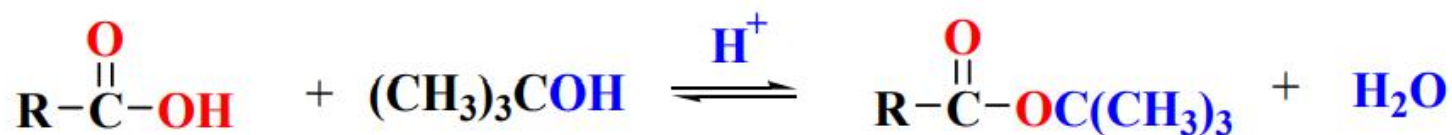
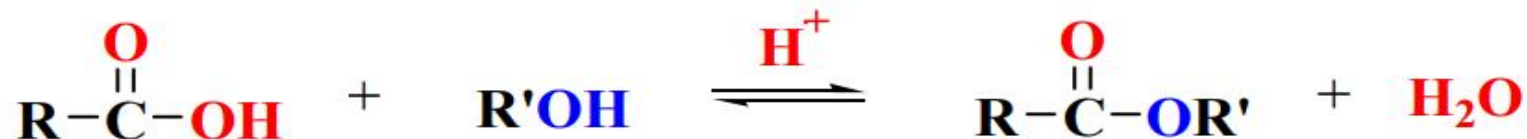
与格氏试剂反应，属于**碱催化下**的开环反应



合成酰氯



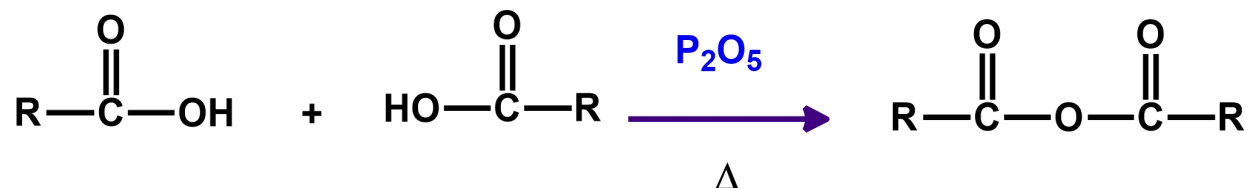
合成酯



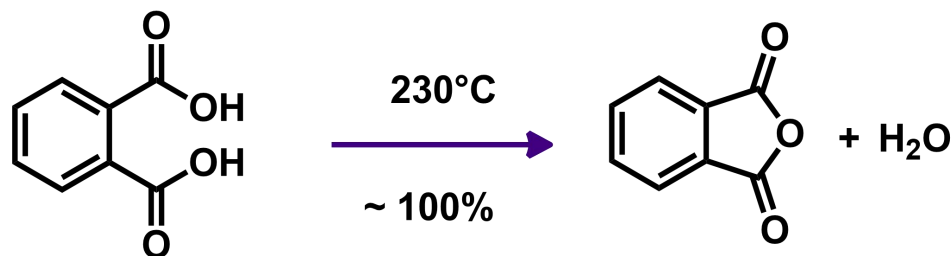
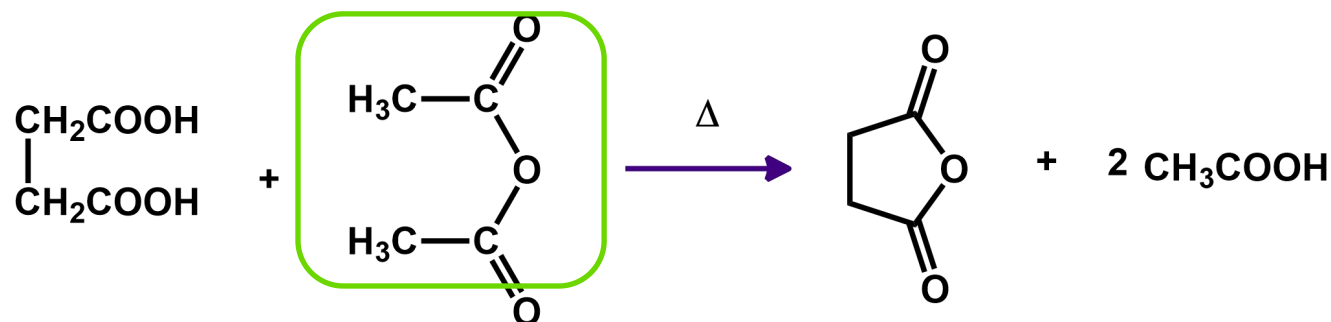
苯甲酸苯酯

合成酸酐

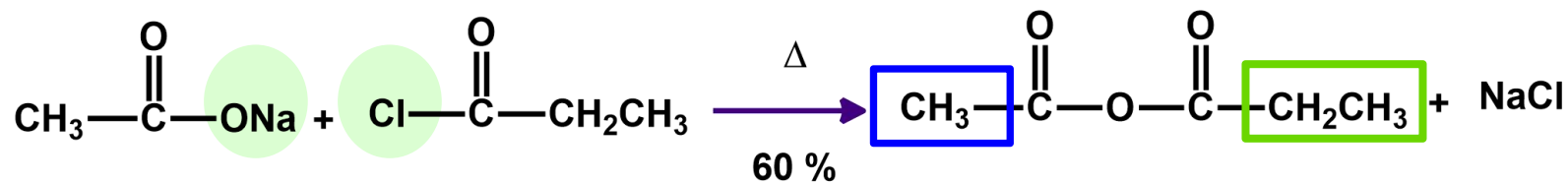
■ 脱水剂



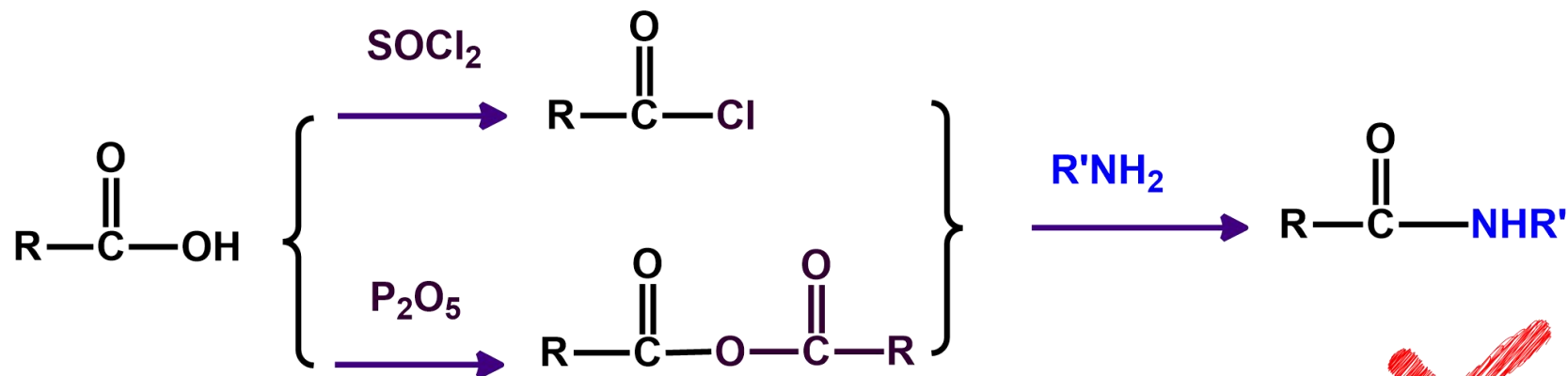
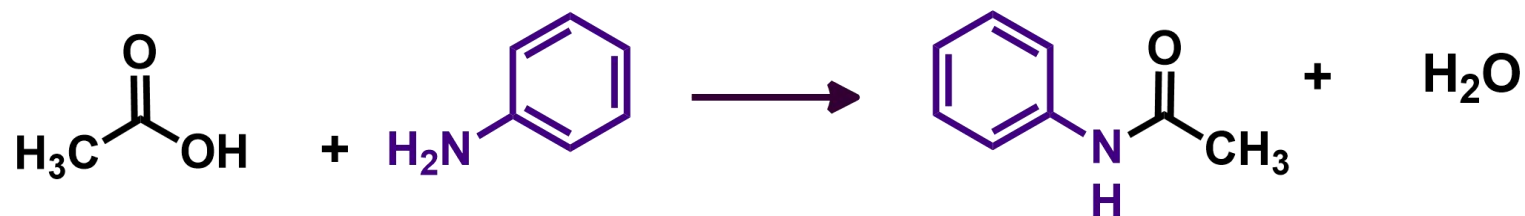
■ 环状酸酐



■ 混合酸酐

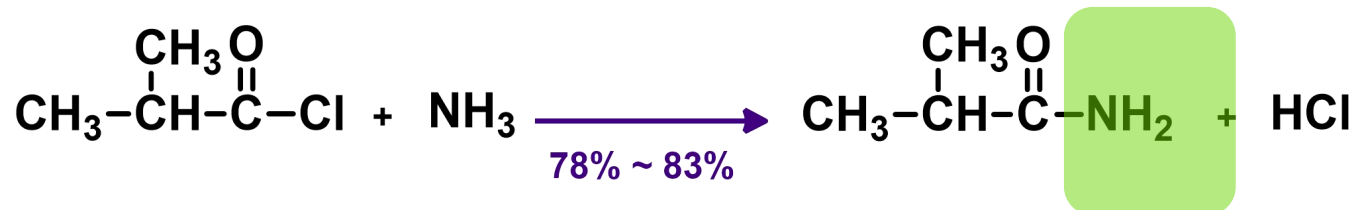
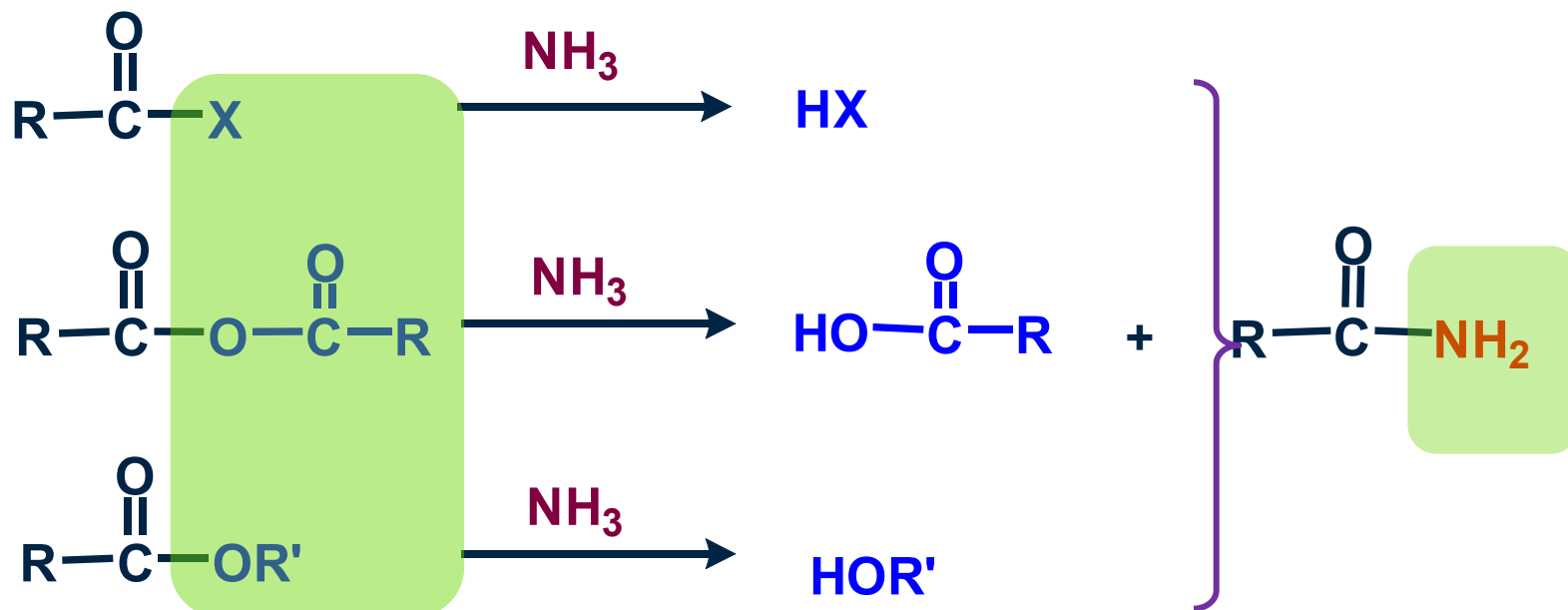


合成酰胺

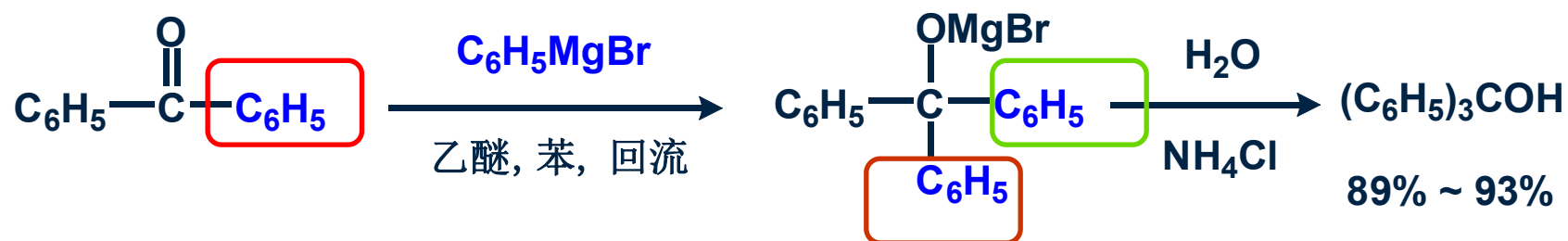


羧酸衍生物

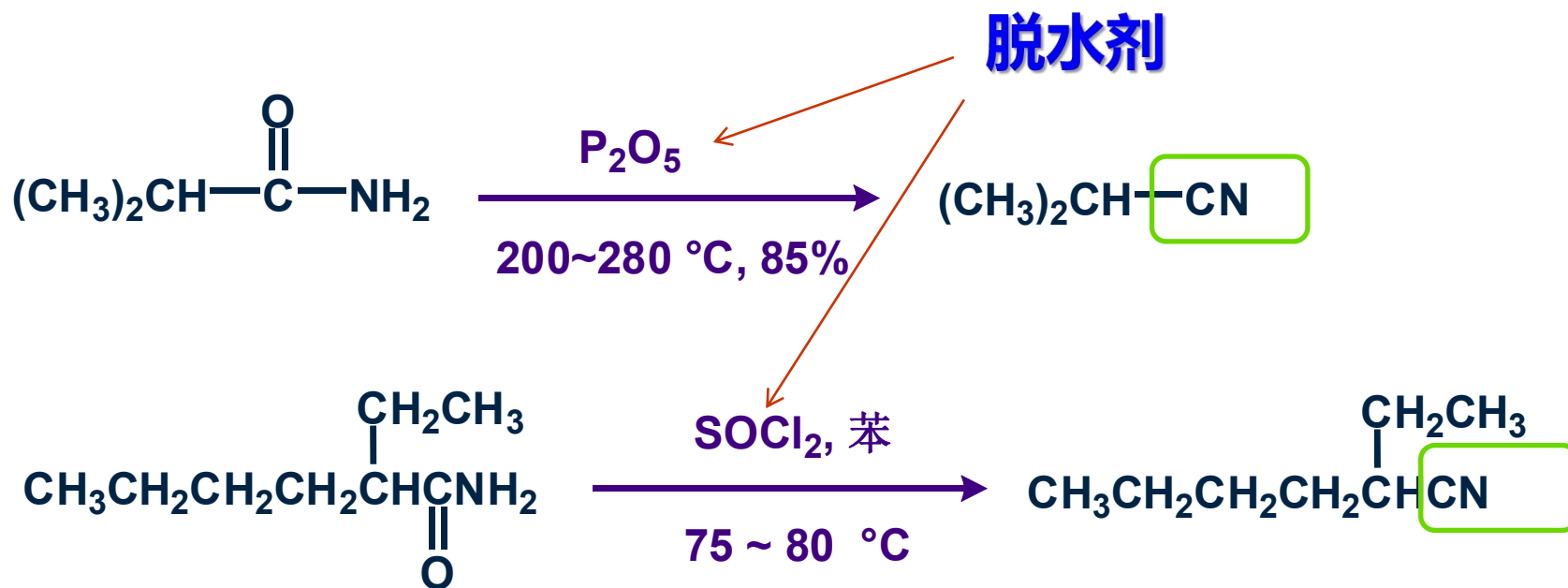
合成酰胺



合成叔醇

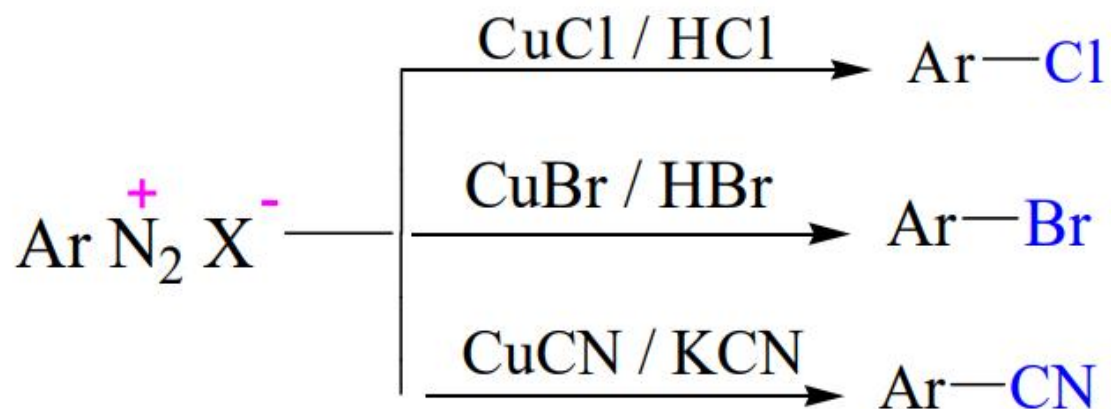


◆ 酰胺脱水--合成腈



苯环上的“置换”反应

在亚铜盐的催化作用下，重氮基受热被置换成Cl, Br、CN

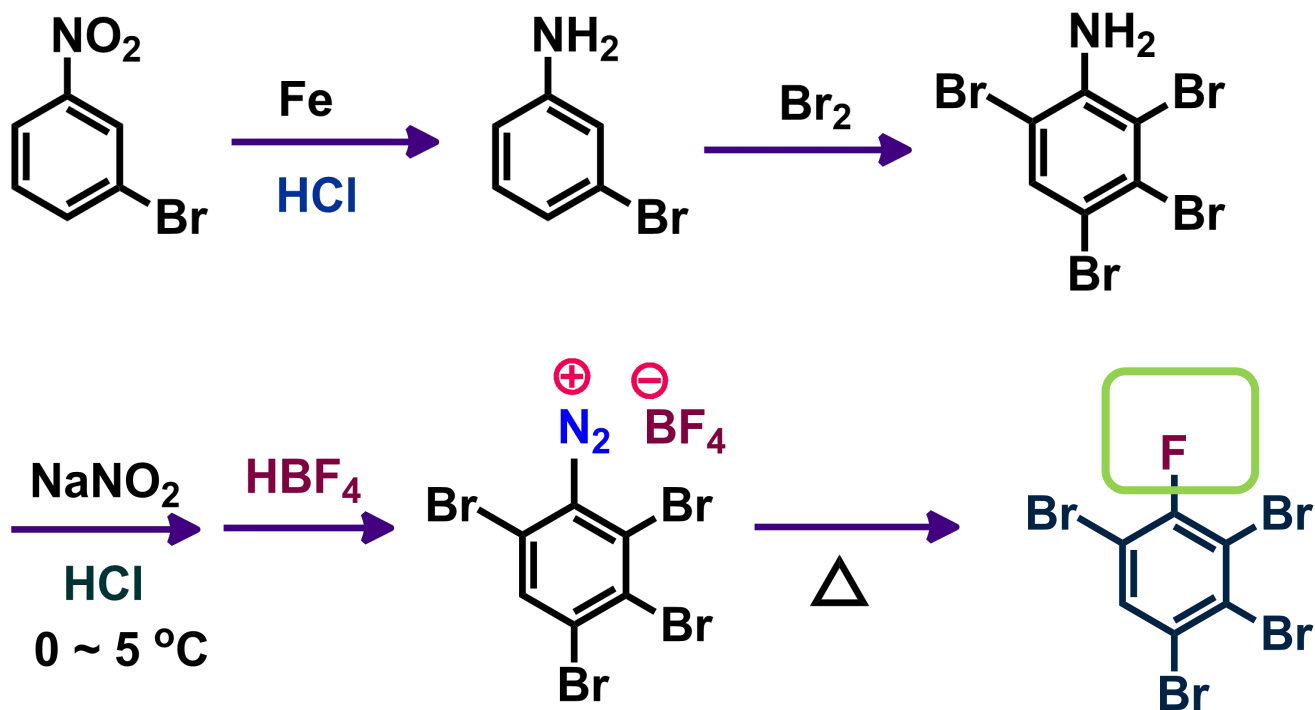


CuX为催化剂用量，需新鲜制备；反应收率高，产物纯度好。

重氮盐失去N₂---应用价值:



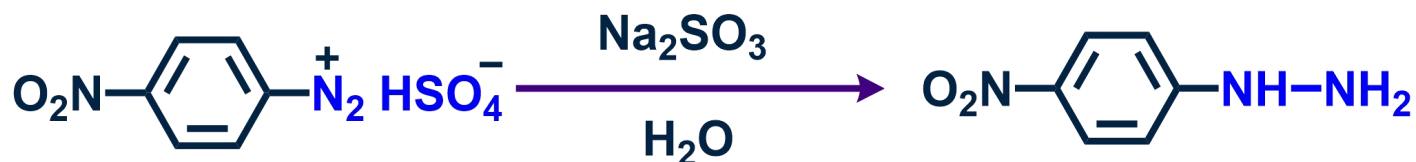
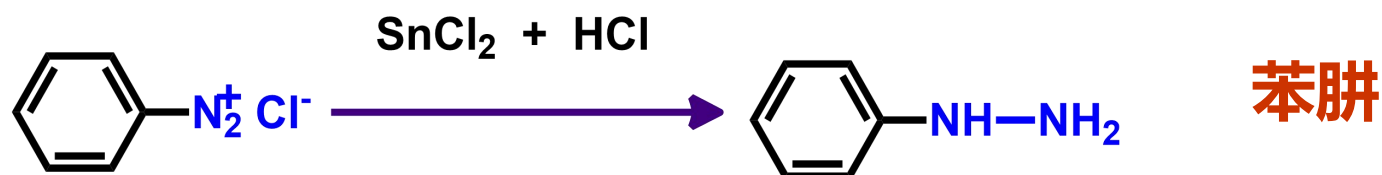
- 在芳环上引入用其他方法**难于引入**的 **-F, -I, -CN, -OH**
- 制备**不能**通过芳环上**亲电取代**反应得到的化合物



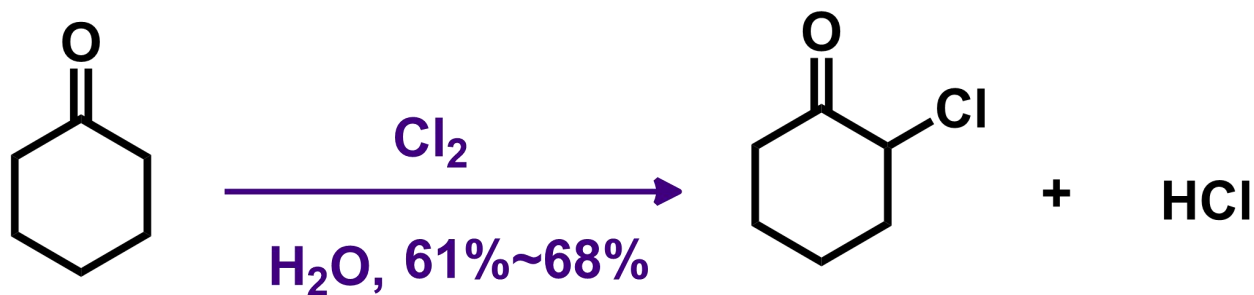
保留氮的反应

(1) -N=N 被还原

使用 $\text{SnCl}_2 + \text{HCl}$, NaHSO_3 , Na_2SO_3 , SO_2 等还原为芳基肼。



α -氢原子卤代反应



三、排序、选择及简答题（机理） 共计20分

- **排序题（4个题，每小题 2 分）**
- **选择题（4个题，每小题 2 分）**
- **简答题（1个题，共4 分）**

酸性强弱

若芳香环上有吸电子基团，则酚酸性增强。

有供电子基酚酸性减弱。如



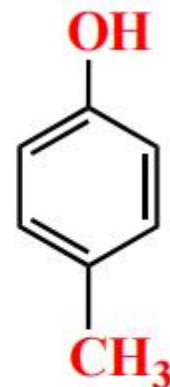
2,4-二硝基苯酚

$pK_a=4.09$



2,4,6-三硝基苯酚
(苦味酸)

$pK_a=0.25$



4-甲基苯酚
(对甲酚)

$pK_a=10.26$

羧酸的酸性和极化效应



酸性比碳酸强



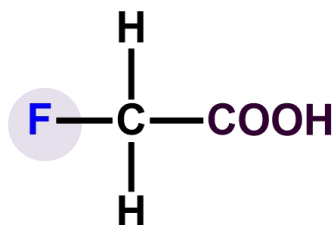
pK_a	
$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{H}$	4 ~ 5
$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{H}$	6.38
 -O-H	10
 -O-H	15.74
$\text{H}-\text{O}-\text{H}$	16 ~
$\text{R}-\text{CH}_2-\text{O}-\text{H}$	19
$\text{R}-\text{C}\equiv\text{C}-\text{H}$	~ 25
$\text{R}-\overset{\text{H}}{\text{N}}-\text{H}$	~ 34
$\text{R}-\text{H}$	~ 50

酸性

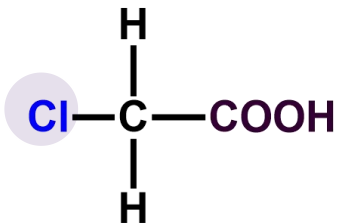
取代基对酸性的影响 (1)



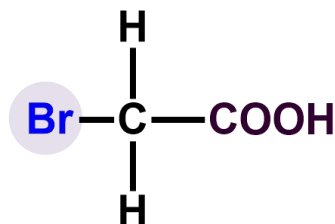
不同原子



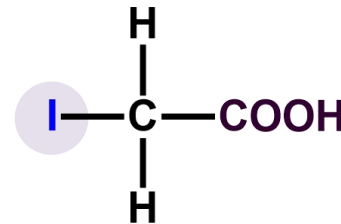
2.66



2.81

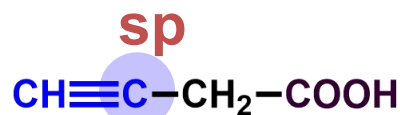


2.87

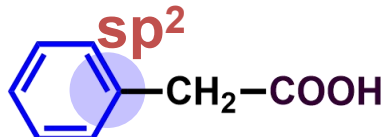


3.31

不同杂化



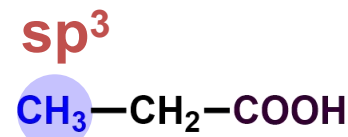
3.32



4.31



4.35

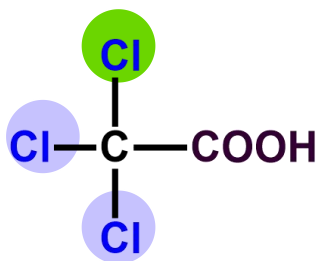


4.82

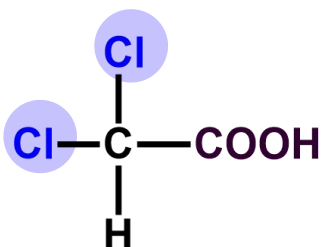
取代基对酸性的影响 (2)



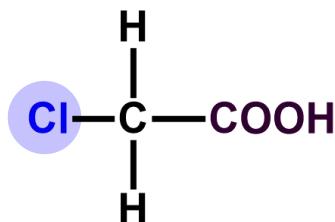
不同数目



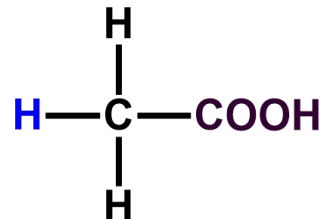
0.70



1.29



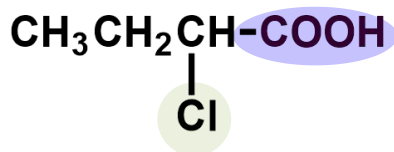
2.81



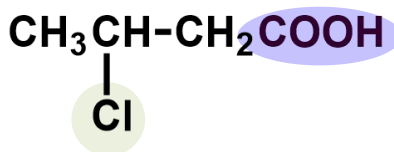
4.75

不同距离

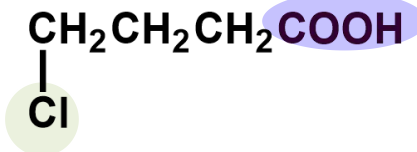
诱导效应随着传递距离的增大而迅速减小。



2.86



4.05

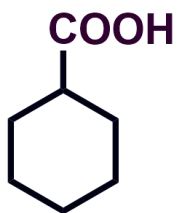
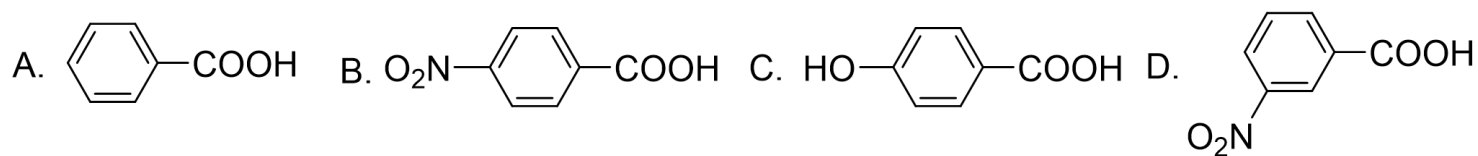


4.57

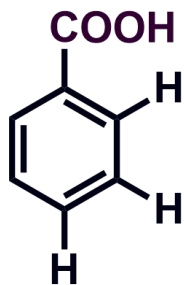


4.82

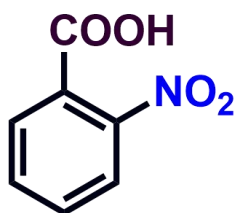
取代基对酸性的影响 (3)



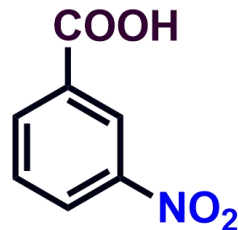
4.87



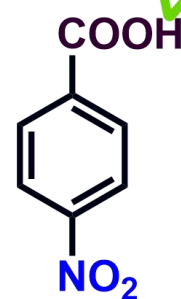
4.20



2.21



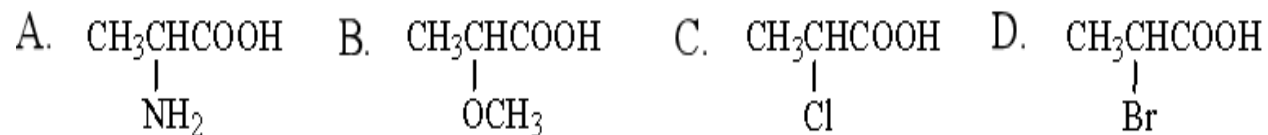
3.49



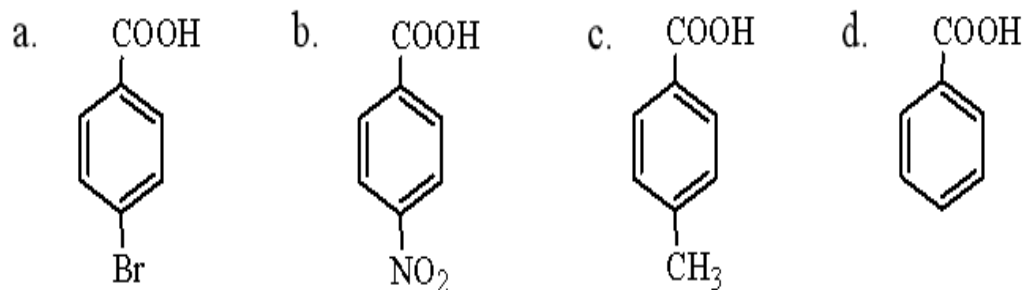
3.42



下列化合物，酸性最强的是 C



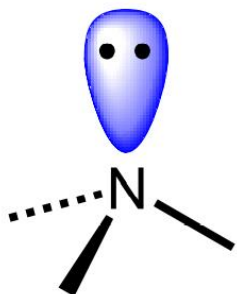
下列化合物，酸性由大到小次序是 B



A. $c > d > b > a$ **B. $b > a > d > c$** C. $a > b > c > d$ D. $b > a > c > d$

碱性强弱

碱性--影响因素



- N上所连取代基的电性--给电子，碱性↑
- 质子进攻N孤电子对的位阻---位阻大，碱性↓

N-所连烷基使碱性增强，↑但质子进攻的空间阻力也增加。↓

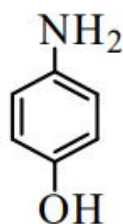


脂肪族 仲胺 > 伯胺 > 叔胺 > 氨

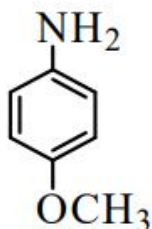
芳香胺



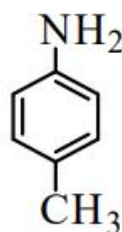
各种对位取代芳胺的 pK_b 值如下：



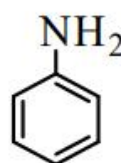
pK_b : 8.5



8.7

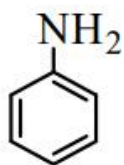


8.9

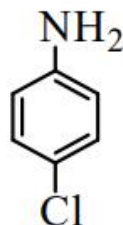


9.3

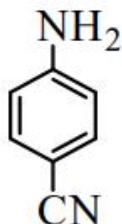
氨基对位连有给电子基团， pK_b 值减小，碱性增强。



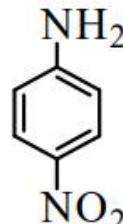
pK_b : 9.3



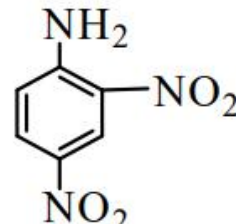
10.0



12.3



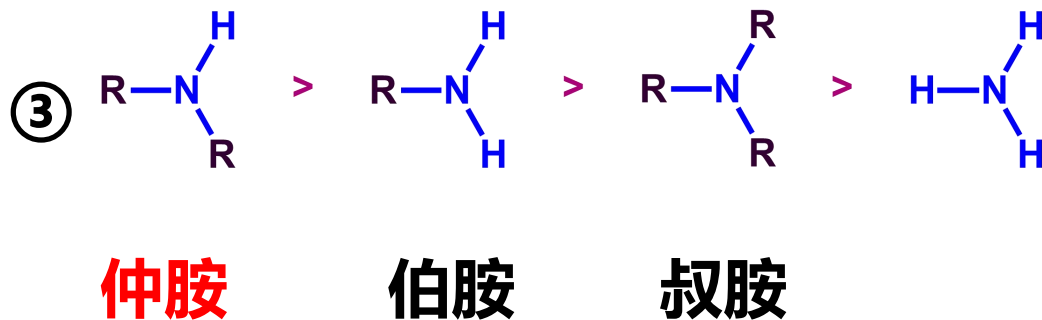
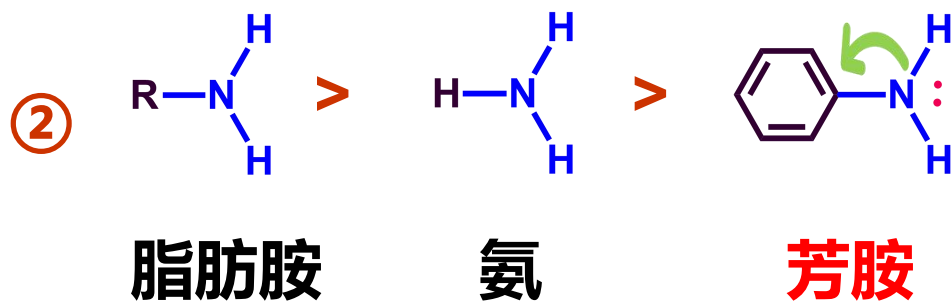
13.0



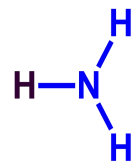
13.8

氨基对位连有吸电子基团， pK_b 值增大，碱性减弱。

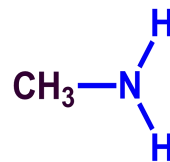
碱性大小



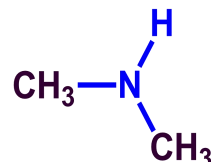
pK_b



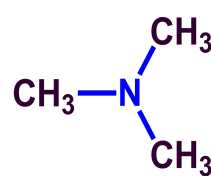
4.7



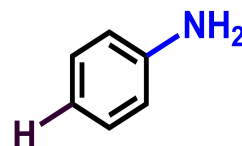
3.4



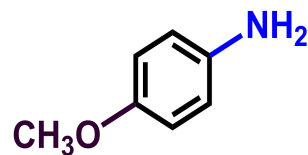
3.3



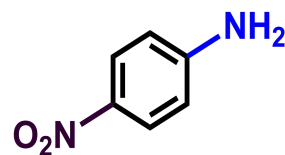
4.3



9.4

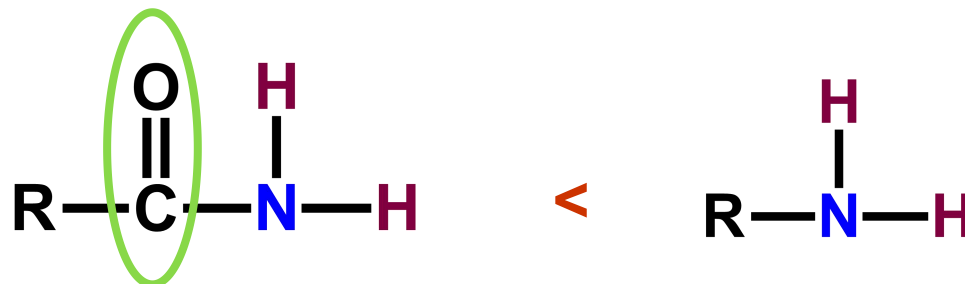


8.7



13.0

酰胺的碱性

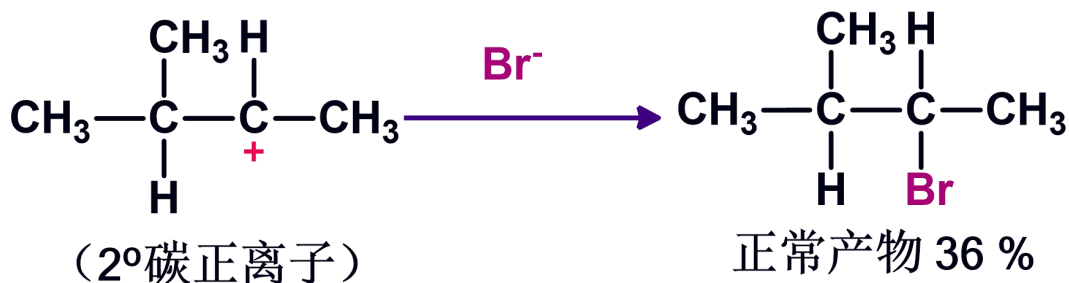
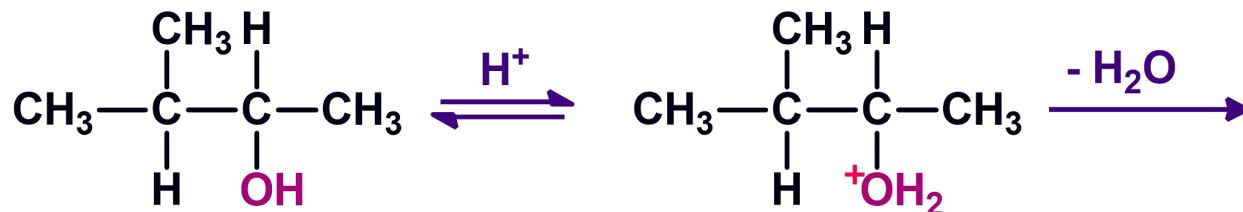


酰基：

-I 效应

-C 效应

卤代过程中的重排



活性强弱

影响 $>\text{C}=\text{O}$ 加成反应的因素：

- 羰基C原子上正电荷密度(电子效应)
- $>\text{C}=\text{O}$ 碳上所连基团（立体位阻）

碳原子上正电荷密度越大，越有利于反应：



羰基碳上所连基团越小，越有利于反应



羰基的反应活性

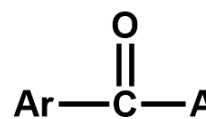
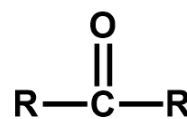
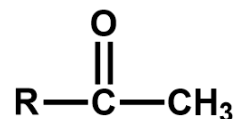
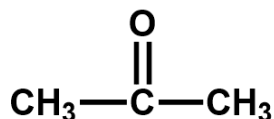
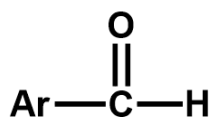
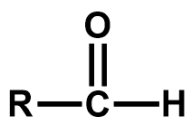
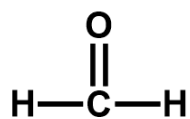
(a)

醛

芳香族醛

酮

芳香族酮

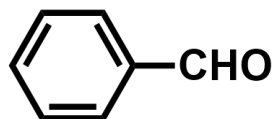


反应活性依次降低

(b)



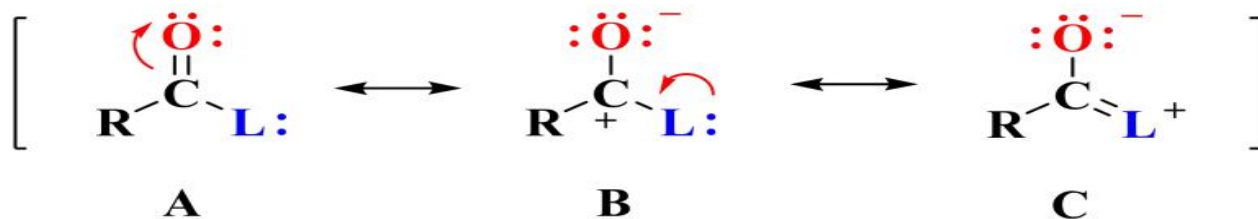
吸电子基
增加活性



供电子基,
降低活性

羰基的反应活性

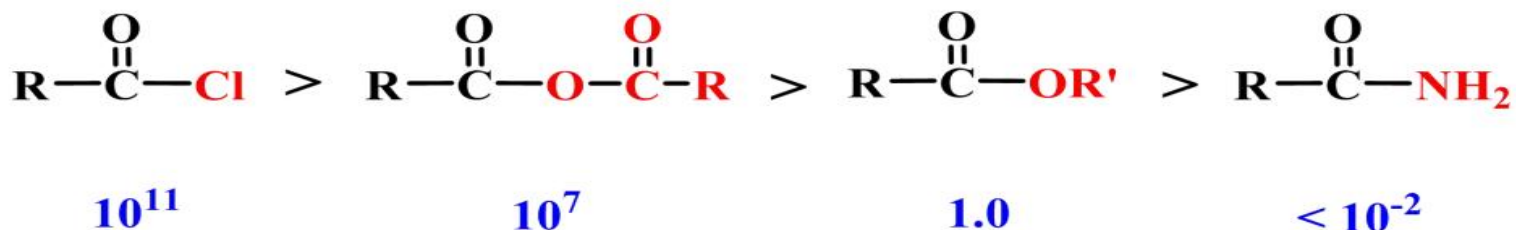
A. 电子效应解释—C原子正电荷密度



共轭作用越强，反应活性越弱；

诱导效应越强，反应活性越强；

水解反应为例：



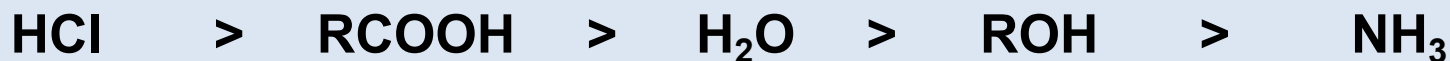
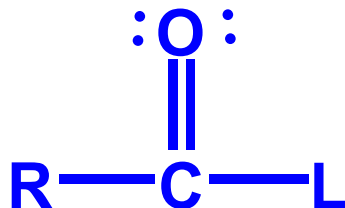
Cl-电负性强，共轭作用弱

N-电负性弱，共轭作用强

羰基的反应活性

B. 酸碱理论——L离去能力分析

酸性：

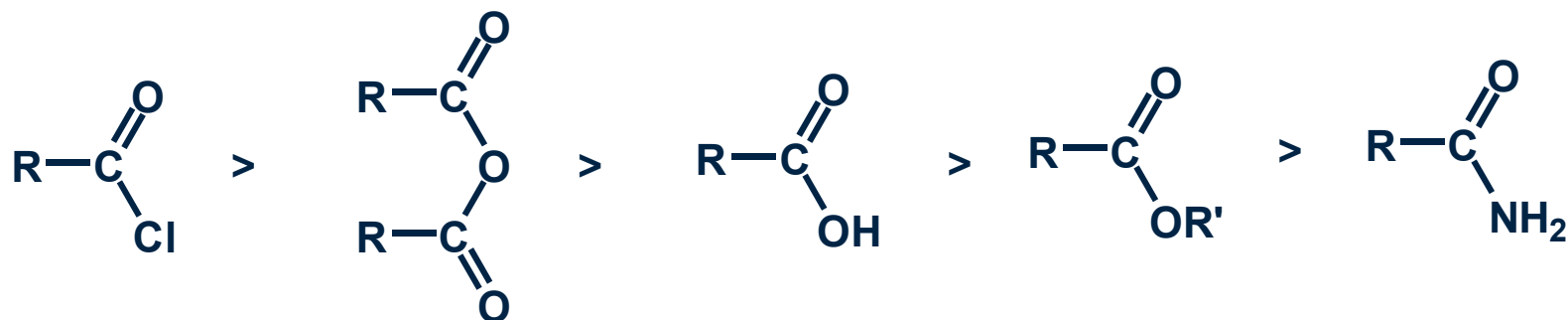


共轭碱碱性：



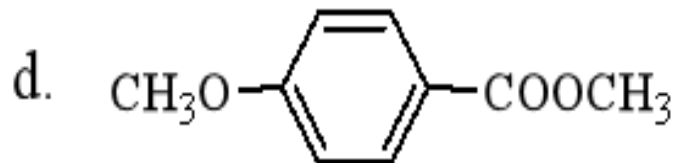
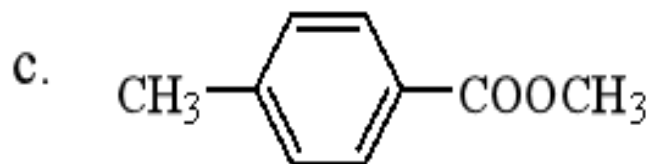
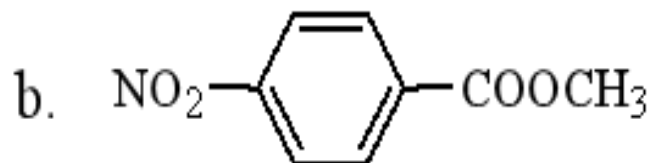
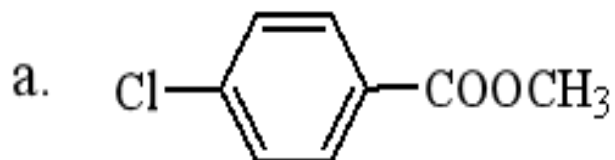
碱性越弱，越容易离去，相应酰基化试剂的活性越大。

羰基的活性次序



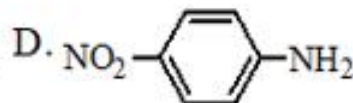
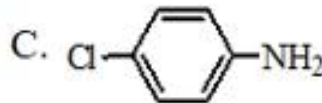
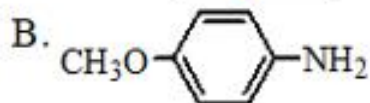
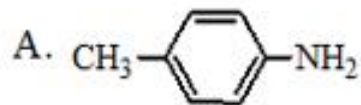
反应活性依次降低

下列酯在碱性条件下，**水解的活性**由大到小： $b > a > c > d$



↵

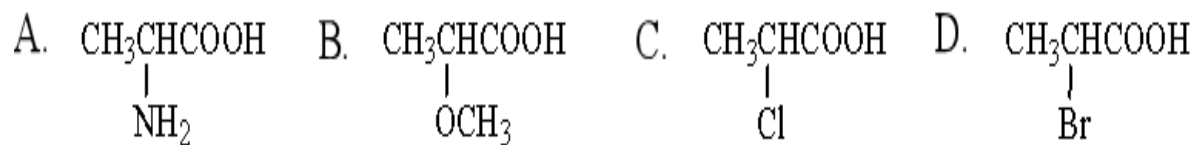
3. 下列化合物中，碱性最强的是 **B** ↵



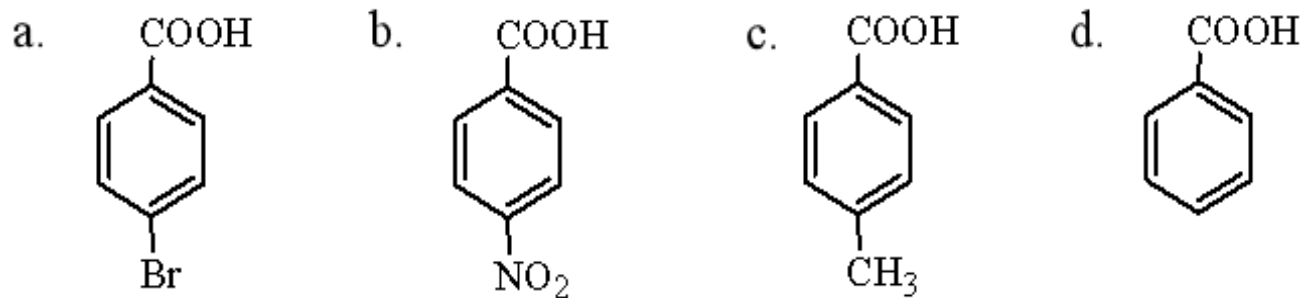
↵

$b > a > c > d$

下列化合物，酸性最强的是 C

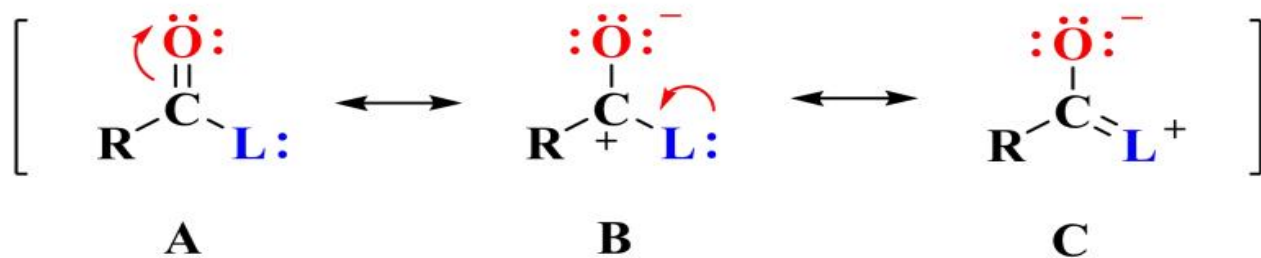
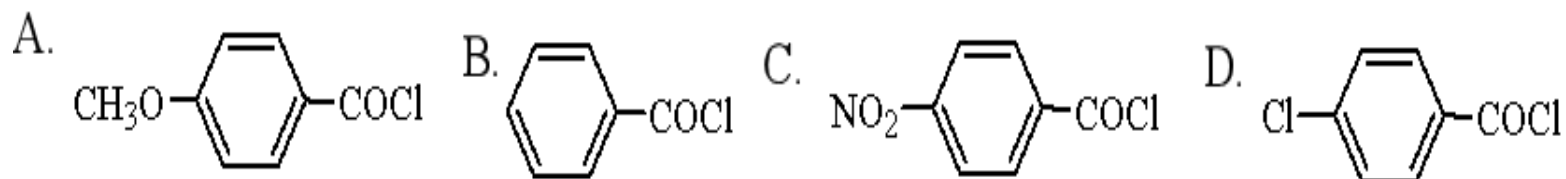


下列化合物，酸性由大到小次序是

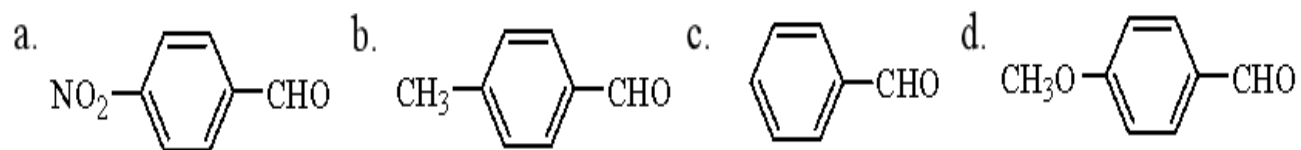


$\mathbf{b > a > d > c}$

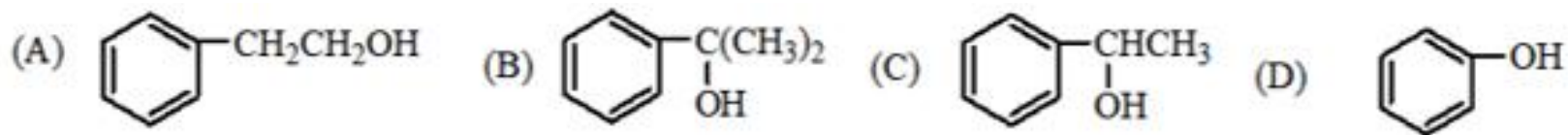
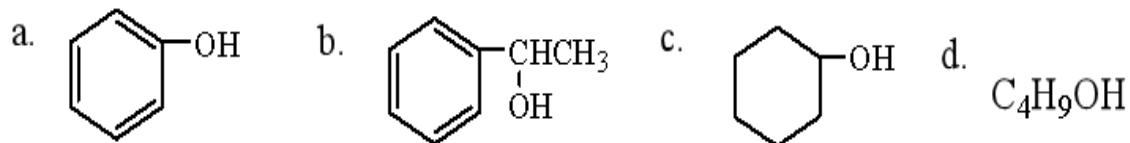
下列化合物氨解时，速率大小排序 C > D > B > A



下列化合物与HCN反应，活性由大到小次序正确的是 **a > c > b > d**

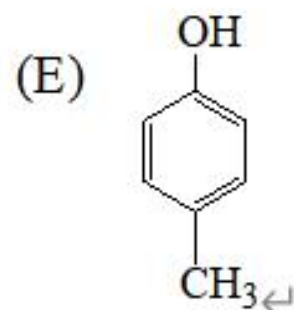
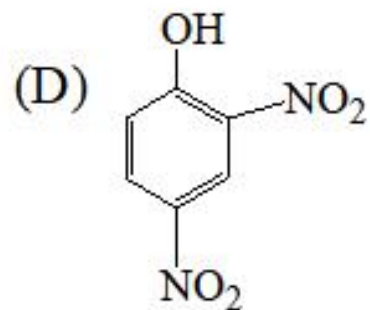
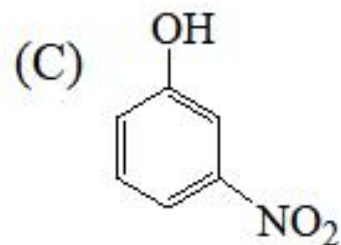
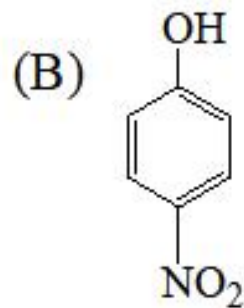
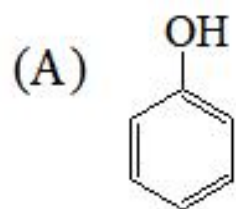


下列化合物脱水反应的反应速率由快到慢次序正确的是 **b > c > d > a**



B > C > A > D

(1) 按酸性强弱←

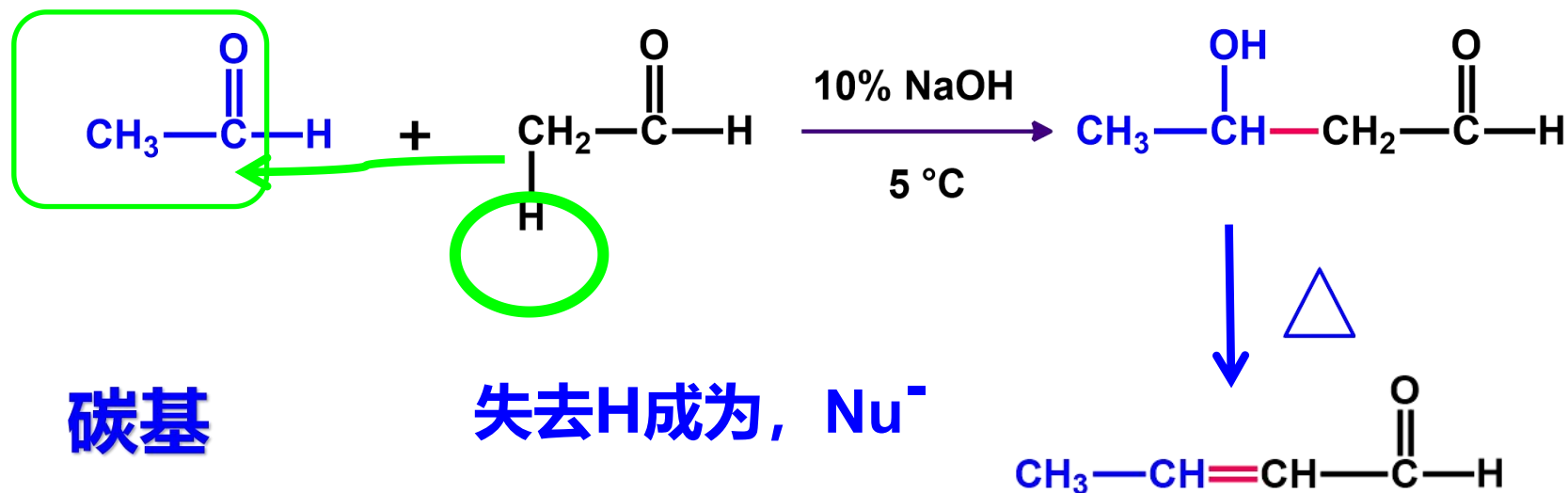


D > B > C > A > E

碳链延长

羟醛缩合:

含有活泼 α -H化合物与碳基,在酸/碱的催化下, 缩合反应



合成: α, β -不饱和醛 (酮)

合成特征: 碳链成倍增长

羟醛缩合:

A. 羟醛缩合

1. 自身羟醛缩合
2. 交叉羟醛缩合
3. 分子内羟醛缩合

B. 克莱森-斯密特缩合

芳醛与含 α -氢醛或酮

C. Perkin (珀金)

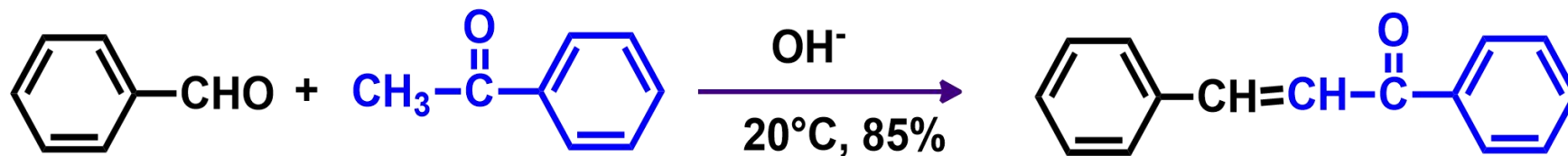
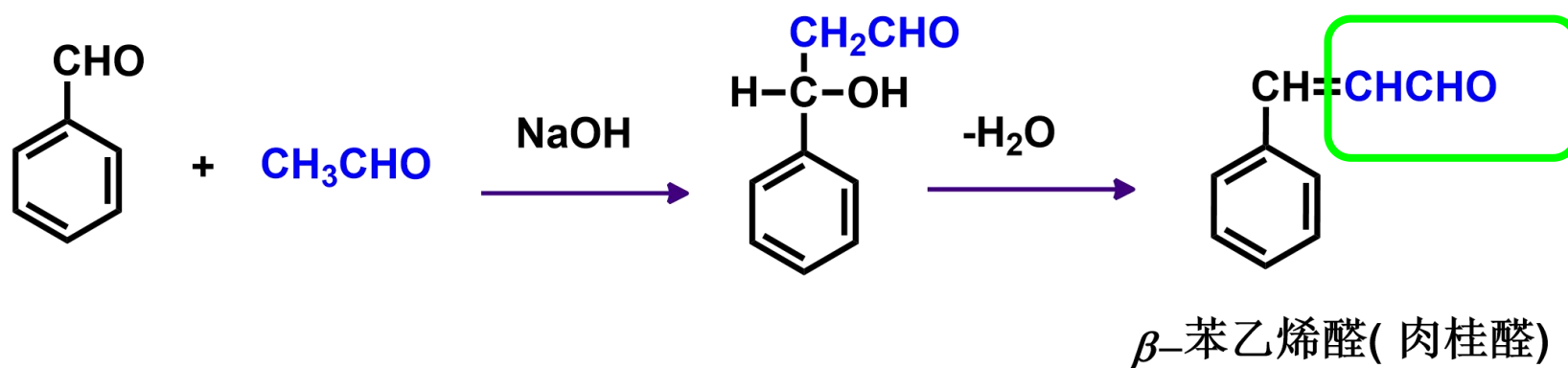
芳醛与脂肪族酸酐

D. Mannich (曼尼希)

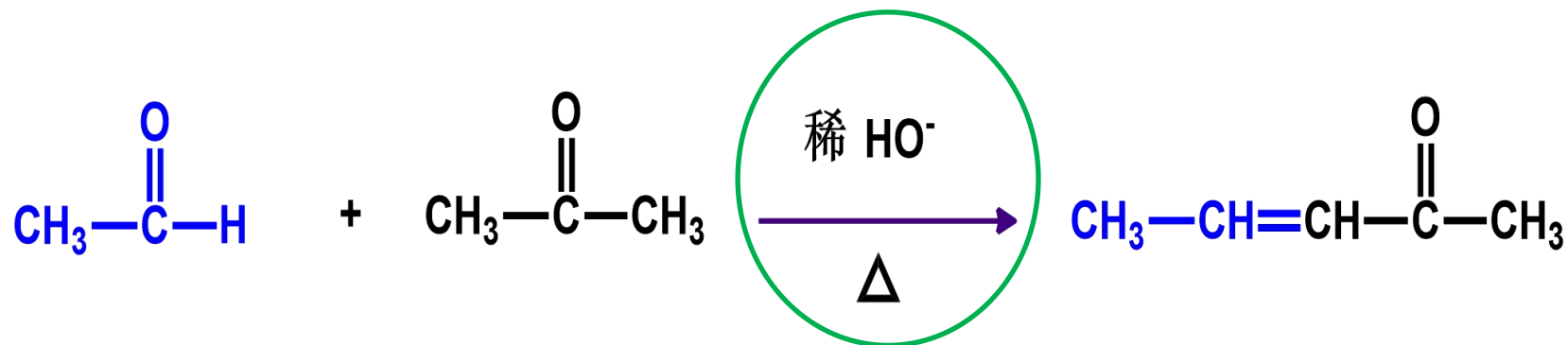
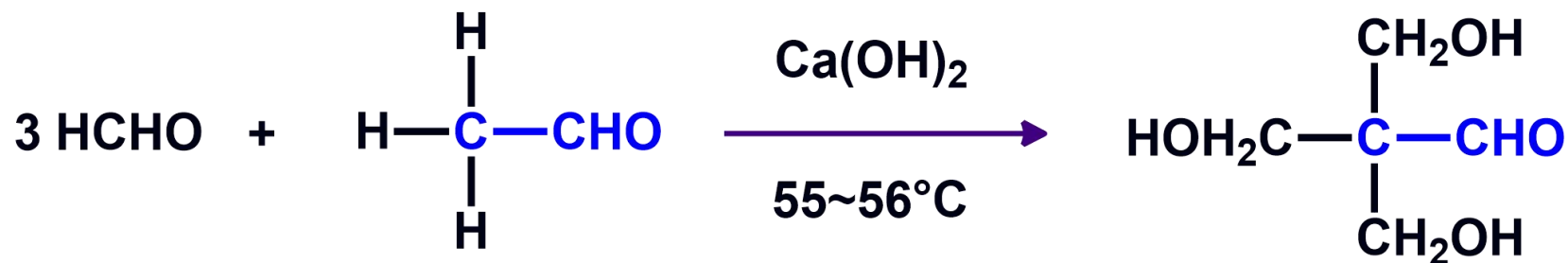
羰基、甲醛、仲胺, **三组份缩合**

含有活泼**H化合物与碳基**,在碱作用下缩合 α , β -不饱和羰基化合物

芳醛与含 α -氢醛或酮在碱性条件下脱水、生成 α,β -不饱和醛或酮。



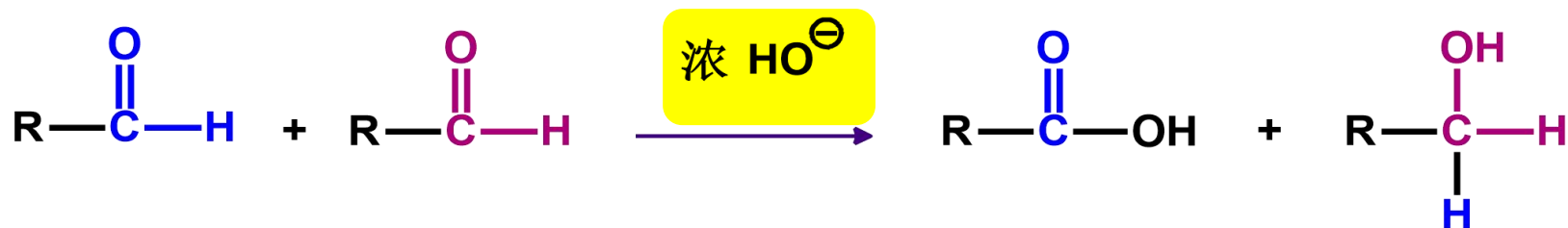
一分子没有 α -H, 另一分子有 α -H--醛酮发生缩合;



Cannizzaro反应(歧化反应)

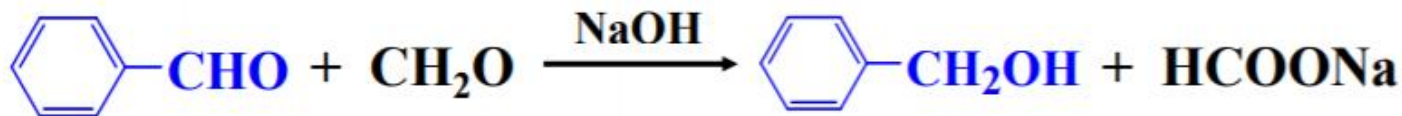
没有 α -氢的醛，在浓碱作用下：

一分子醛被氧化为酸，另一分子被还原成醇

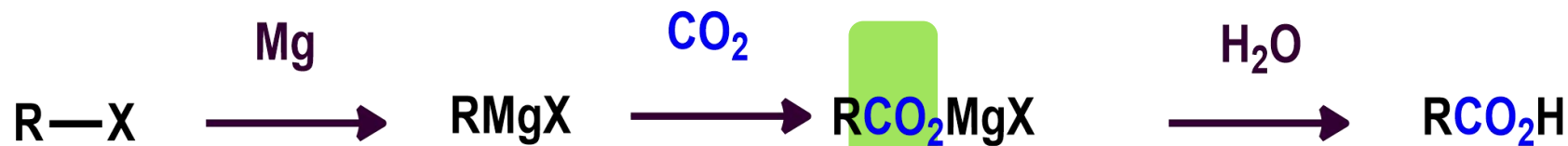


R = 没有 α -H的烃基，芳基

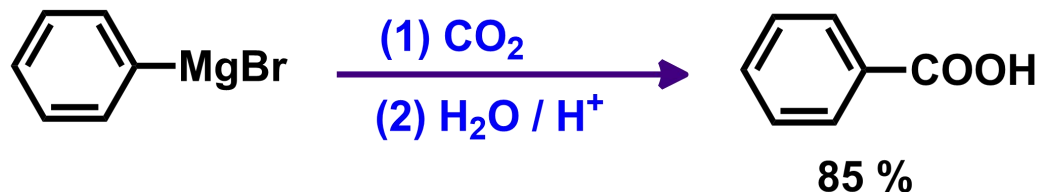
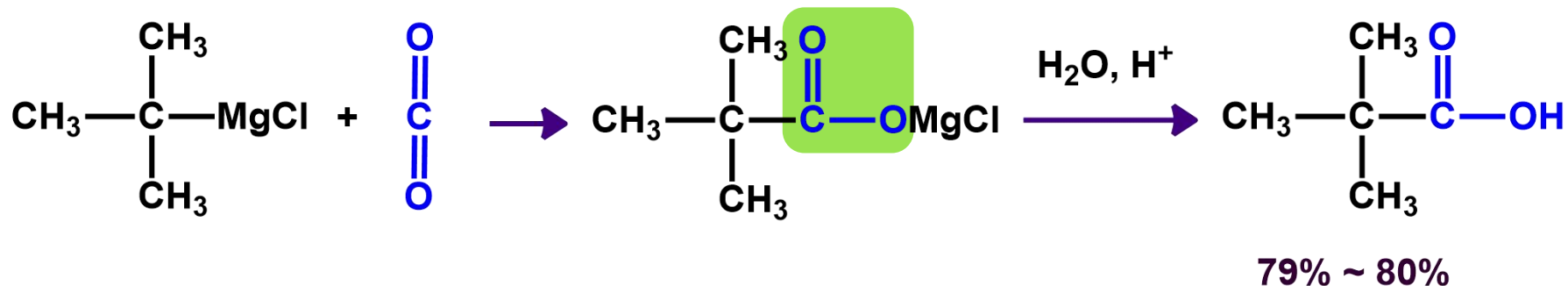
交叉坎尼扎罗反应中，甲醛总是被氧化。



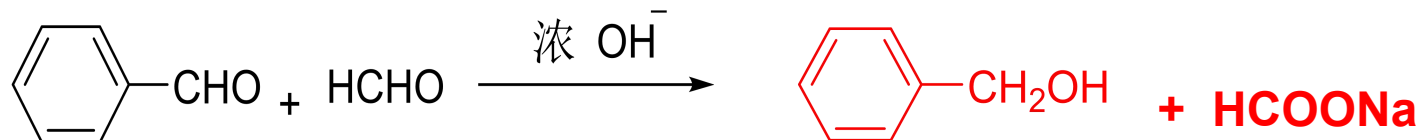
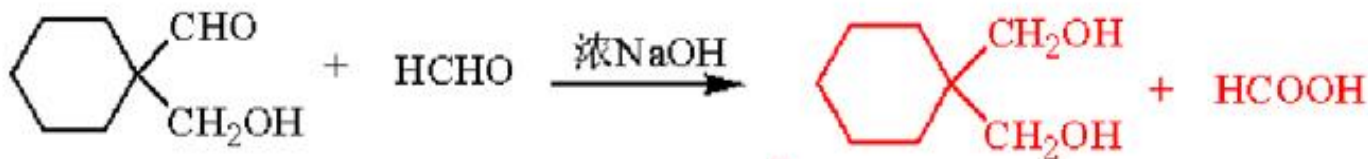
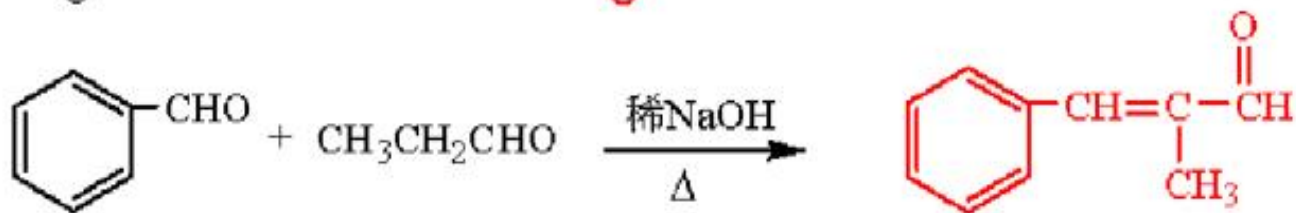
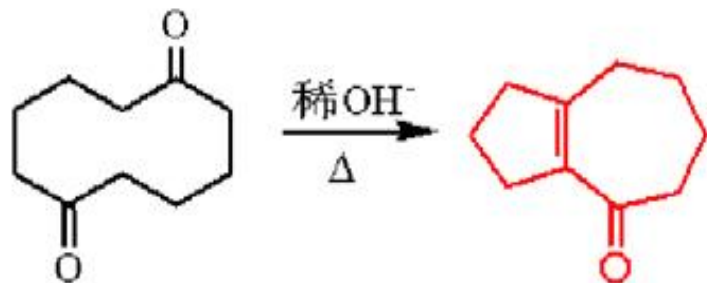
增加一个C--格式试剂与二氧化碳

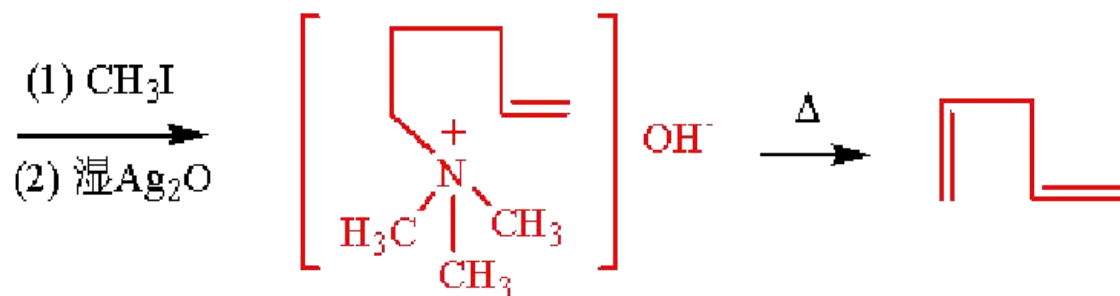


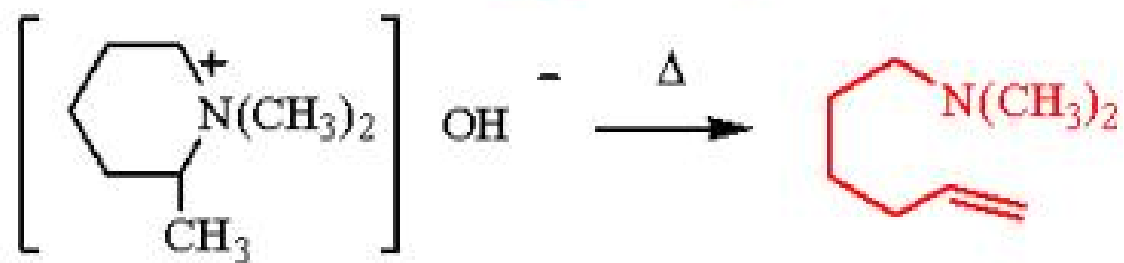
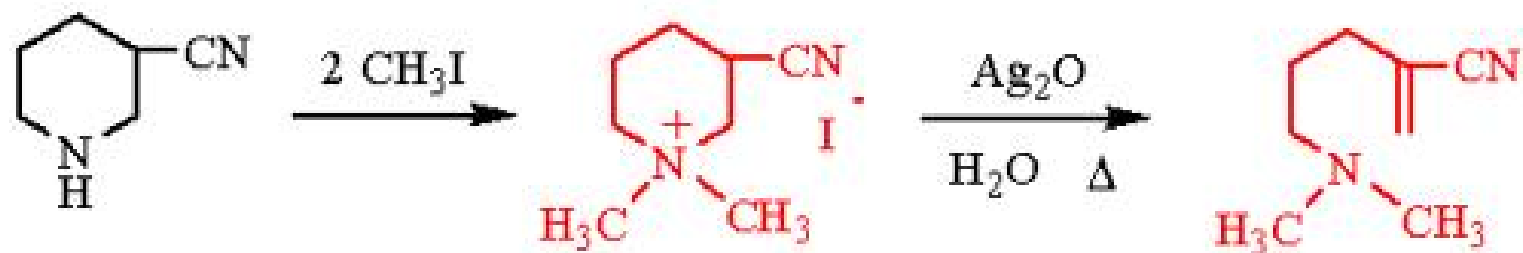
例:

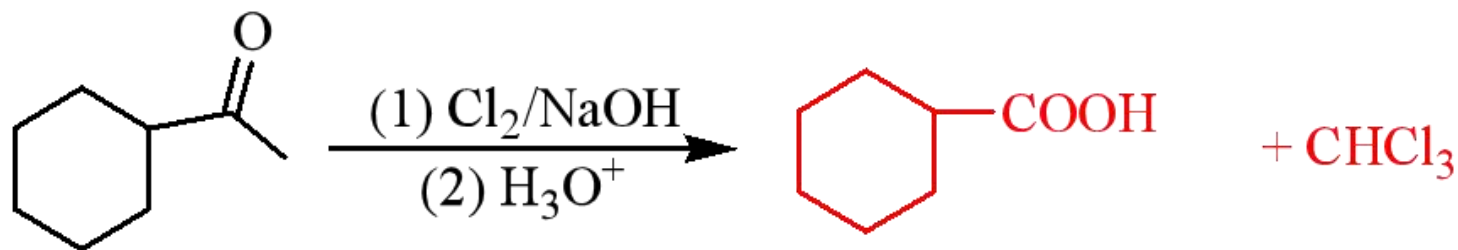
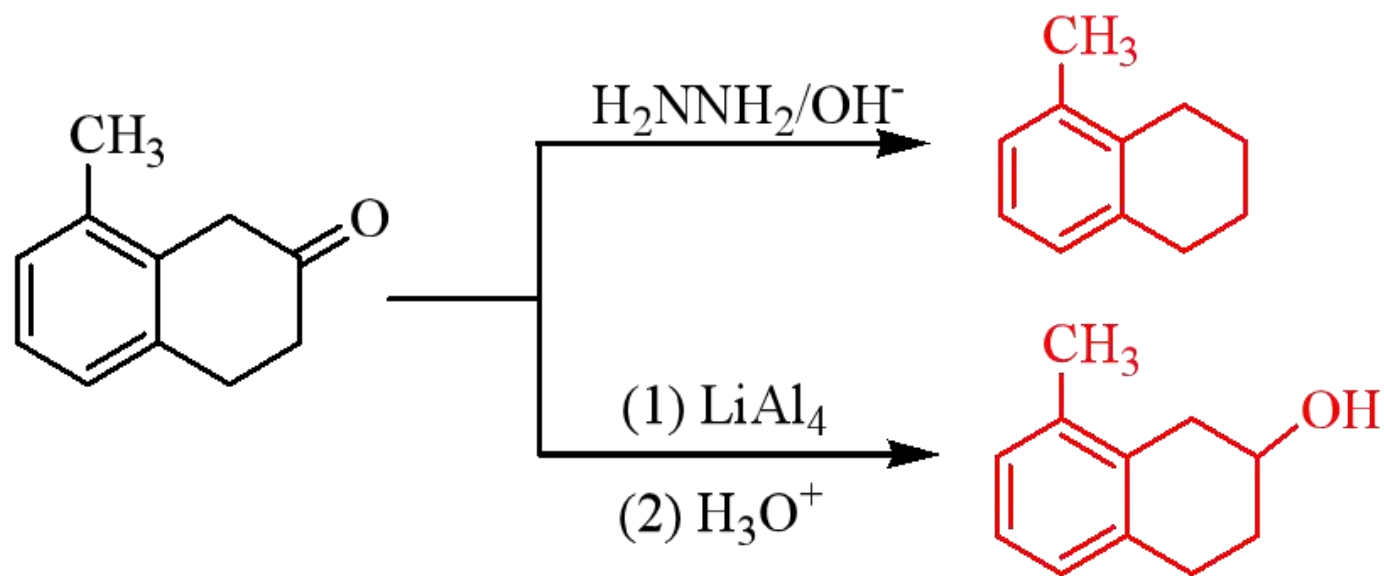
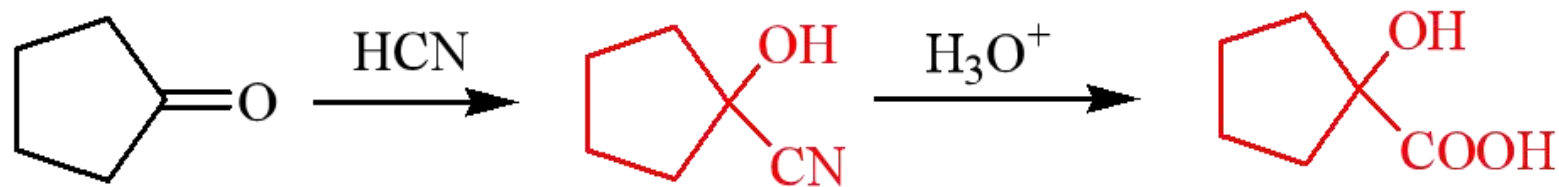


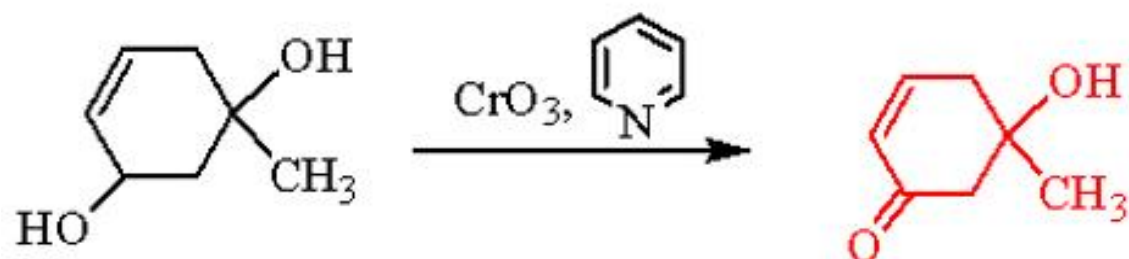
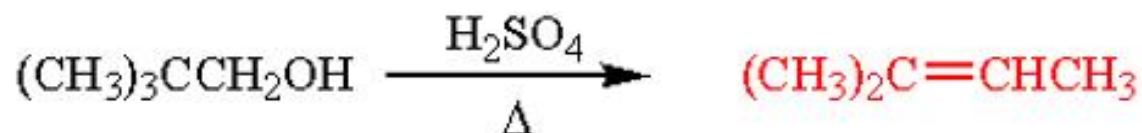
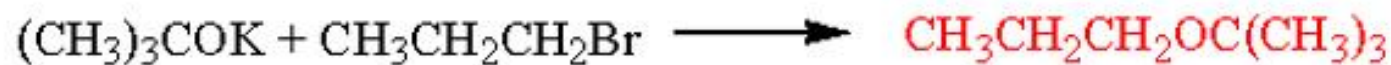
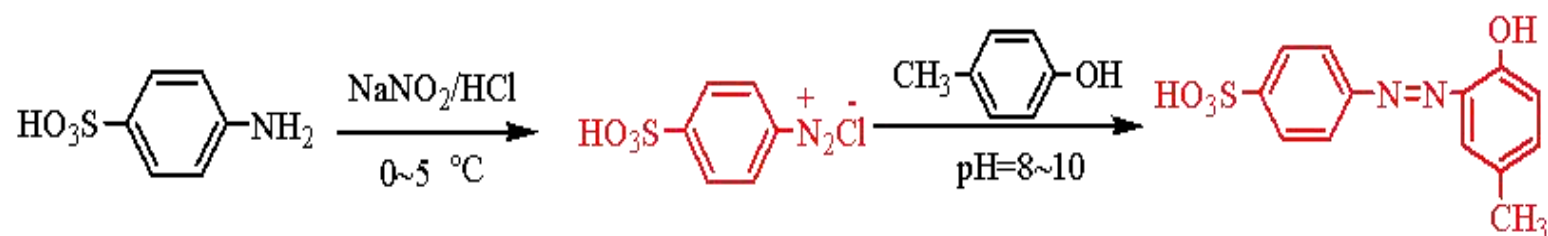
注意条件：稀的NaOH，还是浓的NaOH

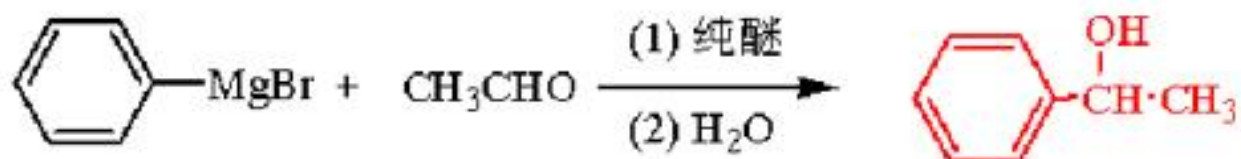
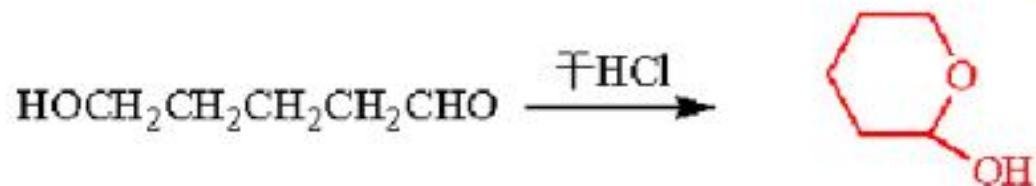
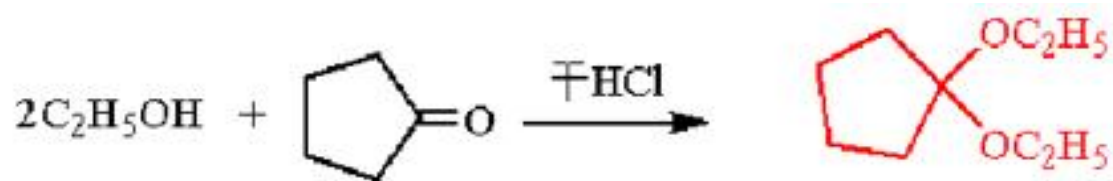
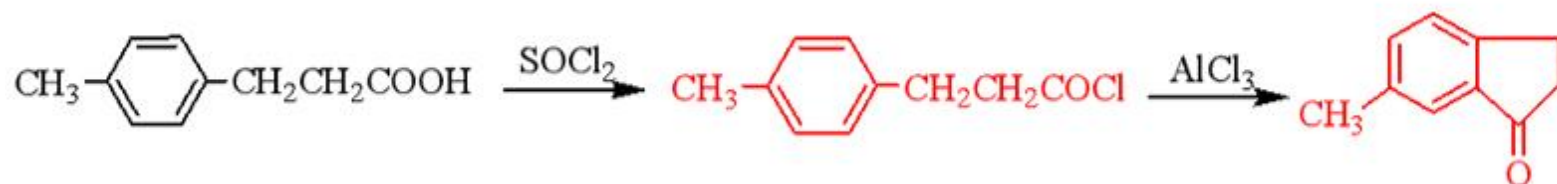
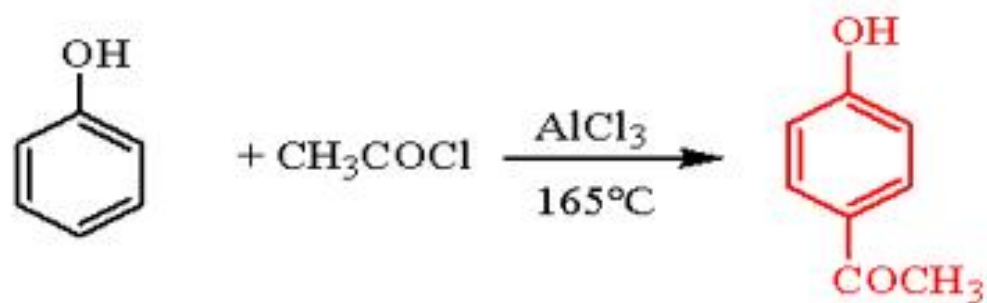






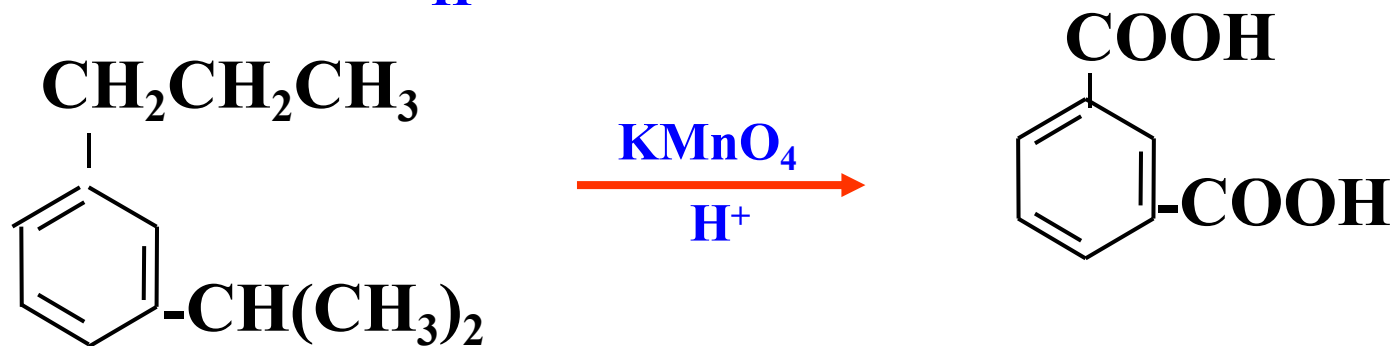
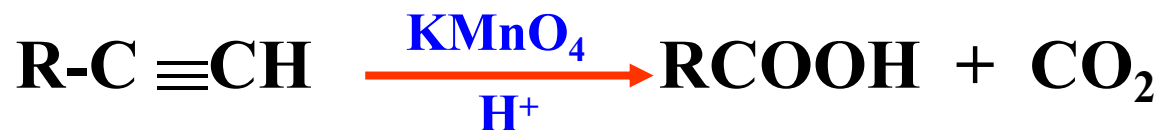
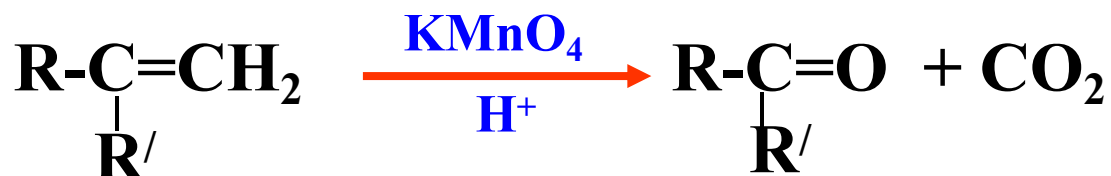
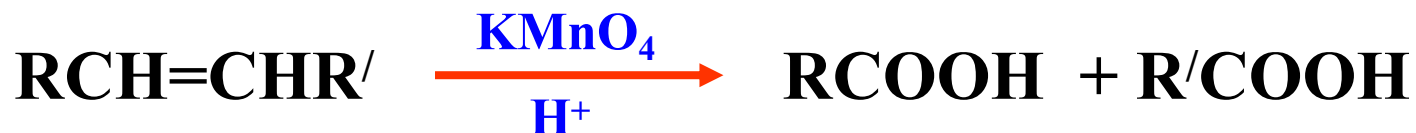




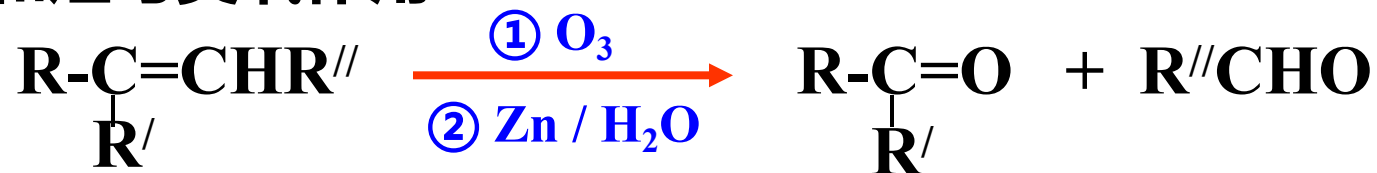


碳链缩短

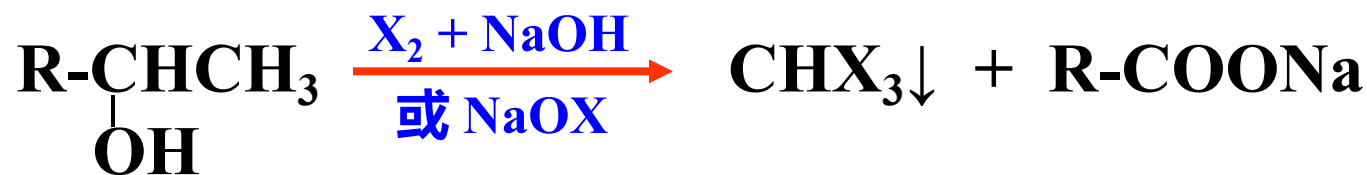
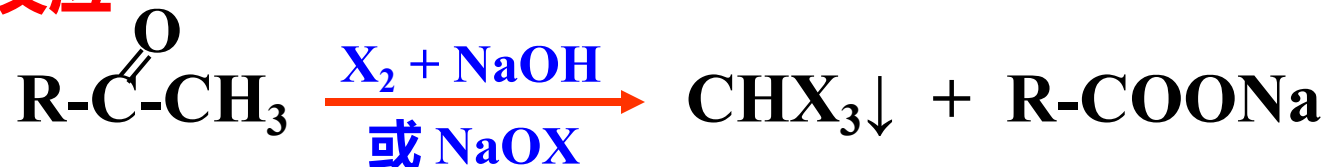
① 不饱和烃的氧化



② 不饱和烃与臭氧作用



③ 卤仿反应

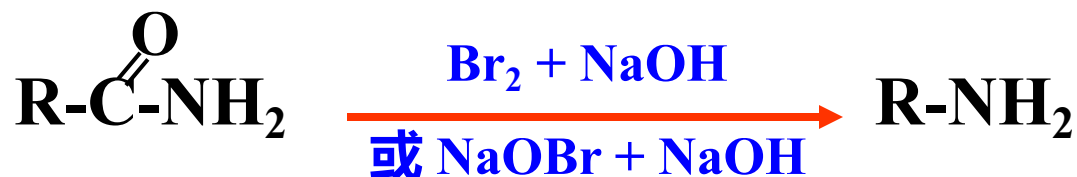


X 为 Cl、Br、I

④ 脱羧反应



⑤ 霍夫曼降解反应 (本次不考核)



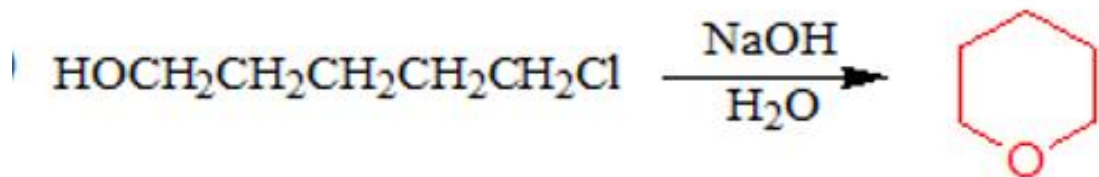
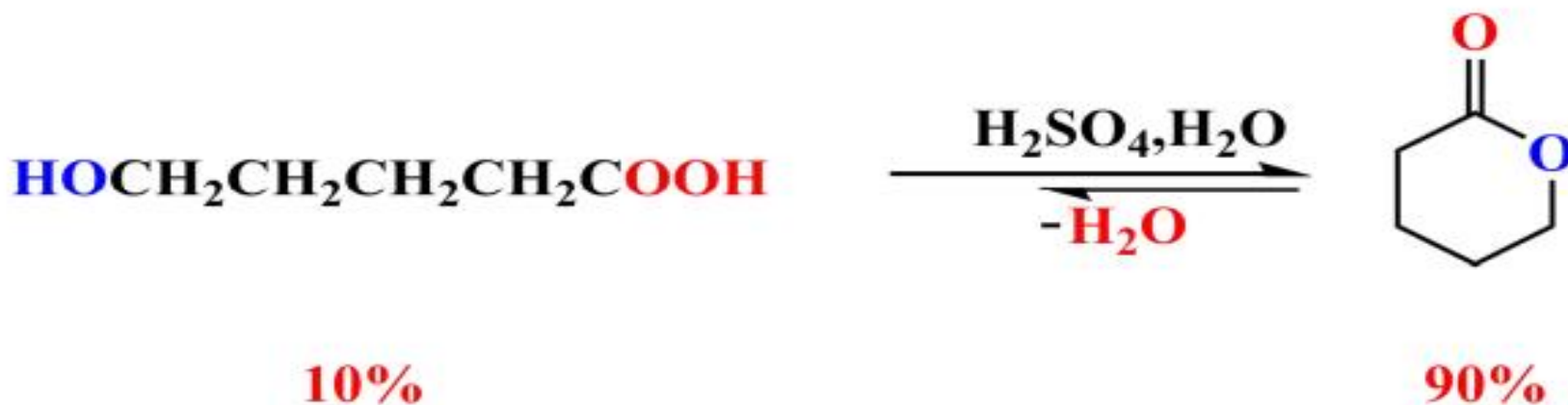
成环延长

环状半缩醛的形成---五/六环



分子内酯化

羟基酸经催化量的无机酸处理，可以通过分子内酯化反应生成环状酯也称为内酯 5元环和6元环的内酯容易形成



分子内成醚

二元酸的受热反应

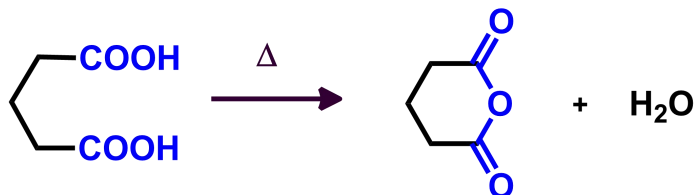
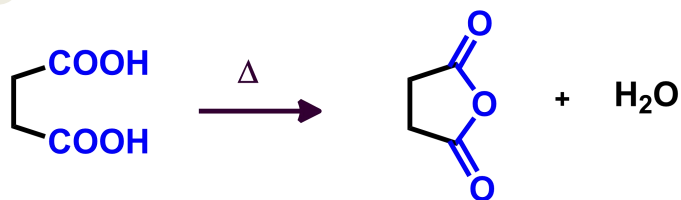


1

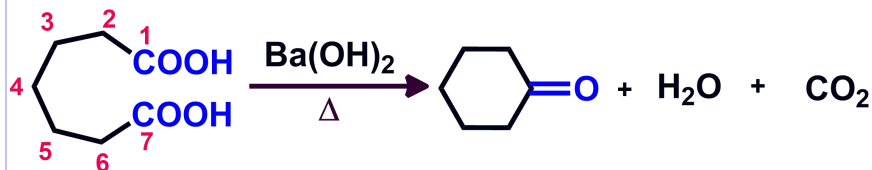
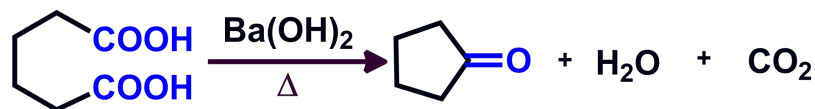


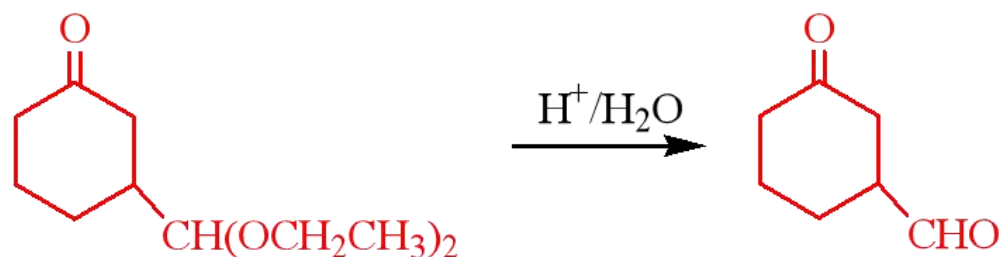
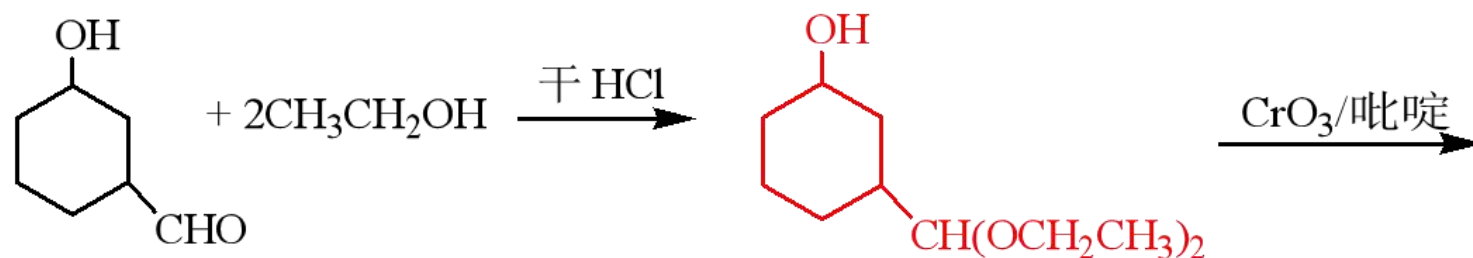
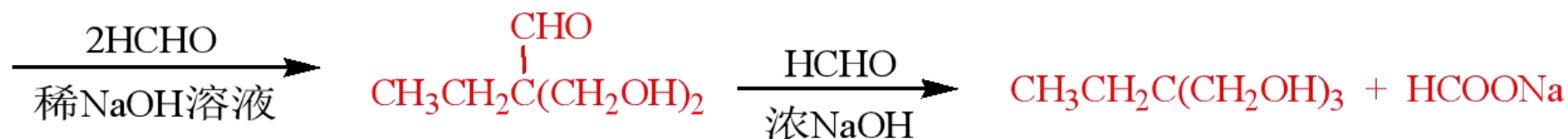
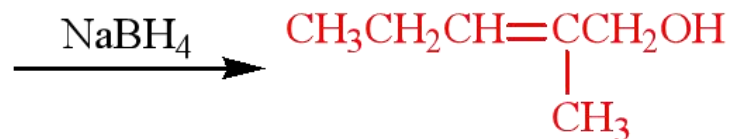
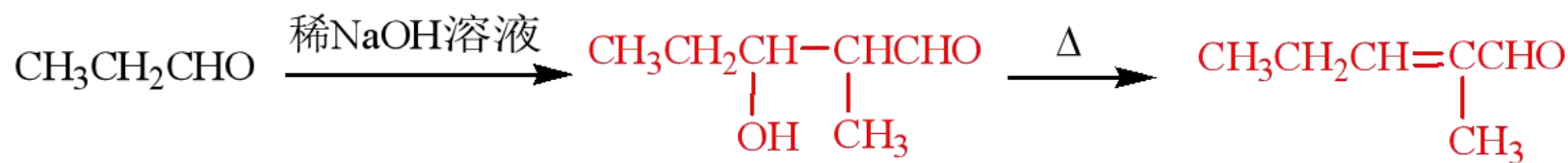
Blanc规律：容易形成五元环或六元环化合物

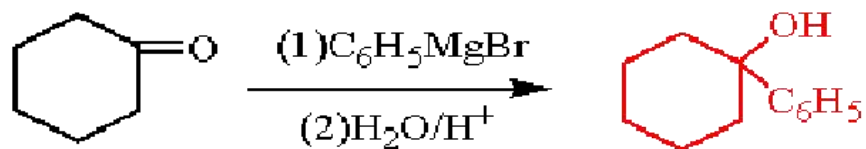
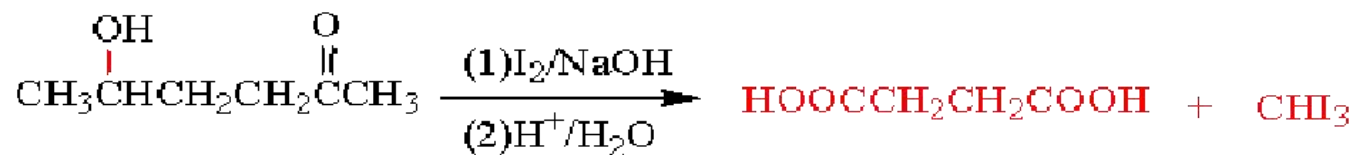
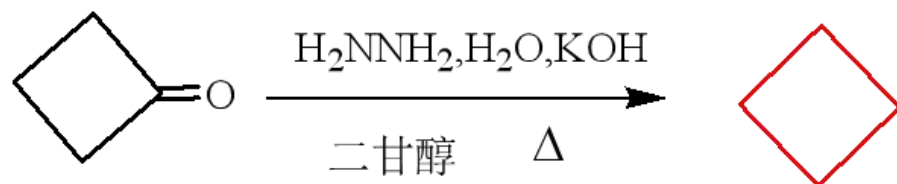
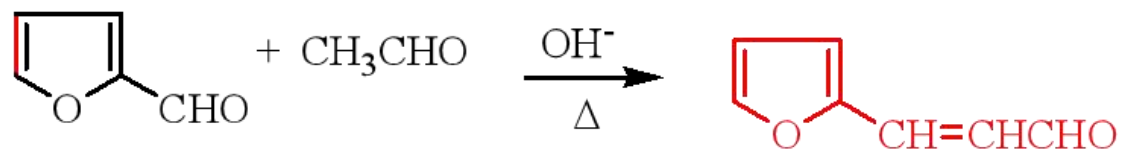
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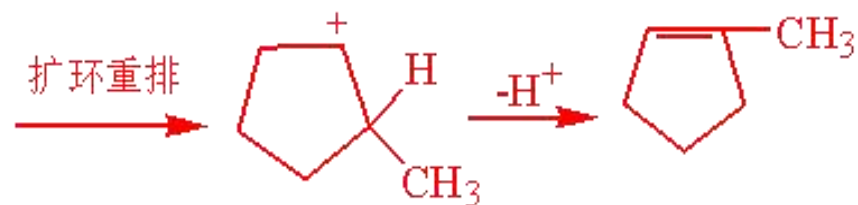
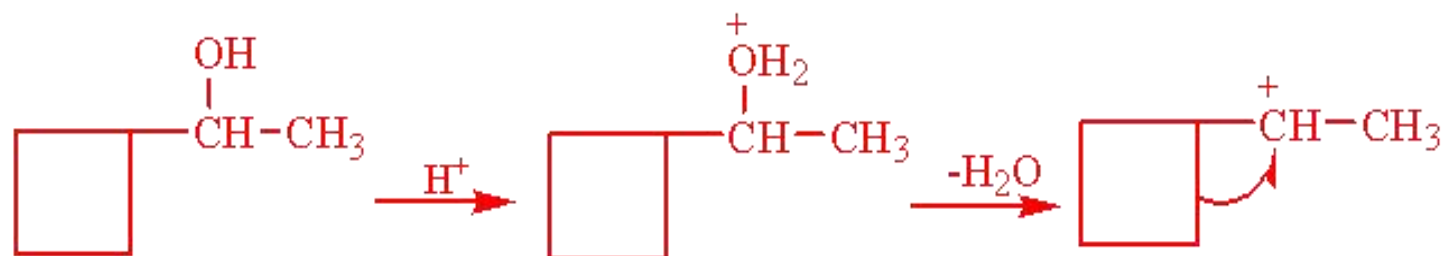
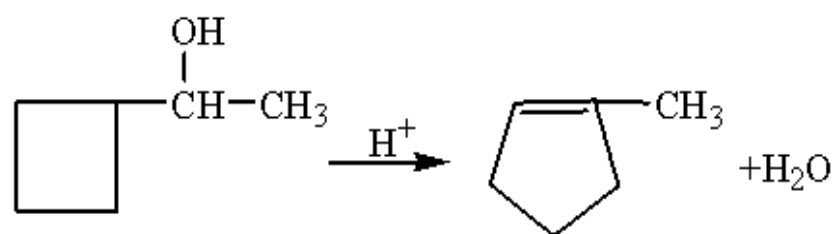
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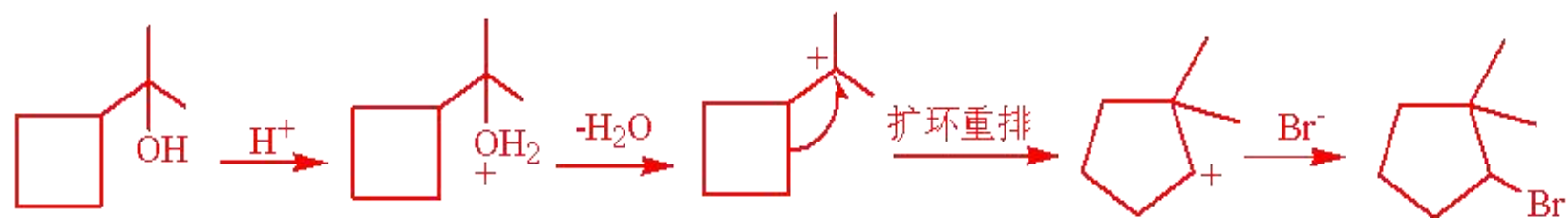
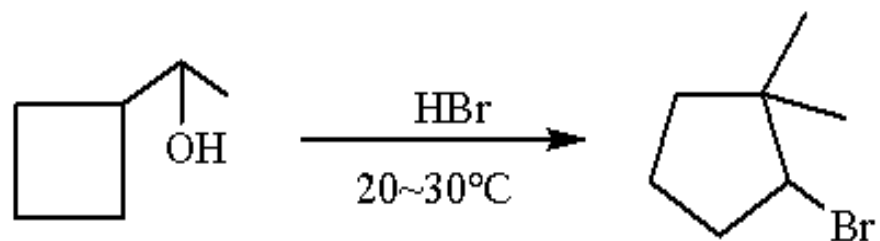


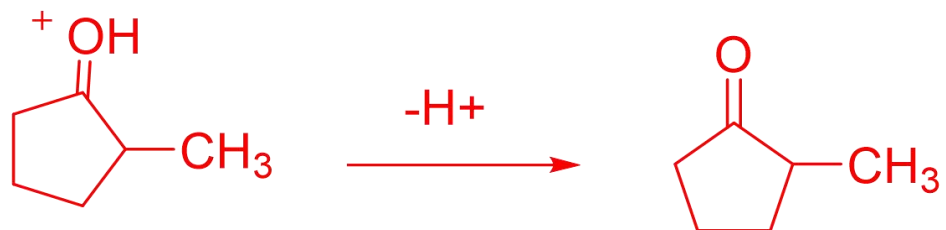
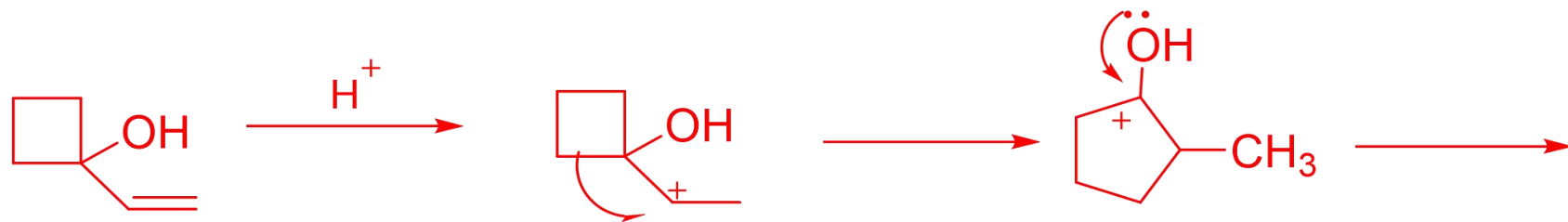
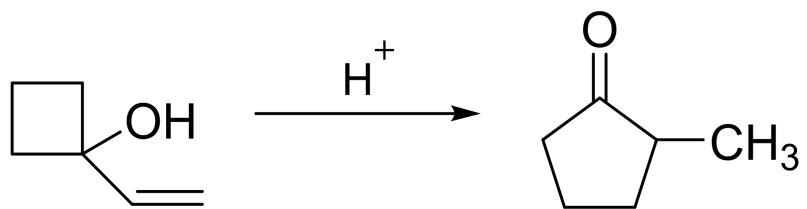


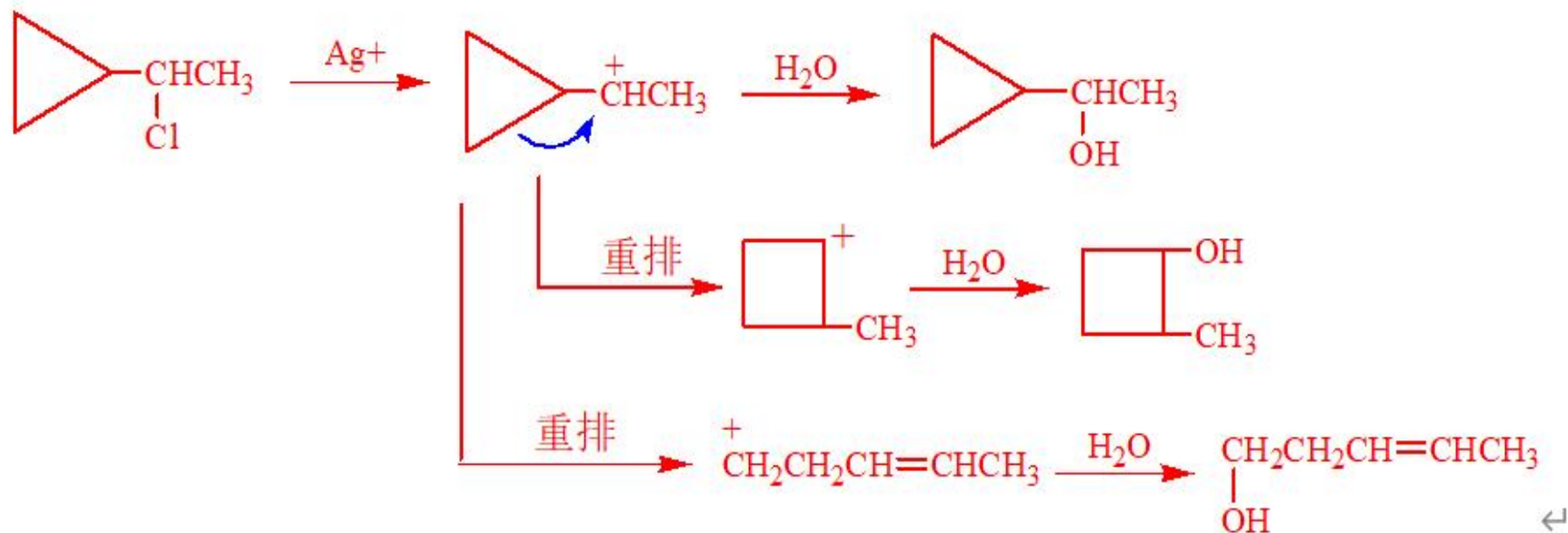
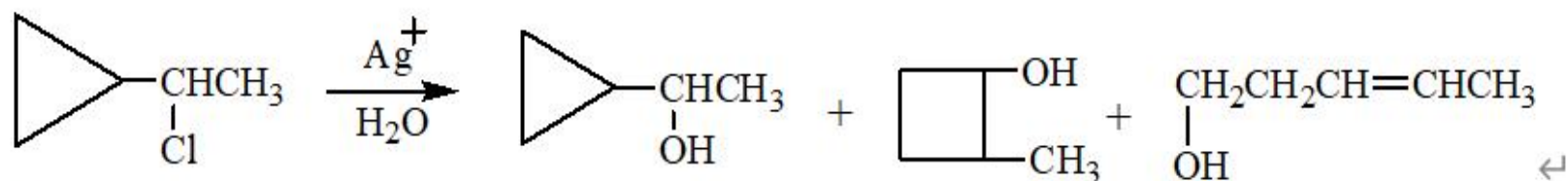


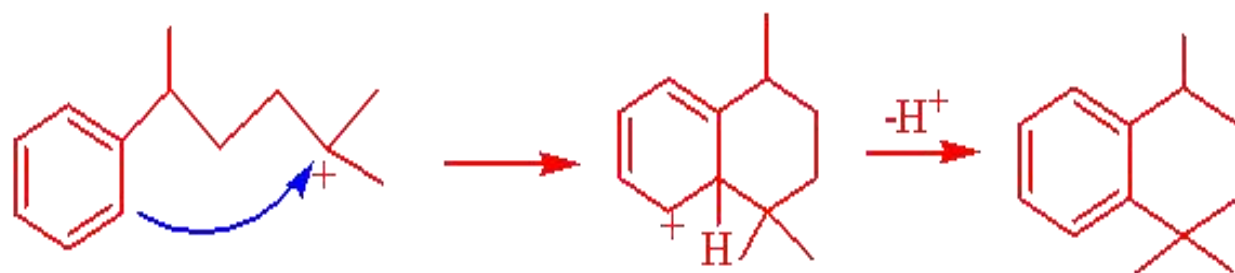
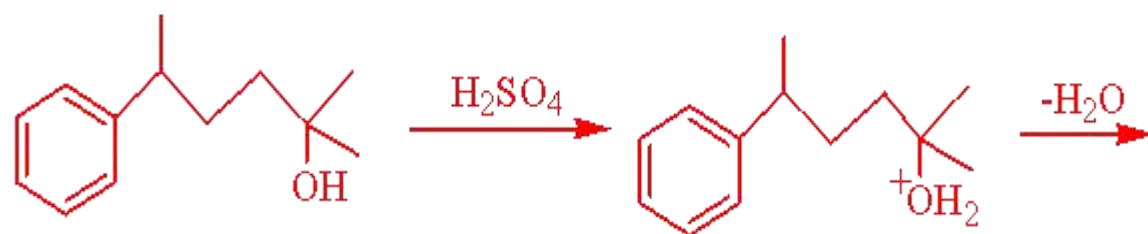
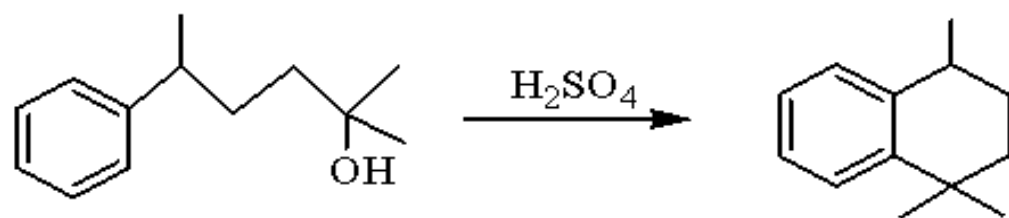
机理题目











四、用简单的化学方法鉴别下列化合物

- 第一小题 6分,**
- 第二题 5 分**
- 共计 11分**

卤代烃与硝酸银作用



该反应用于鉴定卤代烷

反应活性:



叔卤代烃: 立即沉淀

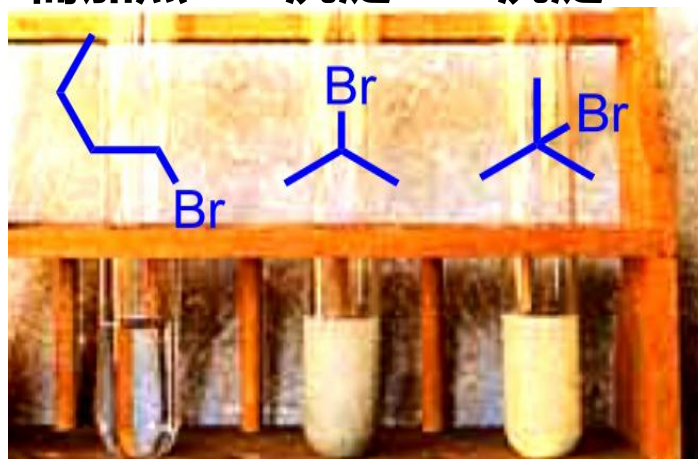
仲卤代烃: 片刻沉淀

伯卤代烷: 需加热

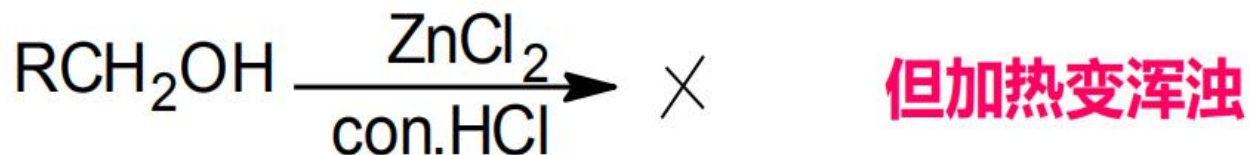
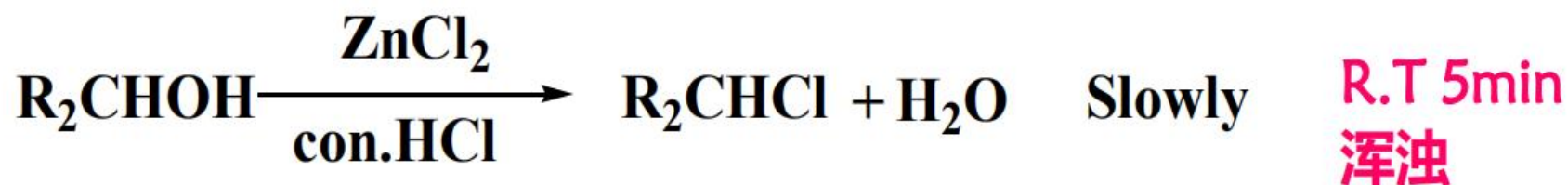
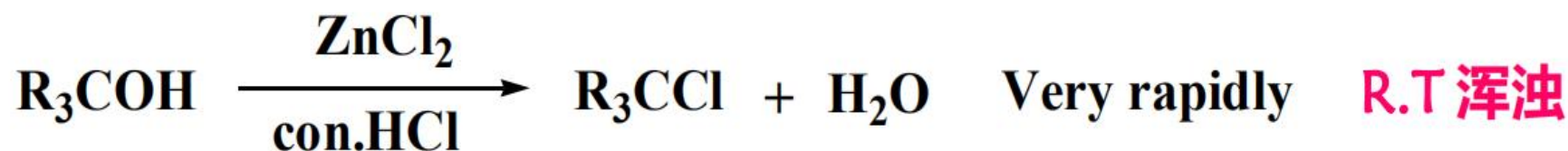
不反应
需加热

片刻
沉淀

立即
沉淀



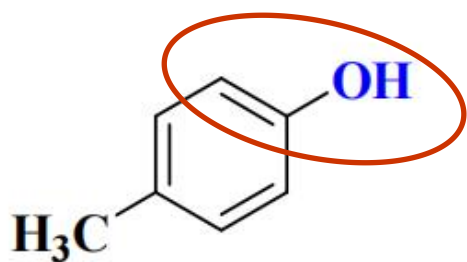
醇与Lucas试剂(无水氯化锌的浓盐酸)反应



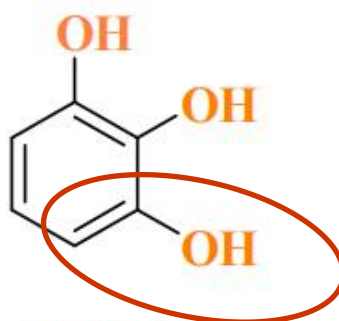
酚--与FeCl₃的显色反应



紫色



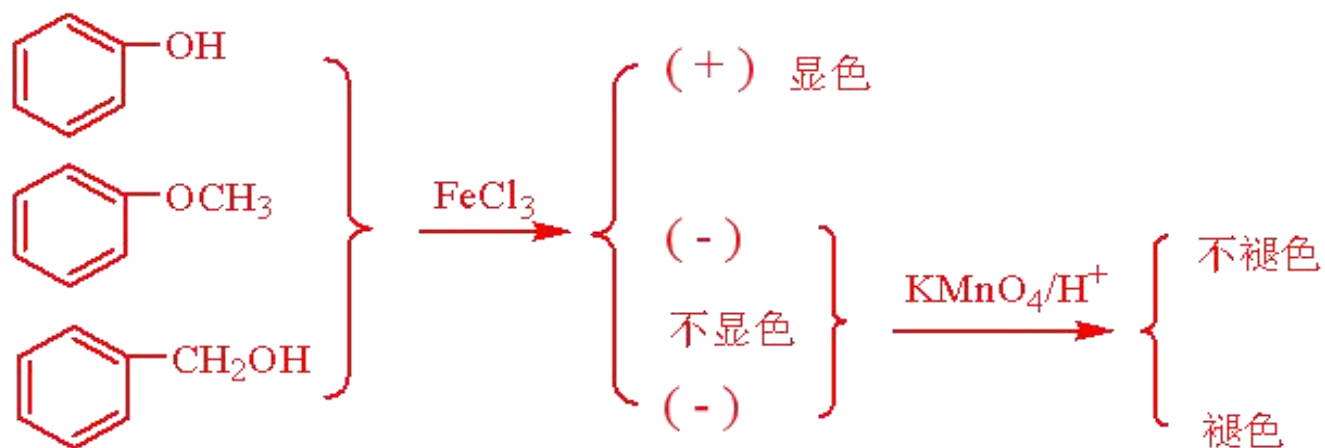
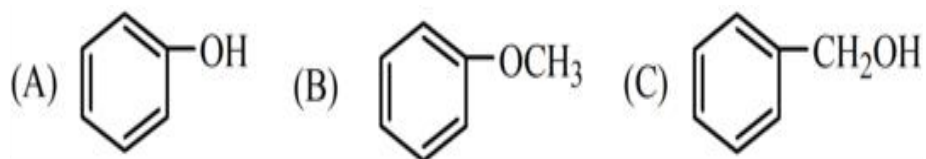
蓝色



淡棕色

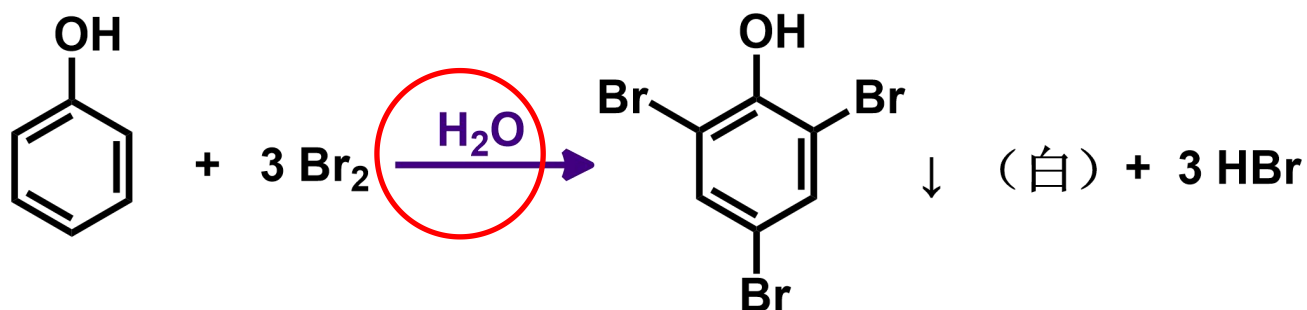
可用于鉴别苯酚、萘酚等含有烯醇结构的化合物。

酚--与FeCl₃的显色反应

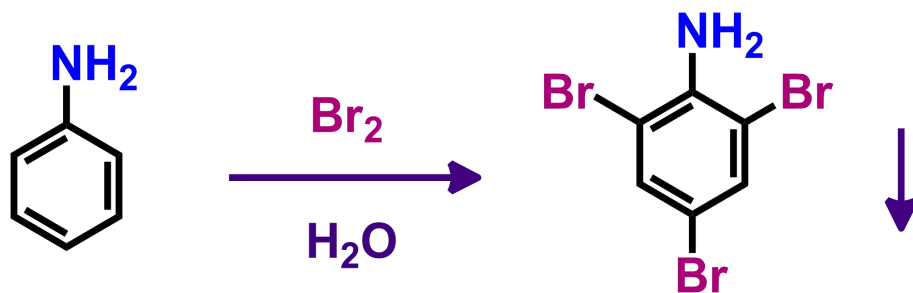


醇可以使 KMnO₄ 褪色

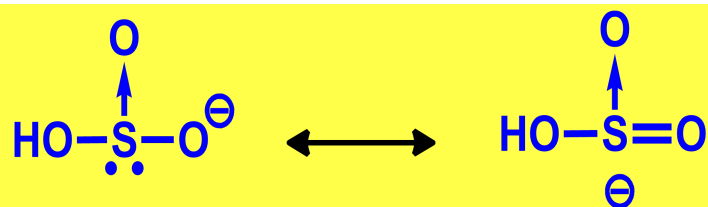
酚--与Br₂反应，生成白色沉淀 (+)



酚--与Br₂反应，生成沉淀 (+)

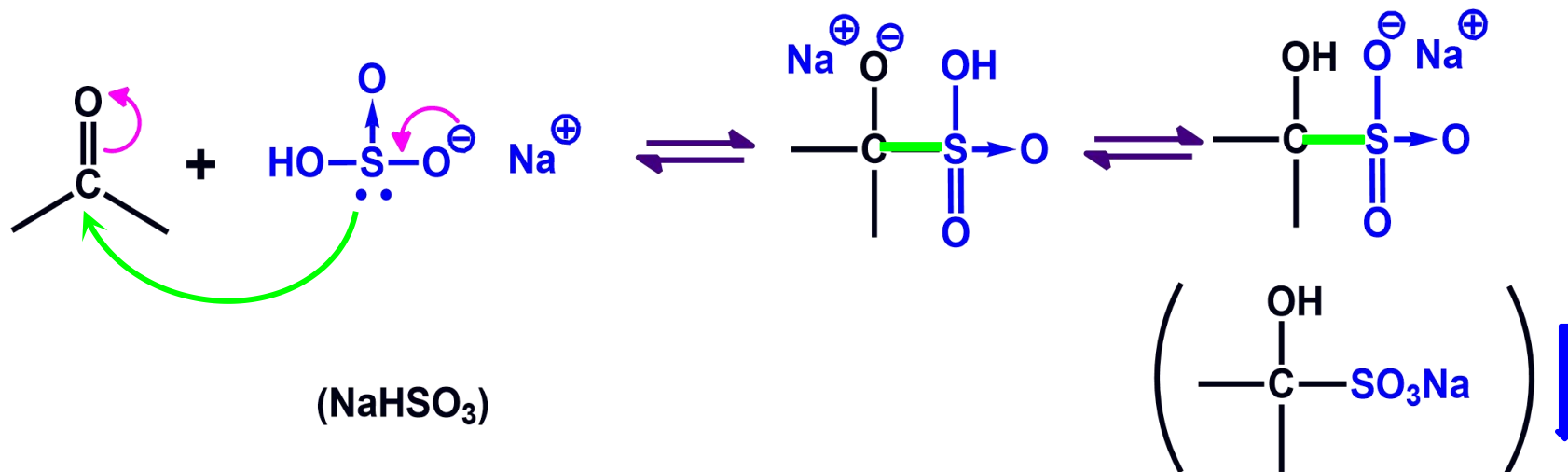


醛酮鉴定——与亚硫酸氢钠的加成



S的电子云密度比O高，
S的原子半径比O大，电子云更加
松散

S的亲核性更大，是S进攻C=O



鉴定：

- ✓ 醛（芳香醛、脂肪醛）
- ✓ 脂肪族甲基酮
- ✓ 8个C以下的环酮

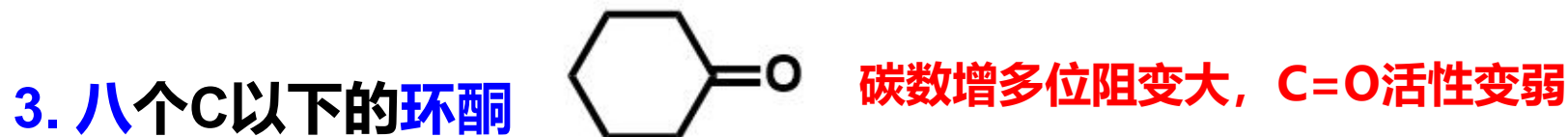
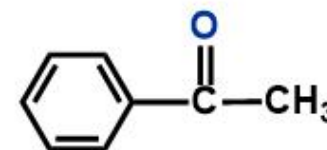
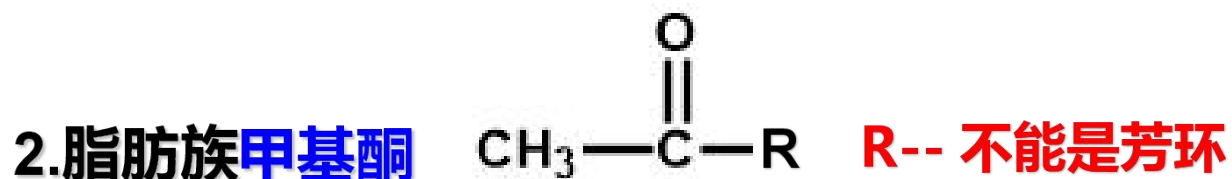
白色结晶物

亚硫酸氢钠鉴定，适应的条件

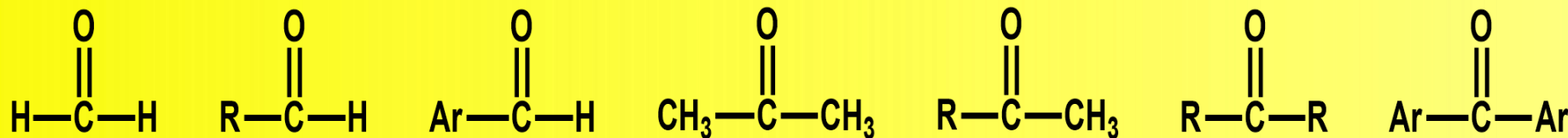


1. 醛（芳香醛、脂肪醛）

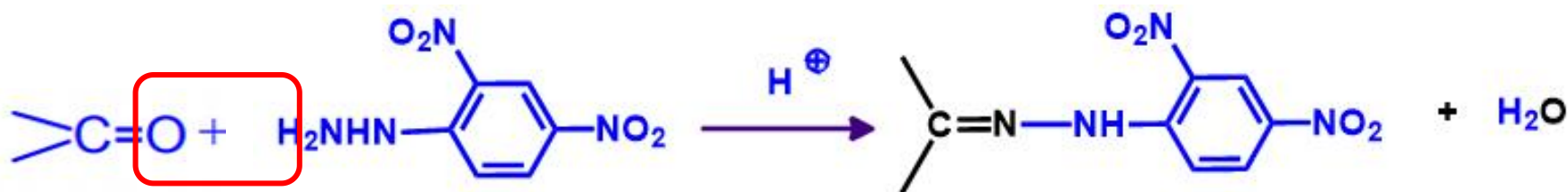
---醛活性高，所有醛可以反应



哪些不能用NaHSO₄鉴定



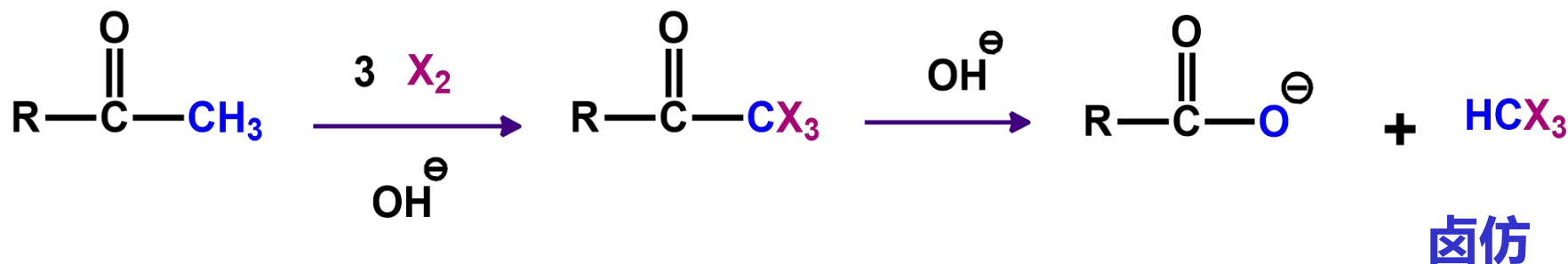
□ 醛酮与2, 4 - 二硝基苯肼反应, 鉴定醛酮



2, 4 - 二硝基苯肼

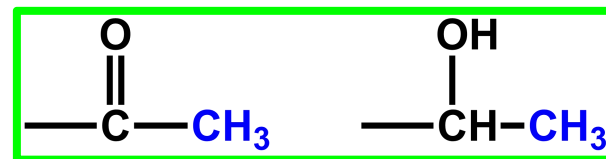
2, 4 - 二硝基苯腙 (黄色固体)

卤仿反应:



应用:

✓ 有机分析: 鉴定甲基酮 (碘仿试验)



✓ 有机合成: 通过甲基酮合成少一个碳原子的羧酸

碘仿反应: 试剂: $\text{I}_2 + \text{NaOH}$

黄色沉淀

托伦试剂与斐林试剂



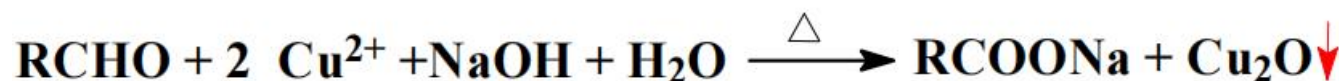
Tollens试剂— $\text{Ag}(\text{NH}_3)_2\text{OH}$ 溶液



银镜

Tollens试剂只与醛作用不与酮作用—鉴别醛；区别酮

Fehling试剂— CuSO_4 与酒石酸钾钠碱溶液的混合液



红色沉淀

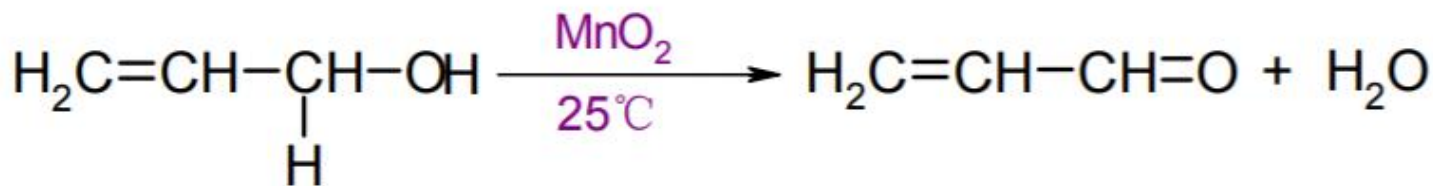
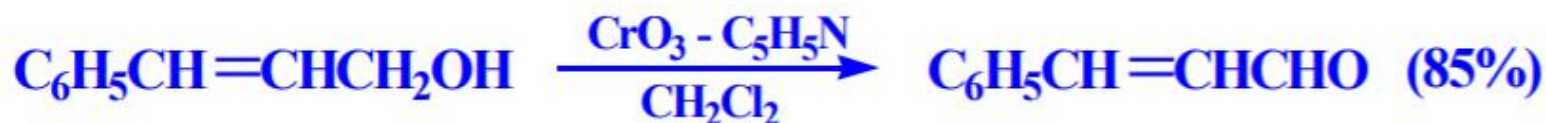
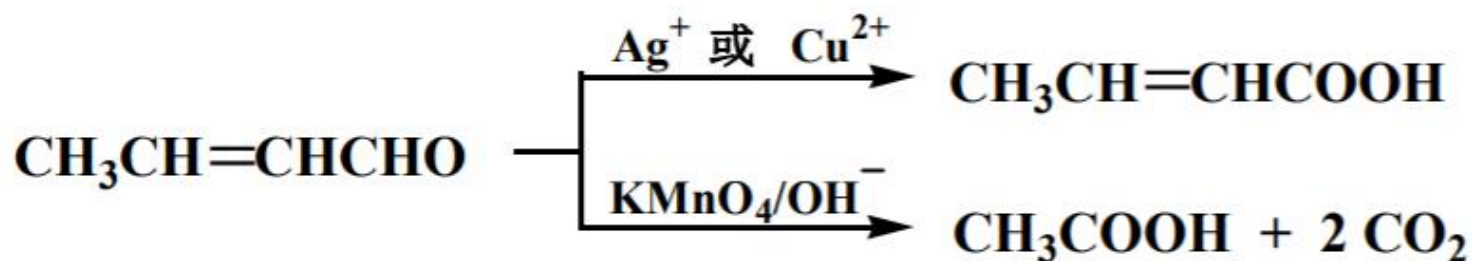
Fehling试剂只与脂肪醛作用，不与芳香醛作用

鉴别：醛是脂肪 Or 芳香醛

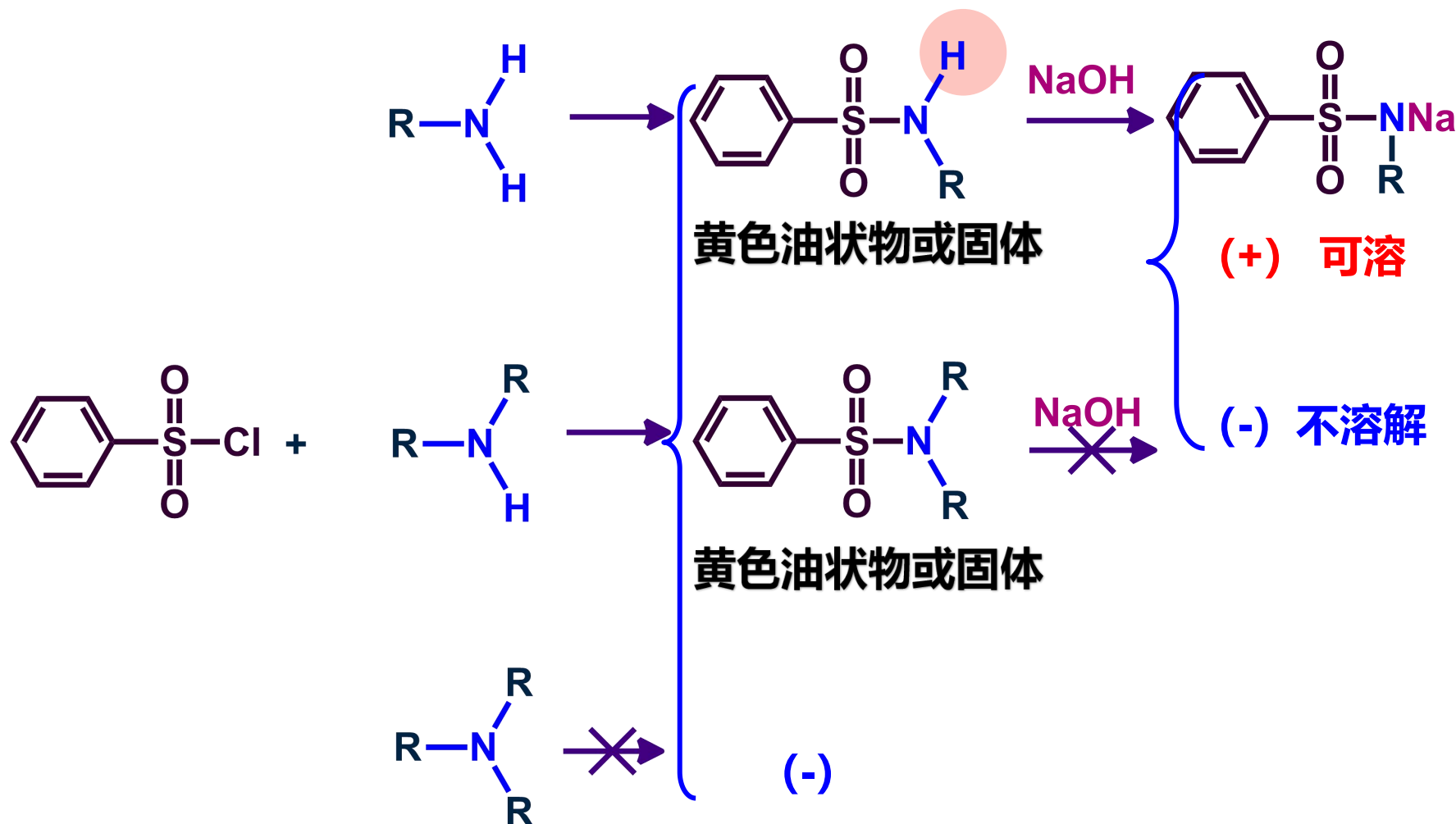
托伦试剂与斐林试剂均是弱氧化剂



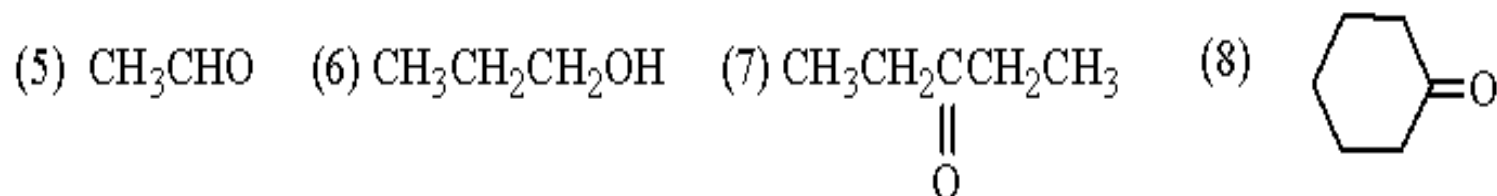
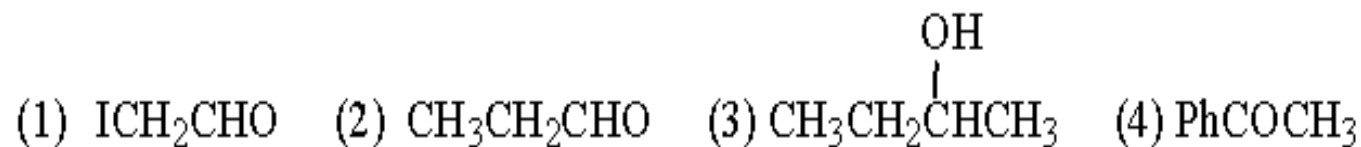
Tollens试剂和Fehling试剂对C=C、C≡C不起反应，因此它们又都可看作是选择性氧化剂。类似的氧化剂还有CrO₃和MnO₂等。



Hinsberg 反应，可以用来分离和鉴别伯、仲、叔胺。



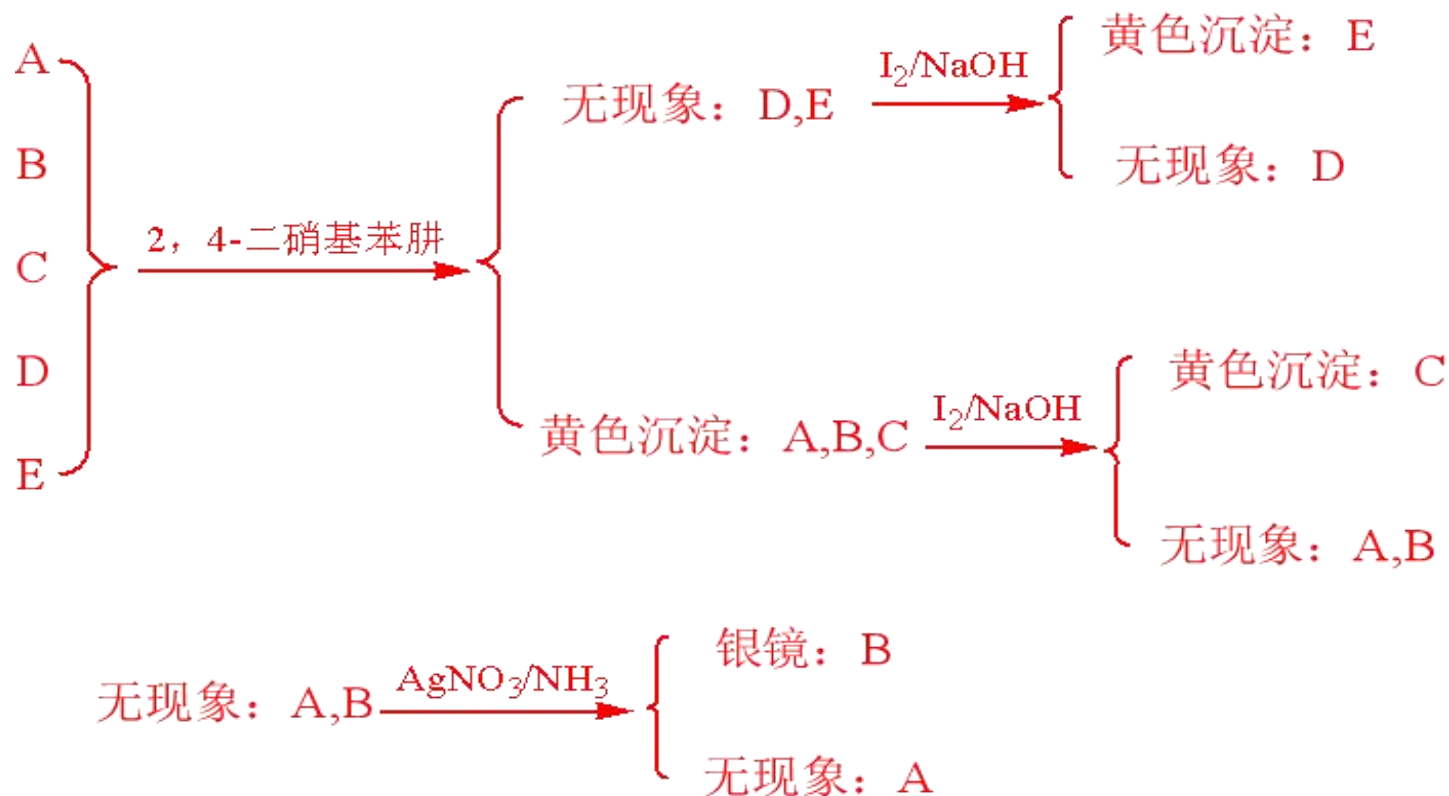
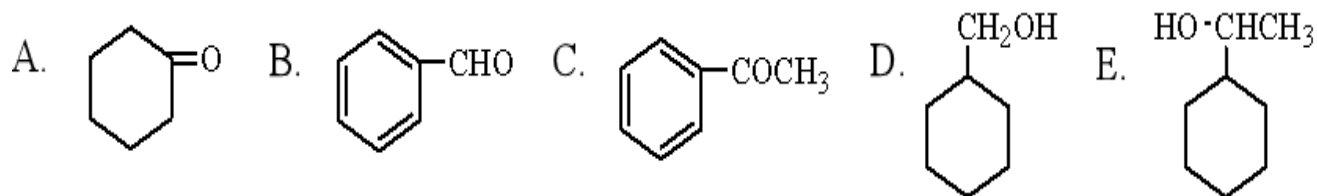
指出下列化合物中，哪个能发生碘仿反应？哪个能与饱和 NaHSO_3 反应？



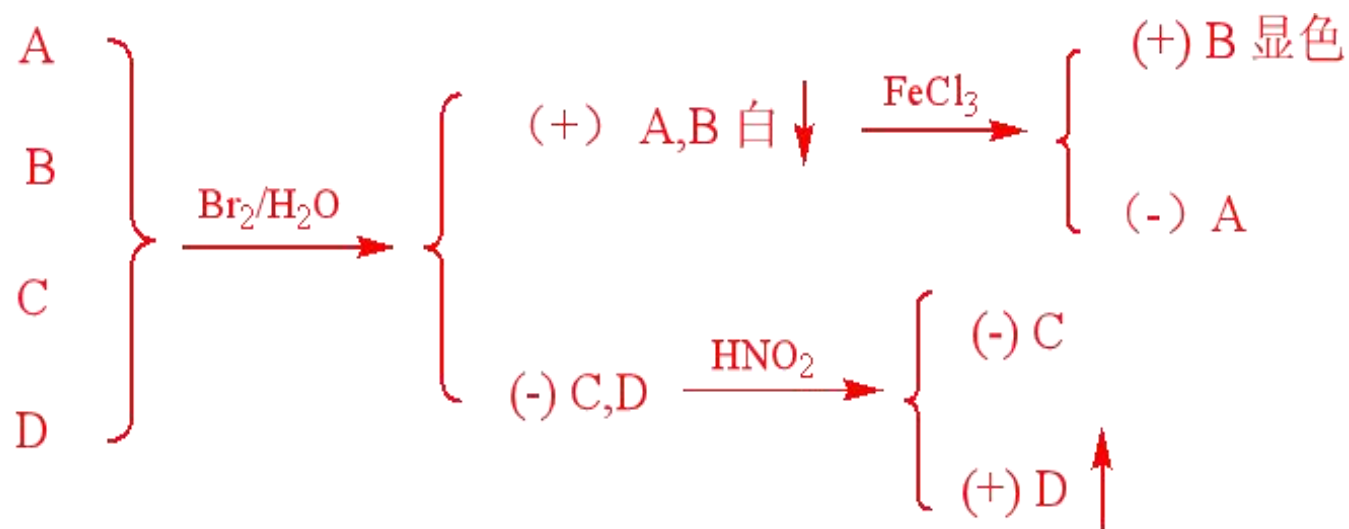
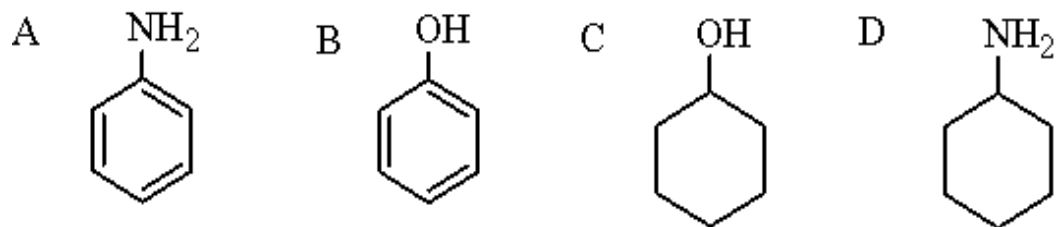
碘仿反应：(1)、(3)、(4)、(5)

饱和 NaHSO_3 反应：(1)、(2)、(5)、(8)

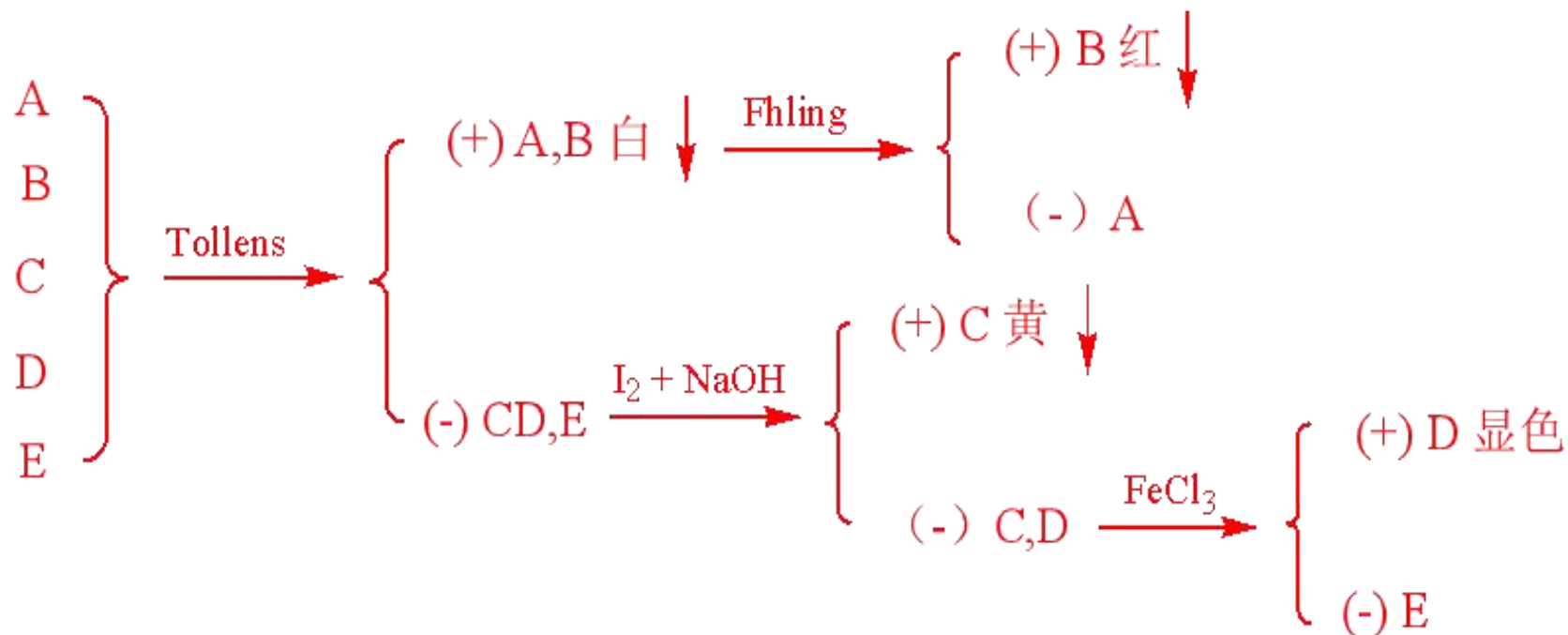
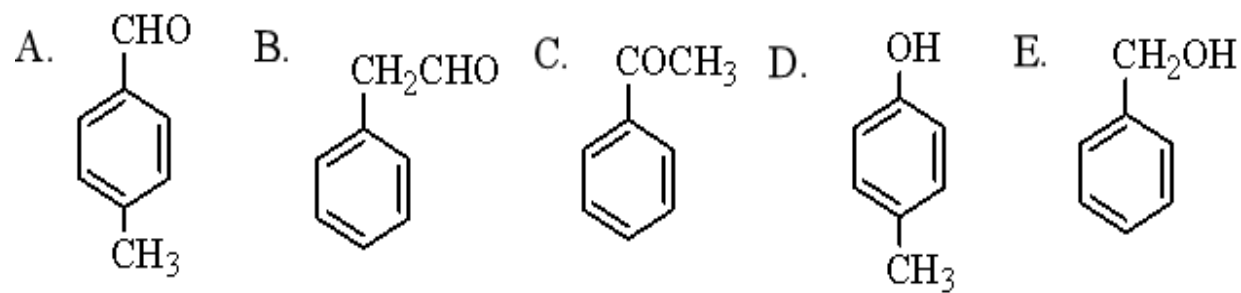
用简单化学方法区别下组化合物

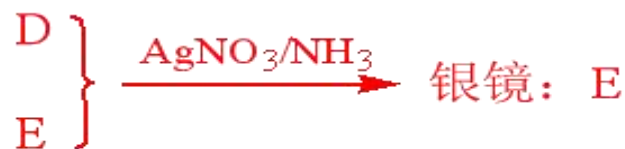
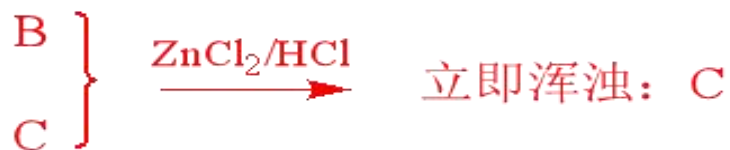
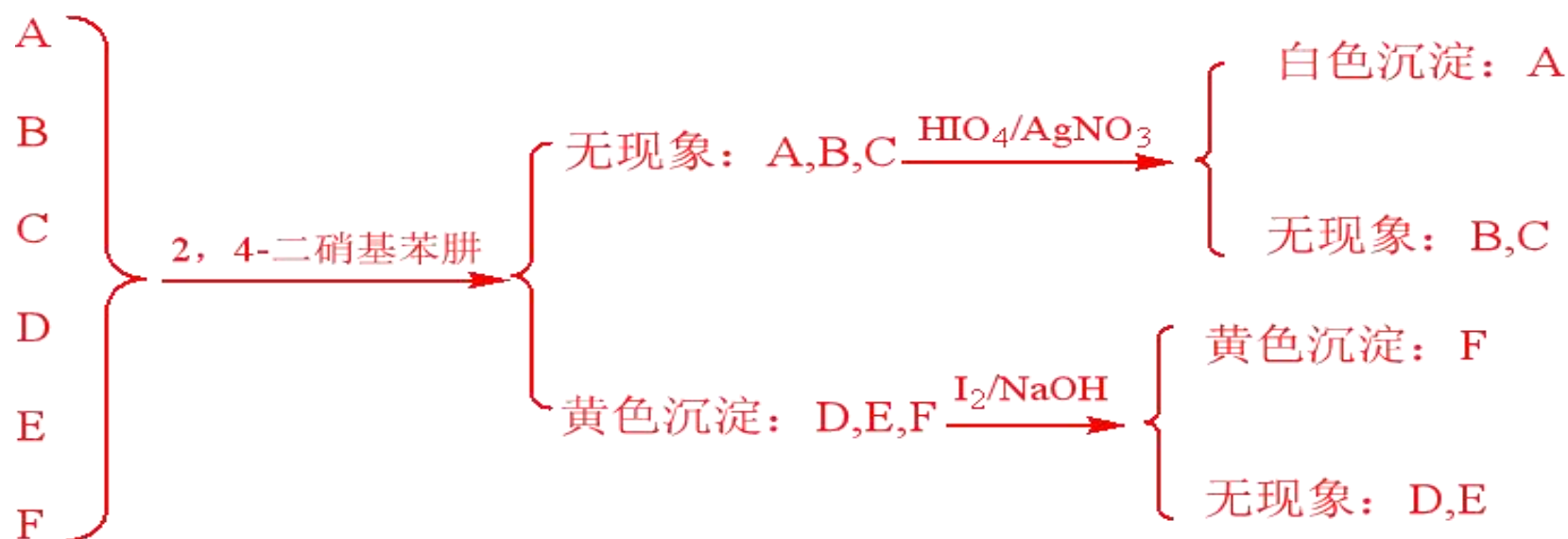
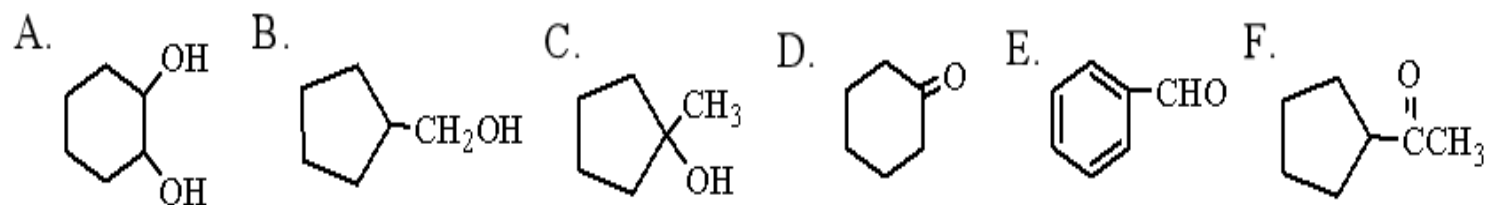


用简单的化学方法鉴别下列化合物：

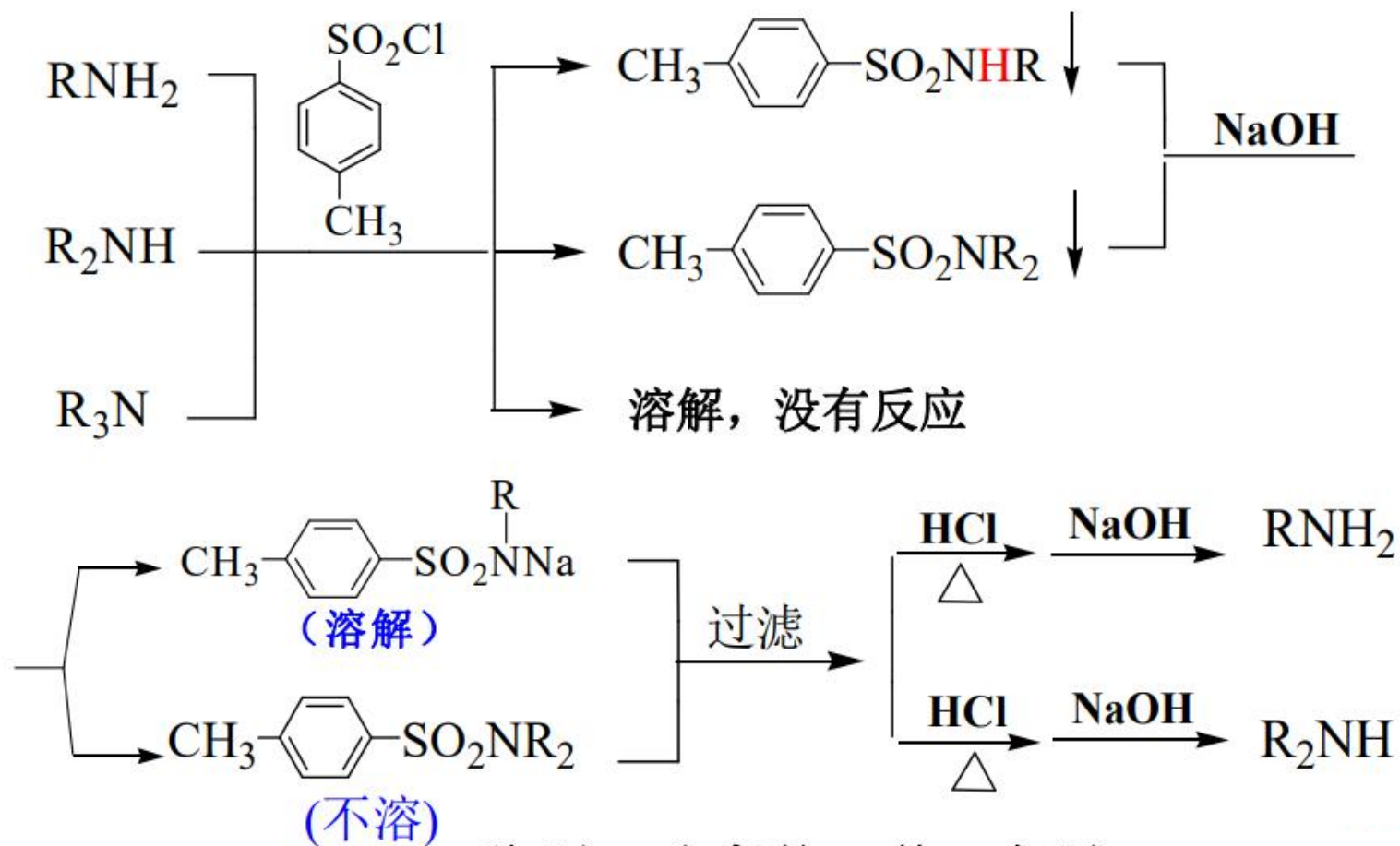


用简单化学方法区别下组化合物





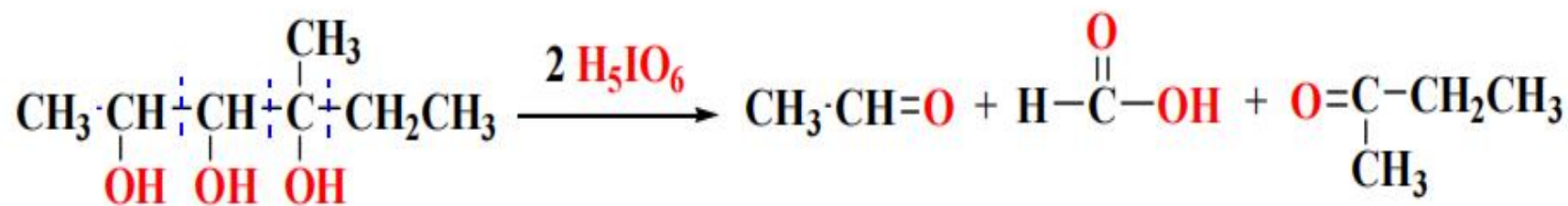
鉴别伯仲叔胺



鉴别、分离伯、仲、叔胺。

五、推测结构【每小题5分，共计10分】

相邻多元醇也可发生这种反应，例如

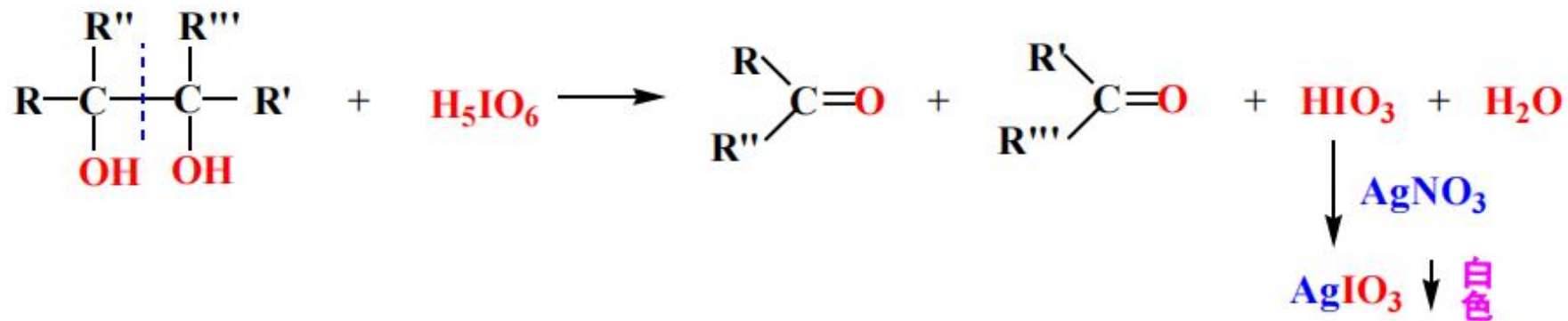


- ◆ 在 H_2O 中反应，可鉴别邻位二醇或邻位多元醇；
- ◆ 反应几乎是定量，由消耗的高碘酸的量，可测定邻二醇的量
- ◆ 根据生成的产物，可推断邻二醇的结构；

邻二醇的氧化



高碘酸/高碘酸钠 ($\text{H}_5\text{IO}_6/\text{NaIO}_4$) 氧化

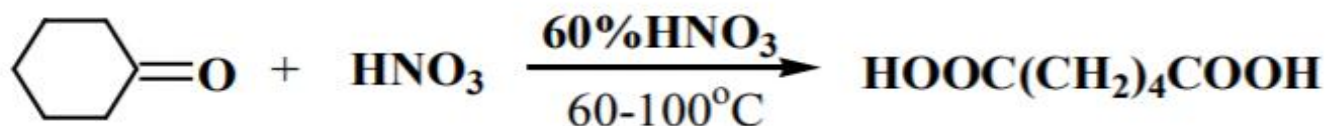


对比：强氧化剂

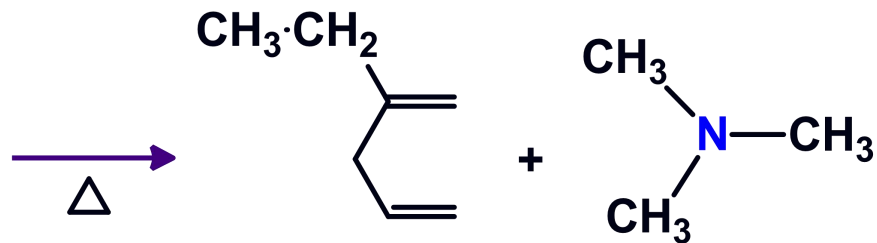
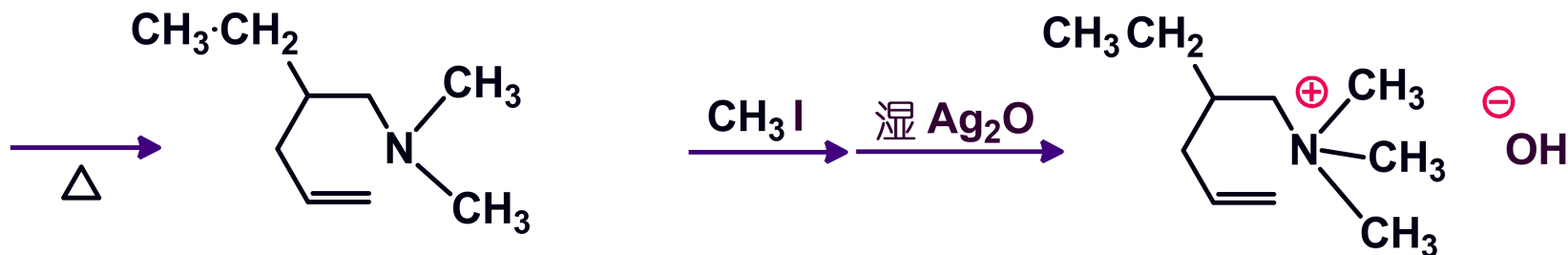
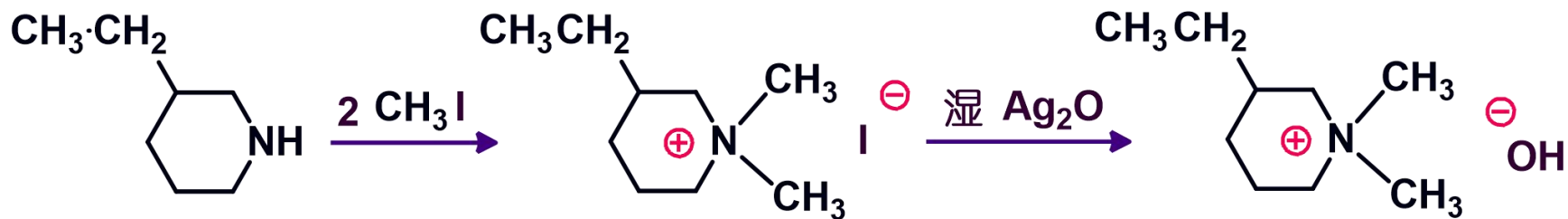
酮 遇一般氧化剂，不容易被氧化；但是酮遇强烈氧化剂时，羰基与 α -碳之间断裂，形成羧酸。



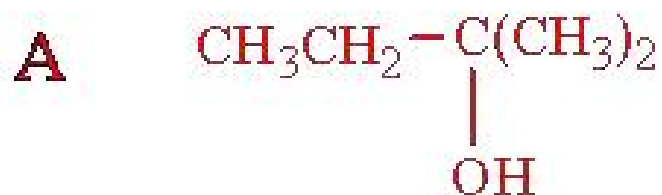
酮氧化时碳碳键断裂，一般地生成混合物，意义不大，但是环酮氧化后可得单一化合物，在制备上有意义。如：



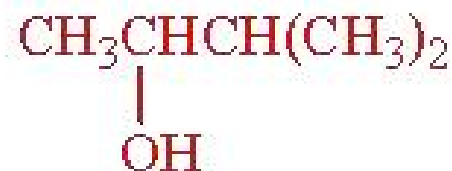
Hofmann消除反应用于结构推断



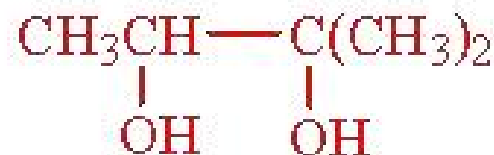
某种化合物 $\text{C}_5\text{H}_{12}\text{O}$ (A) 很易失水成B, B用冷稀 KMnO_4 氧化得 $\text{C}_5\text{H}_{12}\text{O}_2$ (C), C与高碘酸作用得一分子乙醛和另一化合物酮, 试写出A的可能结构和各步反应。



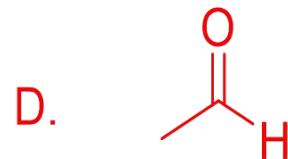
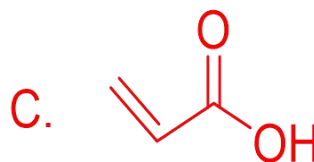
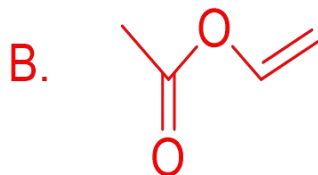
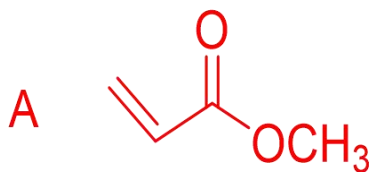
或



C



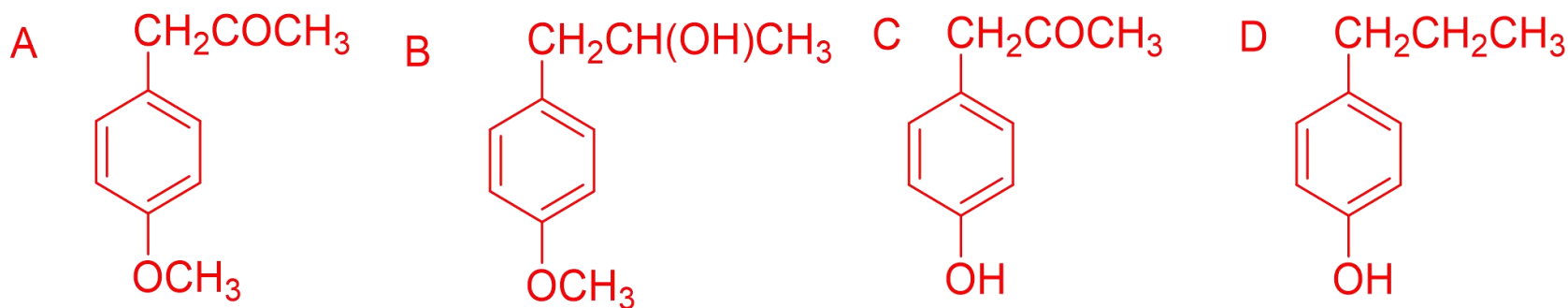
有两个酯类化合物(A)和(B)，分子式均为 C_4H_6O 。(A)在酸性条件下，水解生产甲醇和另外一个化合物 $C_3H_4O_2$ (C)，(C)可使 Br_2-CCl_4 溶液褪色，(B)在酸性条件下水解成一分子羧酸和化合物(D)，(D)可发生碘仿反应，也可以与Tollens试剂反应。试写出(A)~(D)可能的构造式。



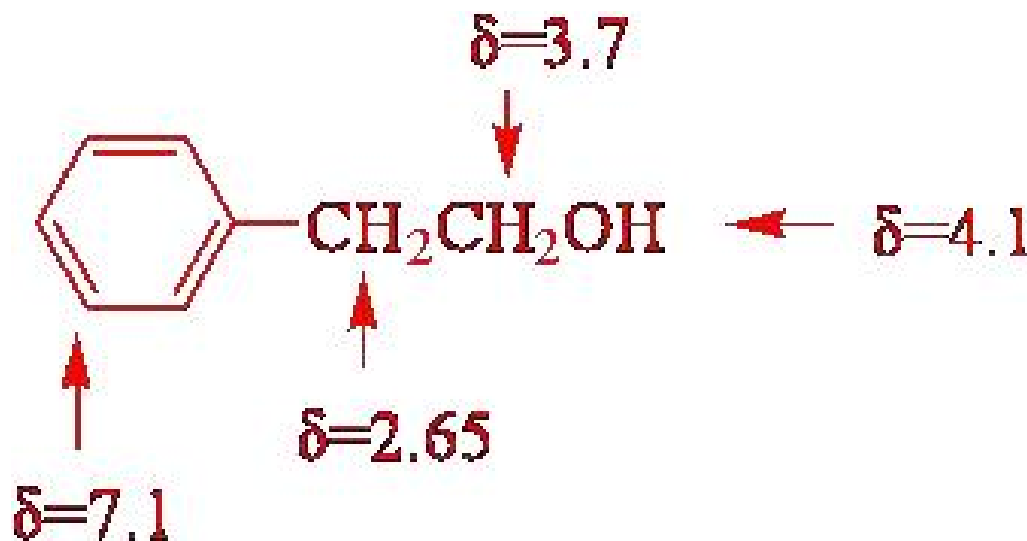
化合物 (A) 的组成为 $C_5H_{10}O$ ；用 $KMnO_4$ 小心氧化 (A) 得到组成为 C_5H_8O 的化合物 (B)。(A) 与无水 $ZnCl_2$ 的浓盐酸溶液作用时，生成化合物 (C)，其组成为 C_5H_9Cl ；(C) 在 KOH 的乙醇溶液中加热得到唯一的产物 (D)，组成为 C_5H_8 ；(D) 再用 $KMnO_4$ 的硫酸溶液氧化，得到一个直链二羧酸。试写出 (A)，(B)，(C)，(D) 的结构式，并写出各步反应式。



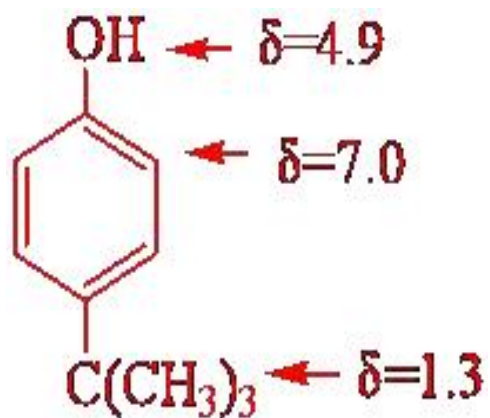
化合物 $C_{10}H_{12}O_2$ (A)不溶于NaOH溶液，能与2,4-二硝基苯肼反应，但不与Tollens试剂作用。(A)经 $LiAlH_4$ 还原得 $C_{10}H_{14}O_2$ (B)。(A)和(B)都进行碘仿反应。(A)与HI作用生成 $C_9H_{10}O_2$ (C)，(C)能溶于NaOH溶液，但不溶于 $NaHCO_3$ 溶液。(C)经Clemmensen还原生成 $C_9H_{12}O$ (D)；(C)经 $KMnO_4$ 氧化得对羟基苯甲酸。试写出(A)~(D)可能的构造式。



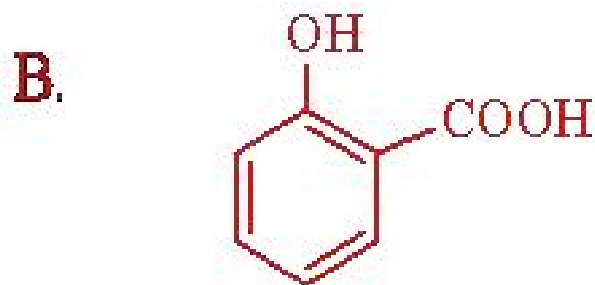
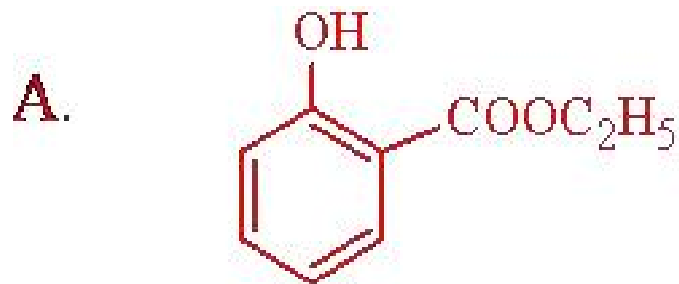
(1) 化合物A是液体，b.p.为220°C，分子式为C₈H₁₀O，IR在3400cm⁻¹和1050cm⁻¹有强吸收，在1600cm⁻¹、1495cm⁻¹和1450cm⁻¹有中等强度的吸收峰，¹H-NMR：δ=7.1（单峰）、δ=4.1（单峰）、δ=3.7（三重峰）、δ=2.65（三重峰），峰面积之比5：1：2：2。推测A的结构。



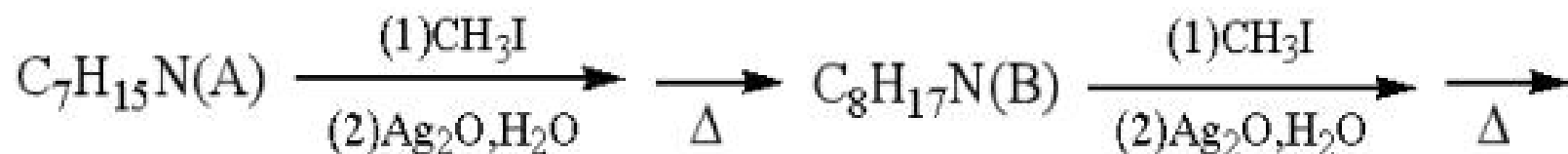
某化合物 $\text{AC}_{10}\text{H}_{14}\text{O}$ ，溶于 NaOH 水溶液但不溶于 NaHCO_3 水溶液，用溴水与A反应得到二溴衍生物 $\text{C}_{10}\text{H}_{12}\text{Br}_2\text{O}$ ，IR在 3250cm^{-1} 处有一宽峰，在 830cm^{-1} 处出有一吸收峰，A的 $^1\text{H-NMR}$ 数据如下： $\delta=1.3$ （单峰、9H）， $\delta=4.9$ （单峰、1H）、 $\delta=7.0$ （多重峰、4H），试推测A的结构。



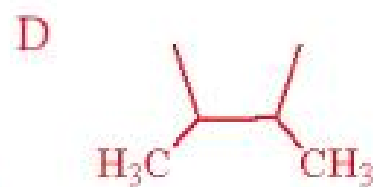
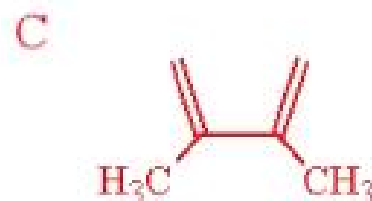
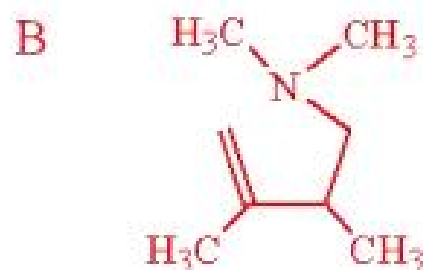
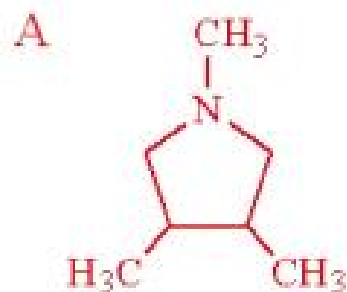
化合物A的分子式为 $C_9H_{10}O_3$ 。它不溶于水、稀HCl、稀 $NaHCO_3$ 溶液，但能溶于NaOH溶液。A与稀NaOH溶液共热后，冷却酸化得一沉淀B，分子式为 $C_7H_6O_3$ ，B能溶于 $NaHCO_3$ 溶液并放出气体，B与 $FeCl_3$ 反应呈现紫色，B在酸性介质中可以进行水蒸气蒸馏。写出A，B的构造式。



2. 根据以下事实，推出A~D的结构。



D的¹HNMR有一组多重峰和一组二重峰，峰面积之比为1：6。

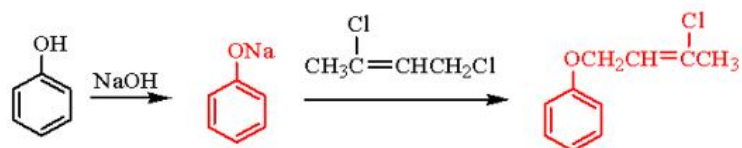
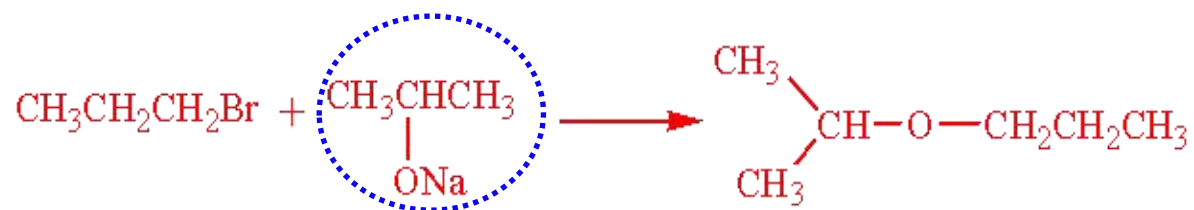
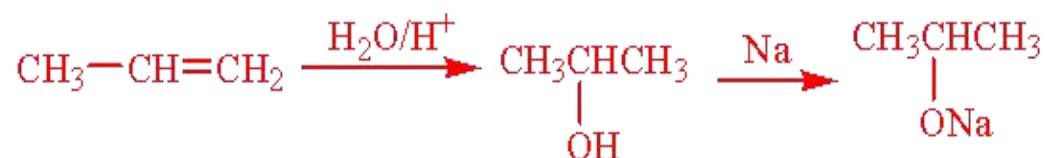


六、合成题

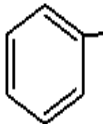
- 有机原料指定，无机试剂任选
- 每小题5分，共计20分

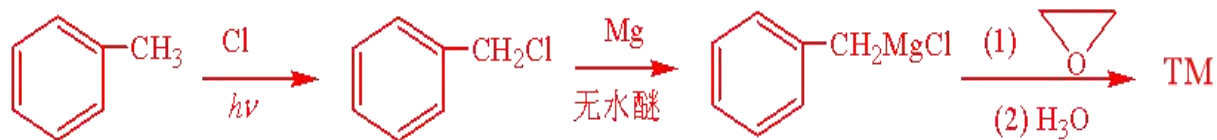
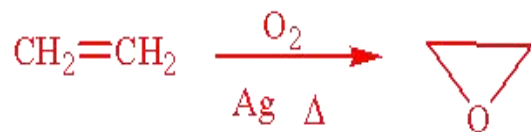
醚的合成

1. 以丙烯为原料合成: $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CH} \\ \diagdown \\ \text{CH}_3 \end{array} \text{---O---} \text{CH}_2\text{CH}_2\text{CH}_3$

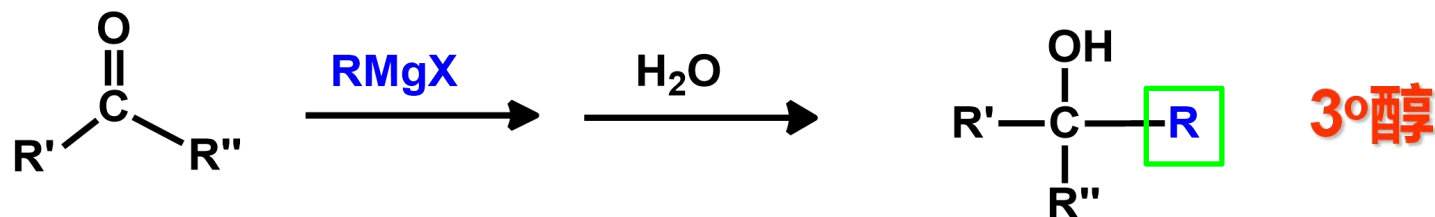
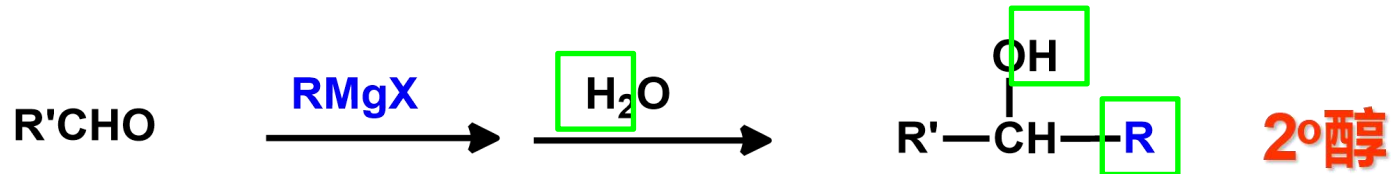
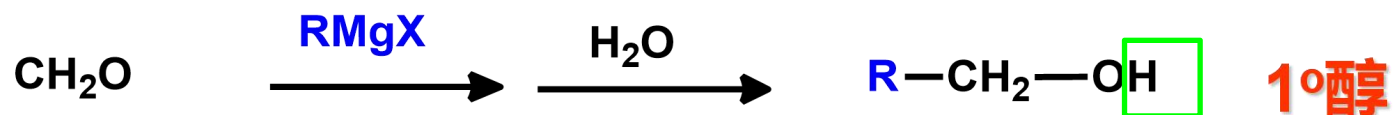
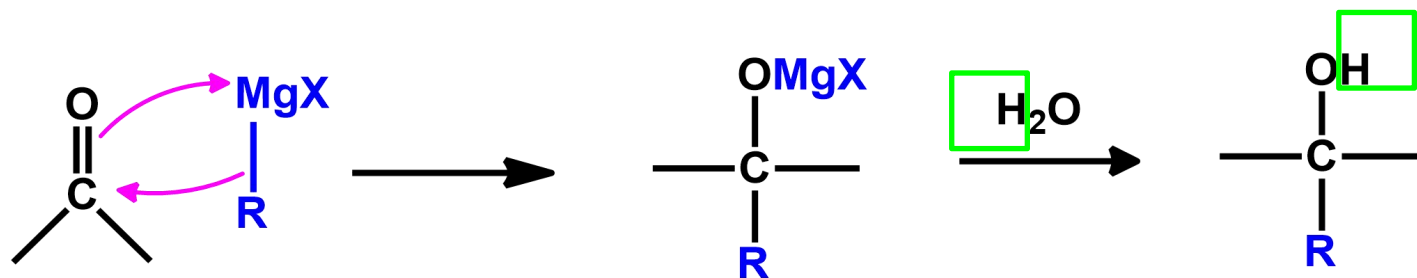


通过环氧乙烷，合成碳链增加两个碳的醇

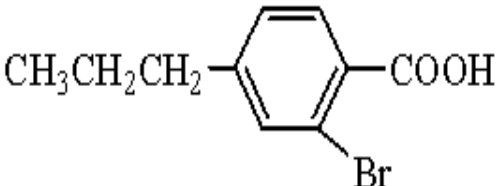
2. 以甲苯和乙烯为原料合成： $\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$



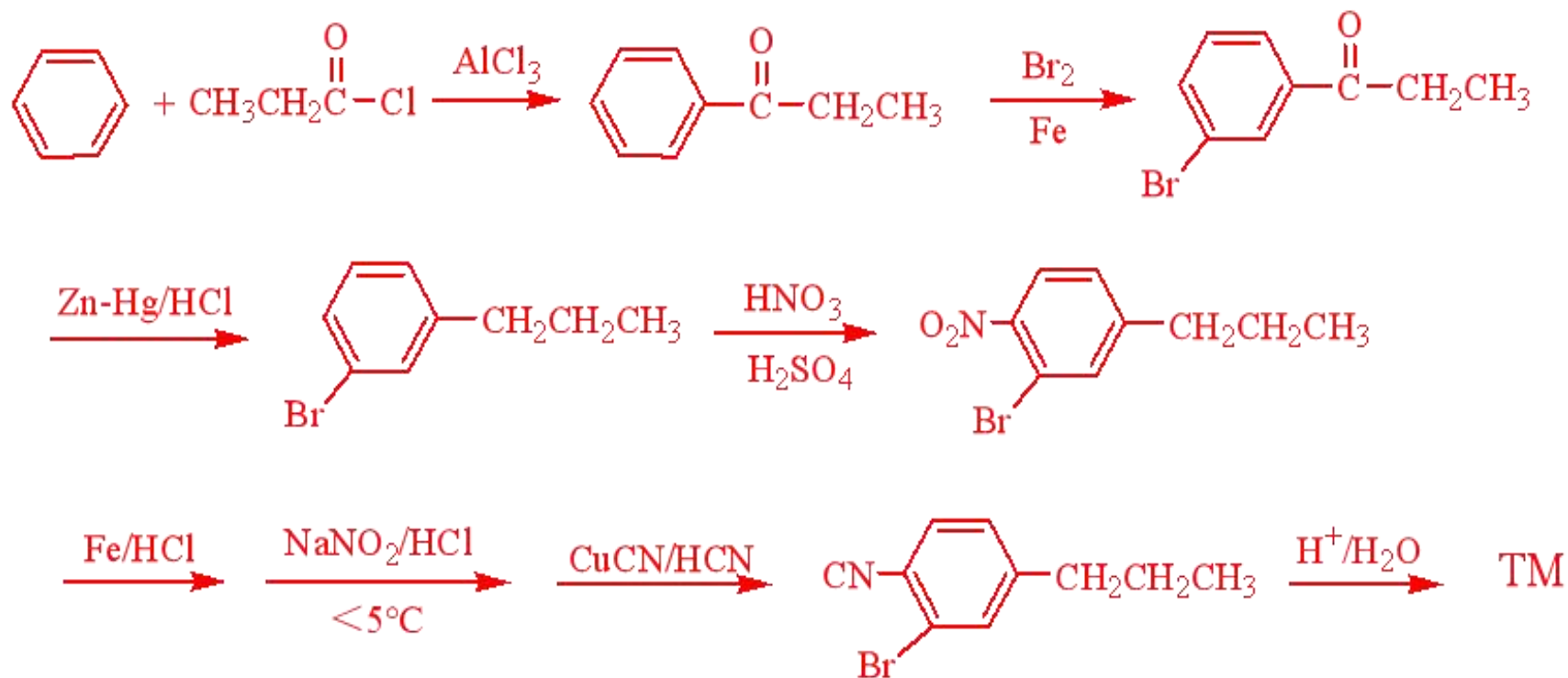
合成各类醇



苯环上多取代，顺序及定位基

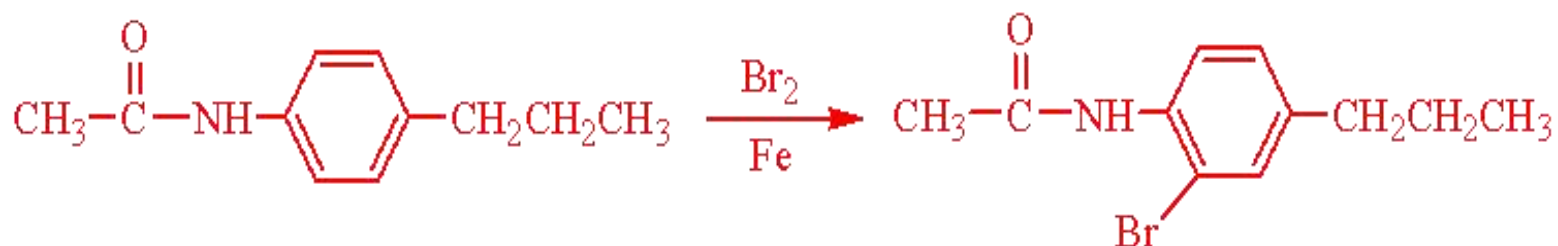
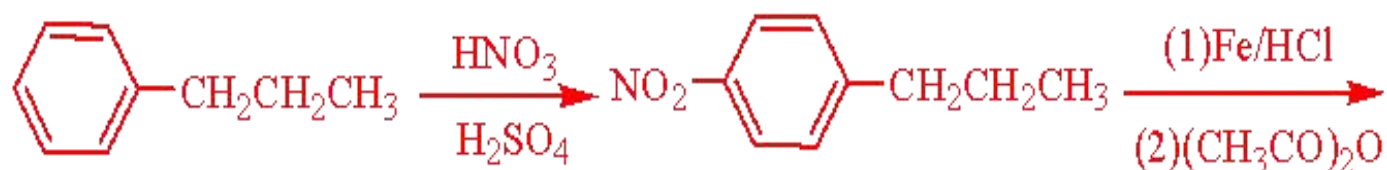
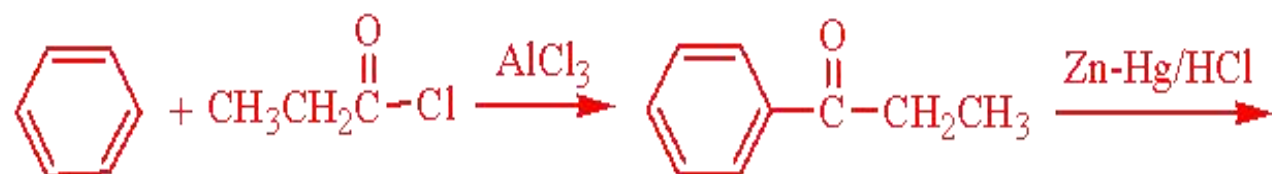
1. 以苯和C₄的有机物为原料合成：

方法1

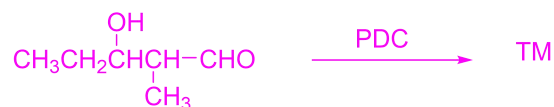
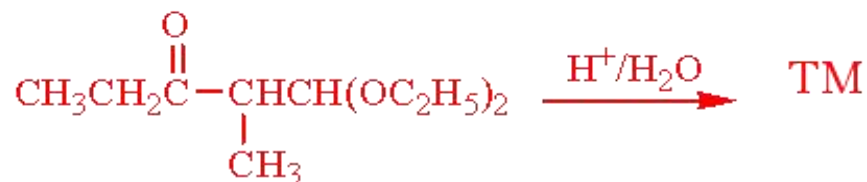
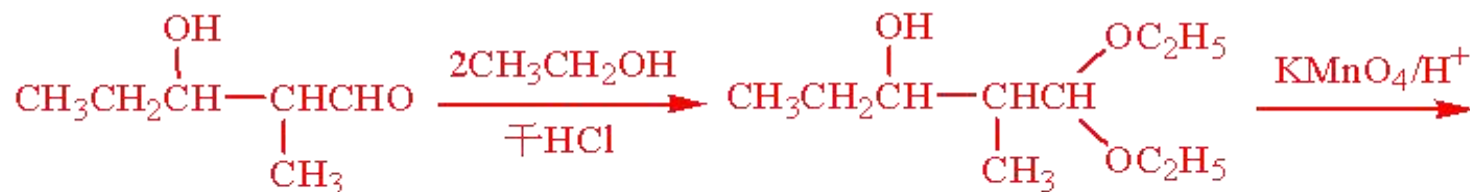
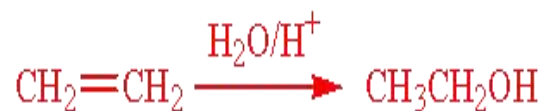
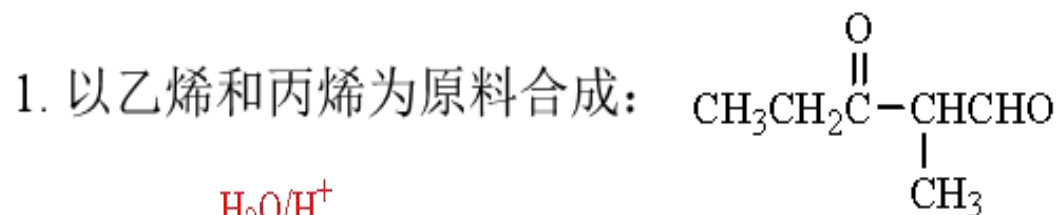


苯环上多取代，顺序及定位基

方法2

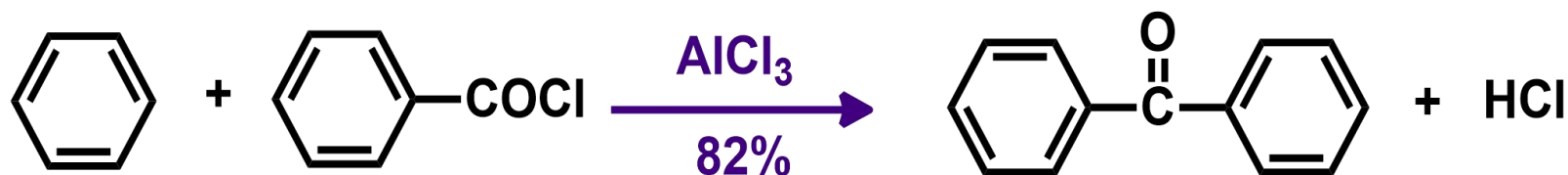


羟醛缩合--合成醛酮



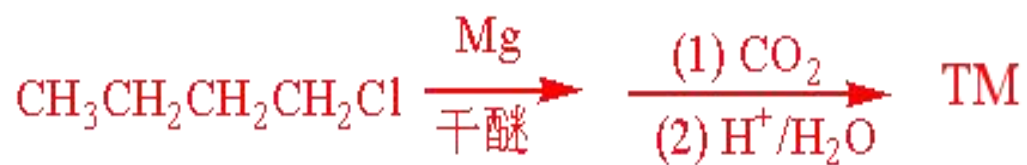
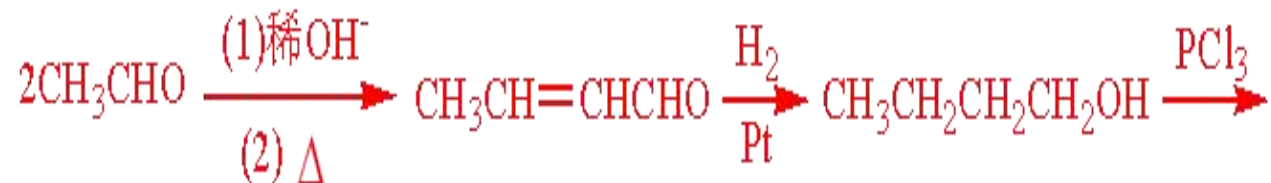
(不保护, 直接选择性氧化, 也可以)

合成酮（通过傅-克，引入酰基）

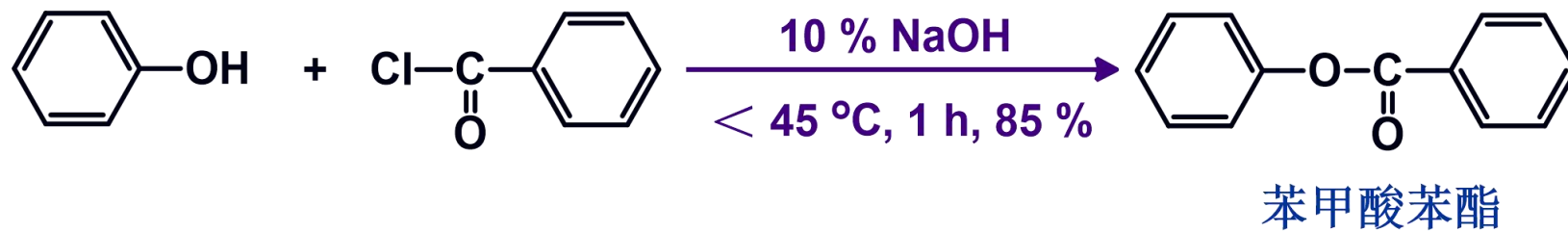
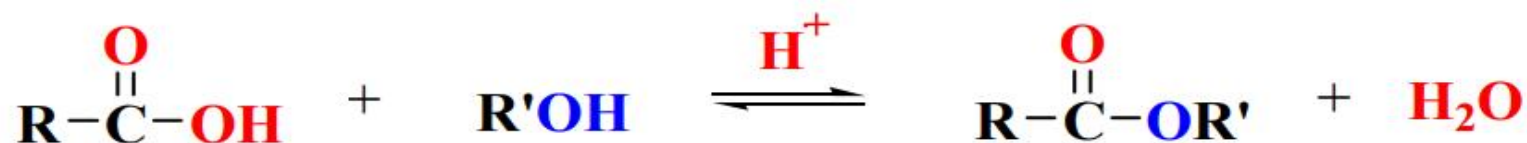


合成酸

3. 以乙醛为原料合成: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$

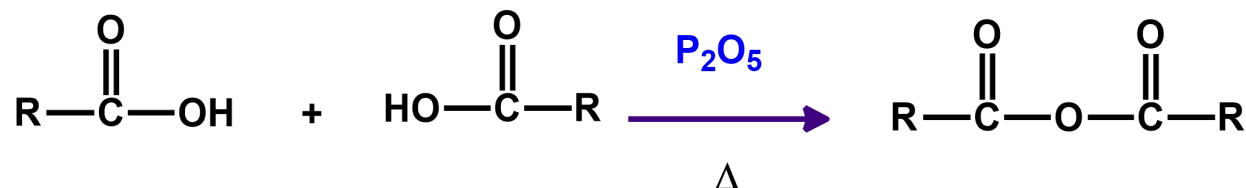


合成酯

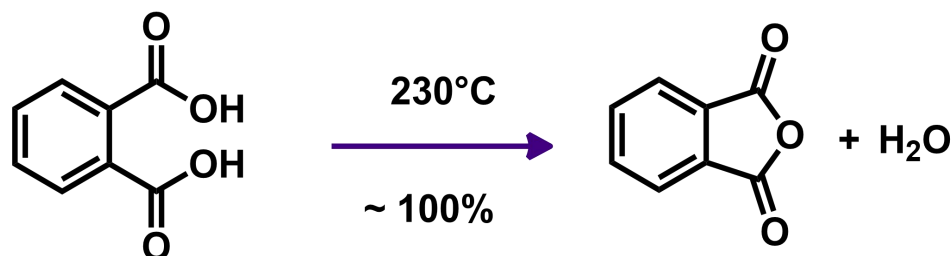


合成酸酐

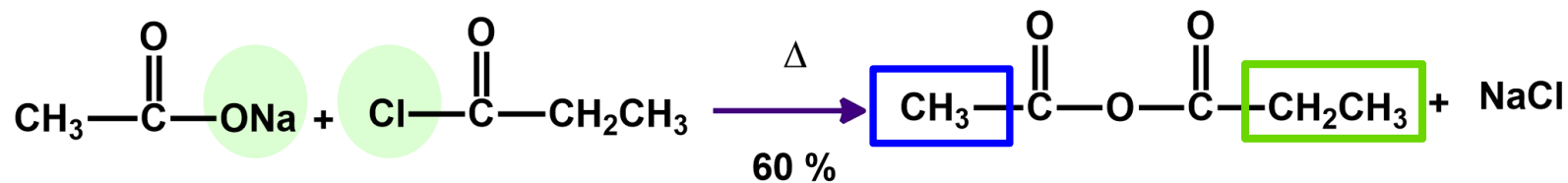
■ 脱水剂



■ 环状酸酐

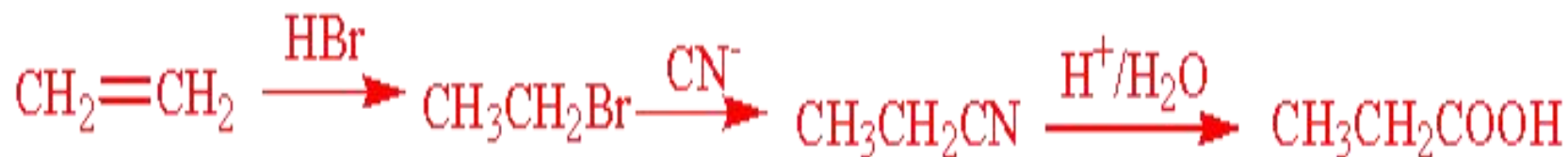


■ 混合酸酐

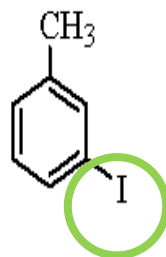


合成酰胺

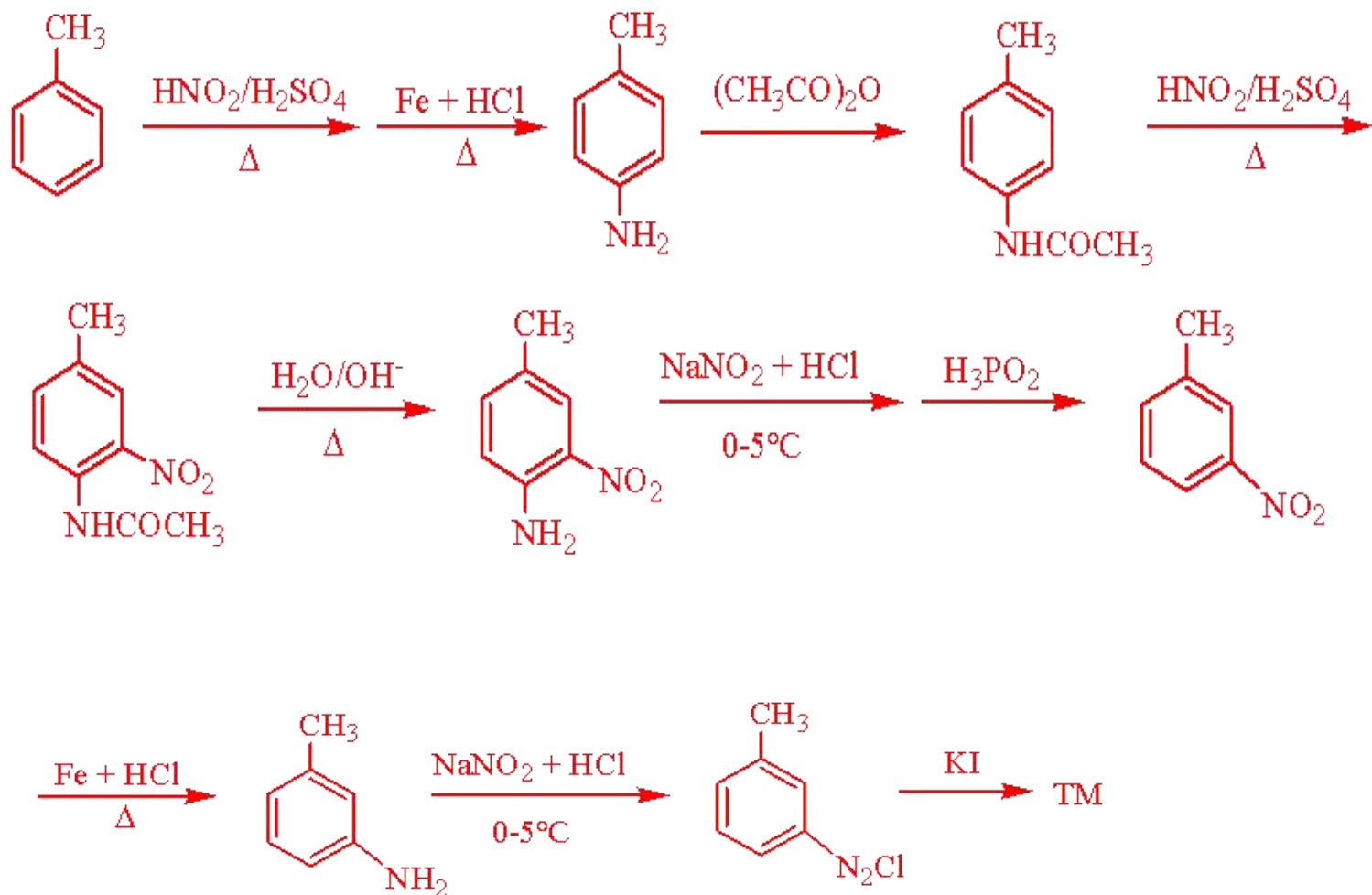
1. 以乙烯为原料合成: $\text{CH}_3\text{CH}_2\overset{\text{O}}{\parallel}\text{C}-\text{NH}(\text{CH}_2\text{CH}_3)$



以甲苯及C<4的有机试剂为原料合成

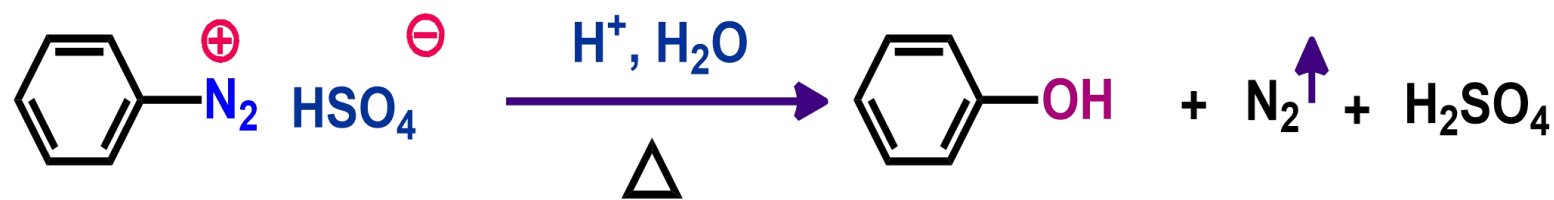


通过重氮盐，制备碘化物



通过重氮盐，制备酚

例：



机理：

